GE VERNOVA

ADMS

16.25.1

OMS Server Configuration Editor User Guide

Copyright

Change History

Date	Change Description
May 31, 2023	ADMS 3.16: • Updated Showing Custom Fields in Grid Tabular Displays. • Added Summary Matrix Tab. • Editorial and formatting updates
Aug 2, 2023	ADMS 3.16.0.2: • Updated Validation Rules Tab.
Oct 3, 2023	ADMS 16.1.0: • Updated Estimated Time to Restore (ETR/ITR) Tab. • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Feb 22, 2024	ADMS 16.5.0: • Updated Call Aggregation Tab.
Apr 1, 2024	ADMS 16.7.0: • Updated document format and styling.
Apr 4, 2025	ADMS 16.16.2: • Added Incident Daily Sequence Number Rollover checkbox to Incident Name Tab.

Contents

1. Introduction **5**
2. General Usage **6**
2.1. Starting the OMS Server Configuration Editor 6
2.2. File Menu Options 6
2.3. Applying Changes to the ADMS Client 6
3. Estimated Time to Restore (ETR/ITR) Tab **7**
4. Time Period Tab **13**
5. Call Options Tab **15**
5.1. Call Action Matrix 15
5.2. Special Call Options 17
6. Chronology Message Configuration Tab **19**
6.1. Chronology Message Types and Message Configuration Table 20
6.2. Chronology Message Test 20
6.3. Chronology Message Types and Arguments List Table 21
7. Case Note Matrix Tab **24**
7.1. Case Note Dispatch Table 25
7.2. Order of Case Note Precedence 25
7.3. Translation of Case Note Codes 26
8. Crew Management Tab **27**
8.1. Customizing the Crew Name 27
8.2. Blocking ETR/ITR Recalculation due to Area Setting ETR 28
9. Incident Ranking Tab **29**
10. Incident Name Tab **30**
11. Order Name Tab **32**
12. Call Aggregation Tab **34**
12.1. Aggregation Field Increment Rules 37
12.2. Call Aggregation Rule Details 38
12.3. Adding and Modifying Rules 39
13. Custom Fields Tab **41**
13.1. Showing Custom Fields in Grid Tabular Displays 43
13.2. Showing Custom Fields in the Incident Update Display 45
13.3. Showing Custom Fields in the Update Area Settings Display 50
13.4. Showing Custom Fields in the Order Worksheet Display 51
14. Order Permissions Tab **55**
15. NonOutage Permissions Tab **57**
16. Incident Symbols Tab **59**
17. Order Symbols Tab **62**
18. Organizations Tab **65**
18.1. Organization Types 67

18.2. Organizations 68
18.3. Organizations Matrix 68
19. Incident Priority Tab 70
19.1. Configuring the Incident Priority 71
19.2. Calculating the Incident Priority 72
20. Validation Rules Tab 74
20.1. Creating an Incident Clearance Validation Rule 75
21. Call Prioritization Tab 78
21.1. Adding a New Call Prioritization Rule 79
21.2. Removing a Call Prioritization Rule 79
22. High Priority Calls Tab 80
23. Planned Outage Tab 81
24. Summary Matrix Tab 83
25. Configuring the Dynamics Instance 85
26. System Configuration 87
26.1. Modifying the Momentary Interruption Setting 87
26.2. Providing Crew Locations 87
26.3. OMS Polygon Groups 88
26.4. Provisioning STORM ETR for Planned Outage 88
27. Adding a Custom Incident Working Status 90
27.1. Defining a Custom Incident Working Status 90
27.2. Using Custom Incident Working Statuses 92
27.3. Integration with the Work Force Management System 92

1. Introduction

The OMS Server Configuration Editor is a support tool provided with ADMS.

The main purpose of this tool is to configure the Outage Management System (OMS) server. It also allows an analyst to set up rules that the system follows when certain events occur, and to configure conditions to meet the needs of individual utilities.

Some of the aspects that can be configured using the OMS Server Configuration Editor include:

- Definition of default Estimated Time to Restore (ETR) values for different outage reasons
- Creation of several event types
- Configuration of chronology messages
- Configuration of case notes
- Configuration of data aggregation from customer calls

The output of the OMS Server Configuration Editor is an XML file (`OmsServerConfig.xml`).

2. General Usage

This chapter describes how to start the editor, use the File menu commands, and save your changes to the configuration.

2.1. Starting the OMS Server Configuration Editor

To start the OMS Configuration Editor, from the Microsoft Windows Start menu, go to All Programs> Eterra> Distribution> Tools> Configuration Tools> OMS Server Configuration Editor.

2.2. File Menu Options

In the OMS Configuration Editor window, the File menu includes the following commands:

- **Open:** Opens an alternate or existing configuration file (`OmsServerConfig.xml`) and a coded translation configuration file (`CodedTranslationConfig.xml`) containing all the codes and definitions (usually found in the %IDMS_ROOT%/Config/Shared folder).
- **Save:** Saves changes to the currently open configuration file.
- **Save As:** Saves all changes to another configuration file.

  The OMS Configuration Editor allows saving the configuration file under different names. However, when it starts, it automatically opens the default configuration file (`OmsServerConfig.xml`). The default file name is defined in the OMS Configuration Editor shortcut.

- **Import ETR Data:** Imports the ETR configuration table from a CSV file.
- **Export ETR Data:** Exports the ETR configuration table to a CSV file.
- **Save XML Schema of ETR Matrix:** Exports the ETR Matrix schema to the `schema0.xsd` file in the specified directory. The schema file can be used by developers to create a C# class for the ETR matrix, and then build the ETR matrix programmatically.
- **Exit:** Closes the application.

2.3. Applying Changes to the ADMS Client

After you use the OMS Server Configuration Editor to change the configuration, those changes must be saved as an XML file.

To apply the changes to the local ADMS client, the XML file must be saved to the correct location, and the file name must be specified in the client dynamics file. The default location and file name are: %IDMS_ROOT%\config\server\OmsServerConfig.xml

3. Estimated Time to Restore (ETR/ITR) Tab

On the Estimated Time to Restore (ETR/ITR) tab, you can define the default Estimated Time to Restore (ETR) or Initial Estimated Time to Restore (ITR) values for different outage reasons.

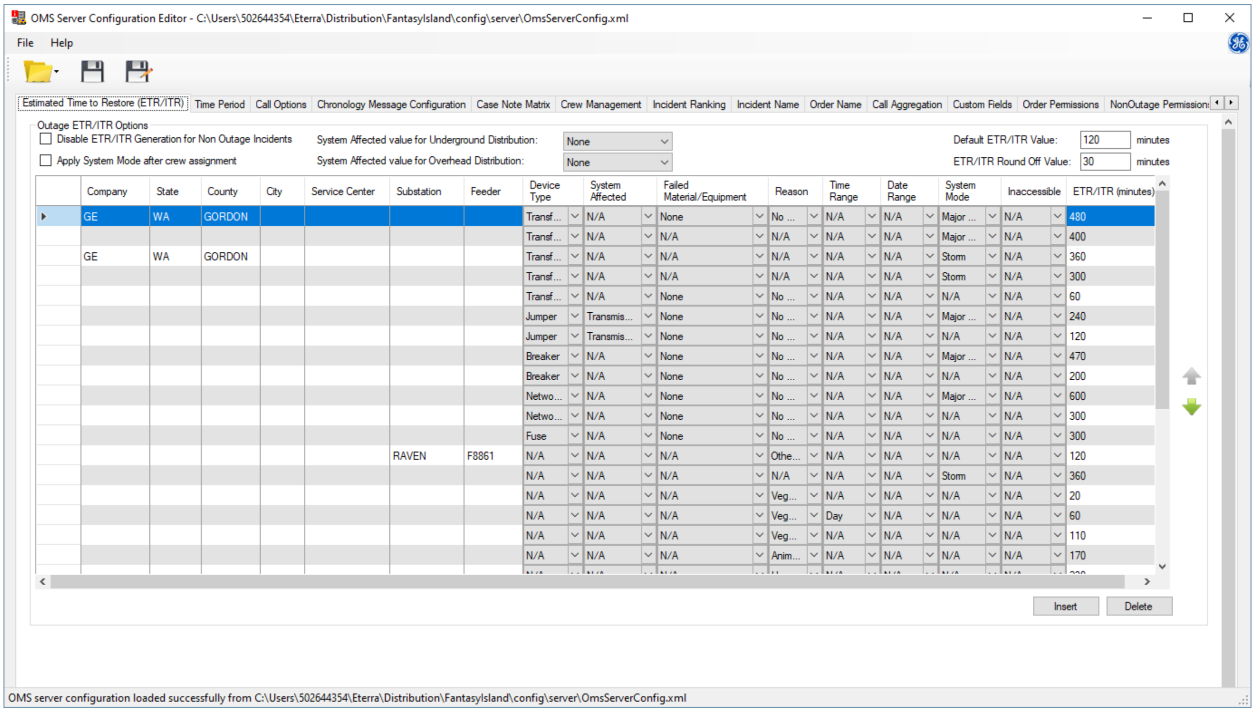


Figure 1. Estimated Time to Restore (ETR/ITR) Tab

When an outage incident is created or an incident change occurs, the system evaluates the incident's ETR/ITR based on the Outage ETR/ITR Options table (ETR/ITR matrix).

The ETR/ITR matrix is implemented as an XML file that is read by the server. It is expected that historical queries are used to periodically update the ETR/ITR matrix (for example, once per year).

Outage ETR/ITR Options Table

The Outage ETR/ITR Options table (ETR/ITR Matrix) is a table of potential outage durations with several fields that can match an incident. When an input triggers an incident's ETR/ITR to be evaluated, the system evaluates each matrix entry from beginning to end. If there is a match to the incident, the system returns the duration for the first row that matches. If no row is matched, the system returns a configurable default ETR/ITR (for example, two hours).

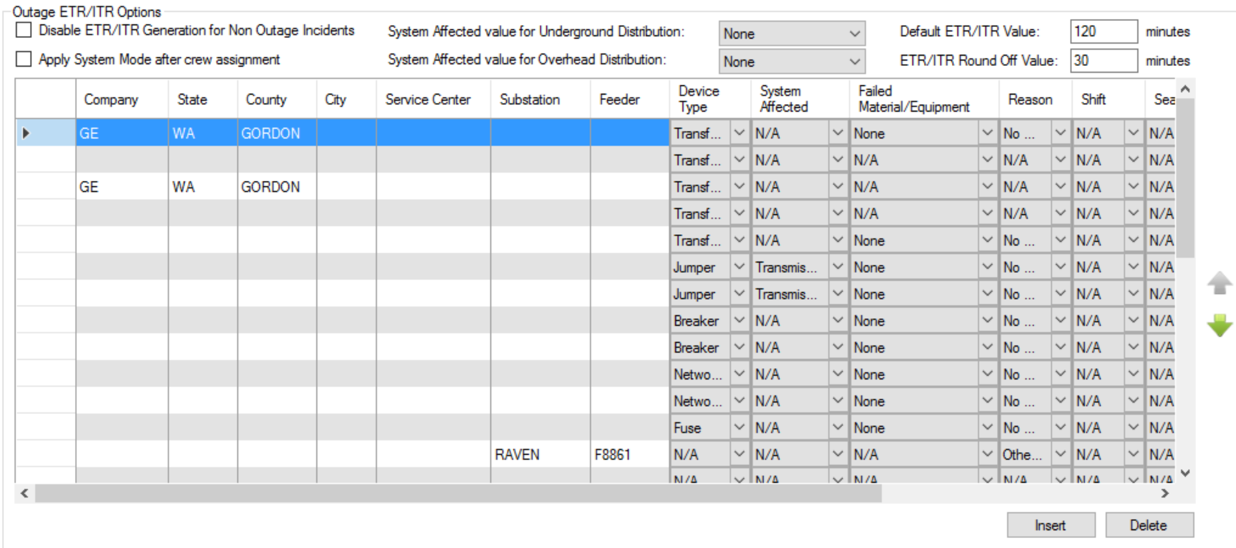


Figure 2. Outage ETR/ITR Options Table

The following settings appear above the table:

> **Note**
>
> The parameter RECALCULATE_ETR_AFTER_CREW_ASSIGNMENT in NetServerOmsConfig_localhost.txt determines whether to recalculate ETR after crew assignment to Incident.
> If RECALCULATE_ETR_AFTER_CREW_ASSIGNMENT = 1 ETR should be recalculated.
> If RECALCULATE_ETR_AFTER_CREW_ASSIGNMENT = 0 ETR should stay without changes.

- **Disable ETR/ITR Generation for Non-Outage Incidents:** If selected, the system does not calculate ETR/ITR for non-outage incidents, and the ITR/ETR fields do not appear on the Properties tab of the Update Incident display for non-outage incidents.
- **Apply System Mode after Crew Assignment:** If selected and System Mode is set for an area, the system will verify that a Crew is assigned before using that row in the ETR matrix.
- **System Affected Value for Underground/Overhead Distribution:** Specifies the default System Affected value for incidents with an underground/overhead device of record.
- **Default ETR/ITR Value:** Sets the default estimated time of restoration for all conditions not configured in the ETR/ITR matrix.
- **ETR/ITR Round-Off Value:** Sets the ETR/ITR round-up value.

  For example, if the incident started at 1:27p.m., with a calculated ITR of 120minutes and an ETR/ITR round-off value of 30minutes, the initial estimated restoration time is 3:30p.m.

Each row in the table has the following fields:

- **Company:** The incident’s company name.

  > **Note**

> Columns for management areas hierarchy levels above Feeder (by default, Service Center, Company, State, County, and City) are populated according to the ManagementAreasHierarchy coded translation item. For more details about configuring coded translation entries, refer to the *Coded Translation Editor User Guide*.

- **State:** The incident's state.
- **County:** The incident's county.
- **City:** The incident's city.
- **Service Center:** The incident's service center.
- **Substation:** The incident's substation name.

  This matches the substation of the device of record for the incident.
- **Feeder:** The incident's feeder name.

  This matches the feeder of the device of record for the incident.
- **Device Type:** Can be set to N/A, None, Cut, Fuse, Jumper, Line Switch, Pole Recloser, Sectionalizer, Breaker, Transformer, Network Protector, or Service Point (if no device is predicted out).
- **System Affected:** Can be set to N/A, None, Transmission Sub, Transmission Lines, Distribution Sub, or Distribution Lines.
- **Failed Material/Equipment:** Can be set to one of the incident material types from the list, N/A, None, or Other.
- **Reason:** Can be any valid incident reason code from the list, No Reason Assigned, orN/A.
- **Time Range:** Can be any defined time range from the Time of Day table on the Time Period tab (Day, Night, Workday, Weekday, Someday, Holiday, orN/A). This matches the time when the incident was created.
- **Date Range:** Can be any defined date range from the Date Range table on the Time Period tab (Winter, Spring, Summer, Fall, orN/A). This matches the date when the incident was created.
- **System Mode:** Can be Major Storm, Storm, Normal, orN/A.

  This matches the current system mode. If set to N/A, a match always occurs. If the other columns are set to N/A, this allows for a default ETR/ITR for each system mode.
- **Inaccessible:** Indicates if the incident site is accessible for the crew. Can be Yes, No, orN/A.
- **ETR/ITR (Minutes):** The estimated time for restoration in minutes.

To add, delete, or move rows (ETR/ITR rules), use the following buttons:

- **Insert:** To insert a row
- **Delete:** To delete a row
- To move a row up
- To move a row down

In addition, the following shortcut menu options are available for the Outage ETR/ITR Options grid:

- **Copy:** Copies the contents of the selected row.
- **Paste:** Pastes the copied row contents into the selected row.
- **Insert:** Inserts a row.
- **Delete:** Delete the selected row.
- **Move Up:** Move the selected row up. The system retains the incident name and increments the ETR count to 1.
- **Move Down:** Move the selected row down. The system does not retain the incident name and resets the ETR count to 0.

Note

The logic for Move Up and Move Down ensures that the ETR count accurately reflects changes only when the incident context is preserved.

Note

Even when two or more entries match, the system returns only the first matching entry. Therefore, it is necessary to carefully sort the matrix entries (from specific to general) to get the desired behavior.

Once an incident is created, split, merged, or updated (this relates to Cause, Failed Material/Equipment, and the Inaccessible flag), the system calculates the duration value from the matrix and the duration value is added to the start time of the incident (for new incidents), or the current time (for updated incidents), and rounded up (extended) to the next largest interval.

The full list of events that trigger ETR/ITR calculations are provided in the following table.

Table 1. Events That Trigger ETR/ITR Calculations

Event	ETR/ITR Calculation (*)
New Incident	
• Area setting does not exist • Incident properties match the ETR/ITR Matrix	ITR = Creation Time + ETR/ITR Matrix value ETR = null
• Area setting does not exist • Incident properties do not match the ETR/ITR Matrix	ITR = Creation Time + Default ETR/ITR value ETR = null
• Area setting exists • ITR Disable option is selected	ITR = null ETR = null
• Area setting exists • Area setting ETR is set	ITR = Area setting ETR ETR = null

Event	ETR/ITR Calculation (*)
• Area setting exists • Area setting ETR is not set	ITR = Current Time + ETR/ITR Matrix value (or Default ETR/ITR value) ETR = null
Dispatching a Crew (**)	
• Incident properties match the ETR/ITR Matrix	ETR = Current Time + ETR/ITR Matrix value
• Incident properties do not match the ETR/ITR Matrix	ETR = Current Time + Default ETR/ITR value
Updating Incident Properties (Cause, Failed Material/Equipment, and Inaccessible)	
• New incident properties match the ETR/ITR Matrix	ETR = Current Time + ETR/ITR Matrix value
• New incident properties do not match the ETR/ITR Matrix	ETR is not recalculated
Splitting an Incident	
• Splitting an incident from an incident	Incident 1 ETR/ITR = ETR/ITR of the initial incident Incident 2 ETR/ITR = Current Time + Incident's ETR/ITR (Default or ETR/ITR Matrix value)
Merging Incidents	
• Merging an incident with one or more incidents	Resulting "master" incident ETR/ITR = ETR/ITR of the "master" incident or the earliest ETR/ITR among the incidents (configurable)
New Area Setting (***)	
• ITR Disable option is selected • Incident ETR is not defined	ITR = null
• Area setting ETR is defined	ITR/ETR = Area Setting ETR
• Area setting ETR is not defined • New incident properties (with the new system mode) match the ETR/ITR Matrix	ITR/ETR = Current Time + ETR/ITR Matrix value
Deleting an Area Setting	
• Area setting is deleted before the area setting ETR • Area setting ETR was defined	ITR/ETR = ITR/ETR that existed before the area setting ETR

Note

(*) ITR can be changed due to an area setting only. ETR is not recalculated for incidents with manually set ETRs. ETR is not recalculated for incidents that are affected by an area setting with an area setting ETR specified. ETR is not recalculated for pending clear and completed incidents.

Note

(**) The ETR recalculation for dispatching a crew is configurable (disabled by default).

> Dispatching an evaluator or the second crew does not trigger the ETR/ITR recalculation.

> **Note**
>
> (***) The ITR/ETR recalculation for a new area setting applies only to incidents with start time after the area setting start time. Disabling incident ETR/ITR overrides by area settings for specific statuses of dispatched crews is configurable. (By default, ETR/ITR overrides by area settings are enabled for all crew statuses.)

4. Time Period Tab

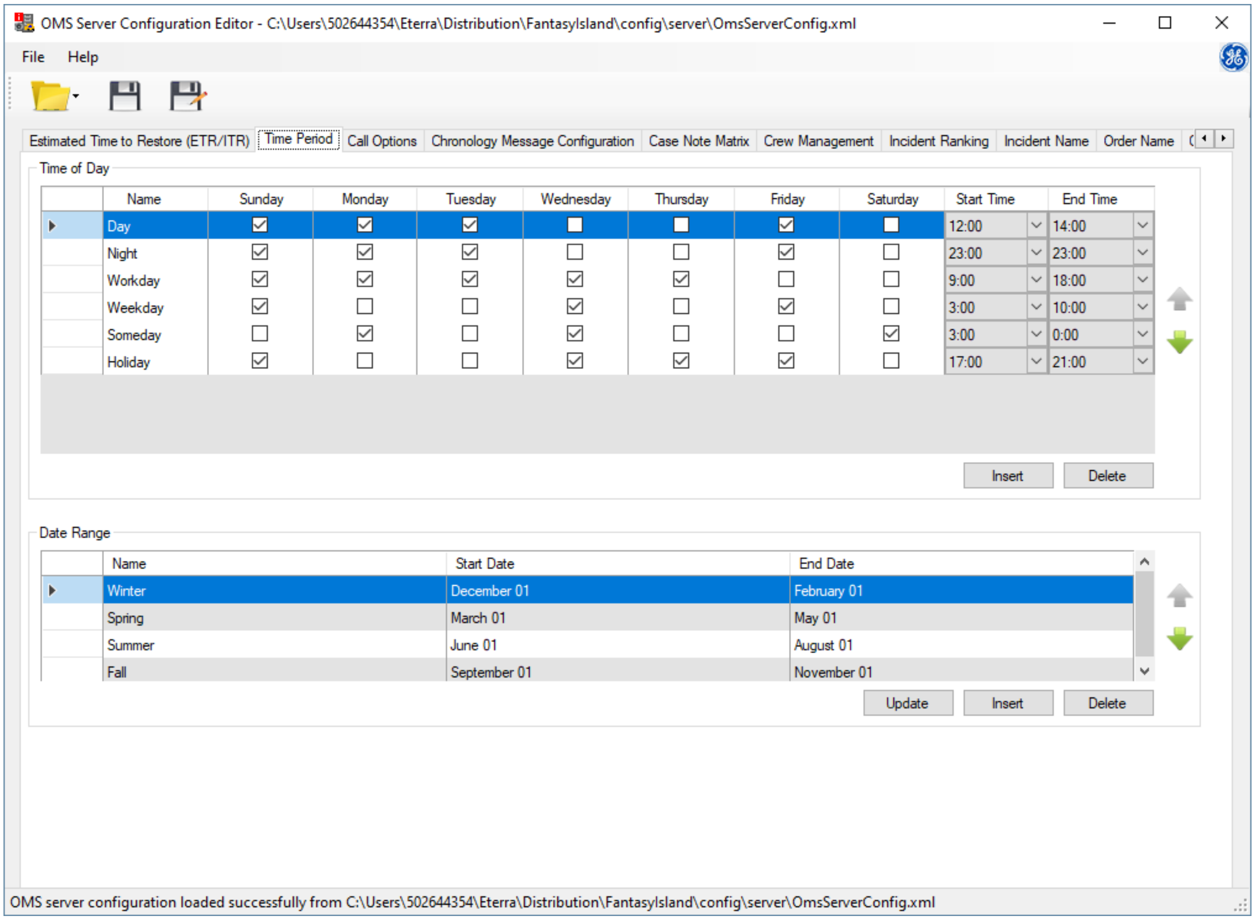


Figure 3. Time Period Tab

Time of Day Table

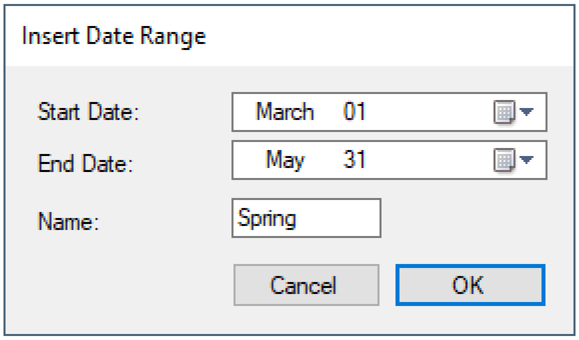
The Time of Day table allows you to configure the start time and the end time for each day or shift type (for example, Holiday or Workday) to use in the Time Range field on the Estimated Time to Restore (ETR/ITR) tab. A time range set each day from midnight to midnight equals to N/A.

To configure each time range, set flags for the corresponding weekdays, and select the start and end times from the list.

Date Range Table

The Date Range table allows you to configure the start time and the end time for each date range or season (for example, Summer or Winter) to use in the Date Range field on the Estimated Time to Restore (ETR/ITR) tab. A date range set from January 1 to December 31 equals to N/A.

To configure date ranges, use the Update, Insert, or Delete buttons.


Insert Date Range
Start Date:
March 01
End Date:
May 31
Name:
Spring
Cancel
OK

5. Call Options Tab

The Call Options Tab allows you to configure OMS system actions for customer and location calls. Depending on the call type, the system creates an outage incident, non-outage incident, or an unassociated call.

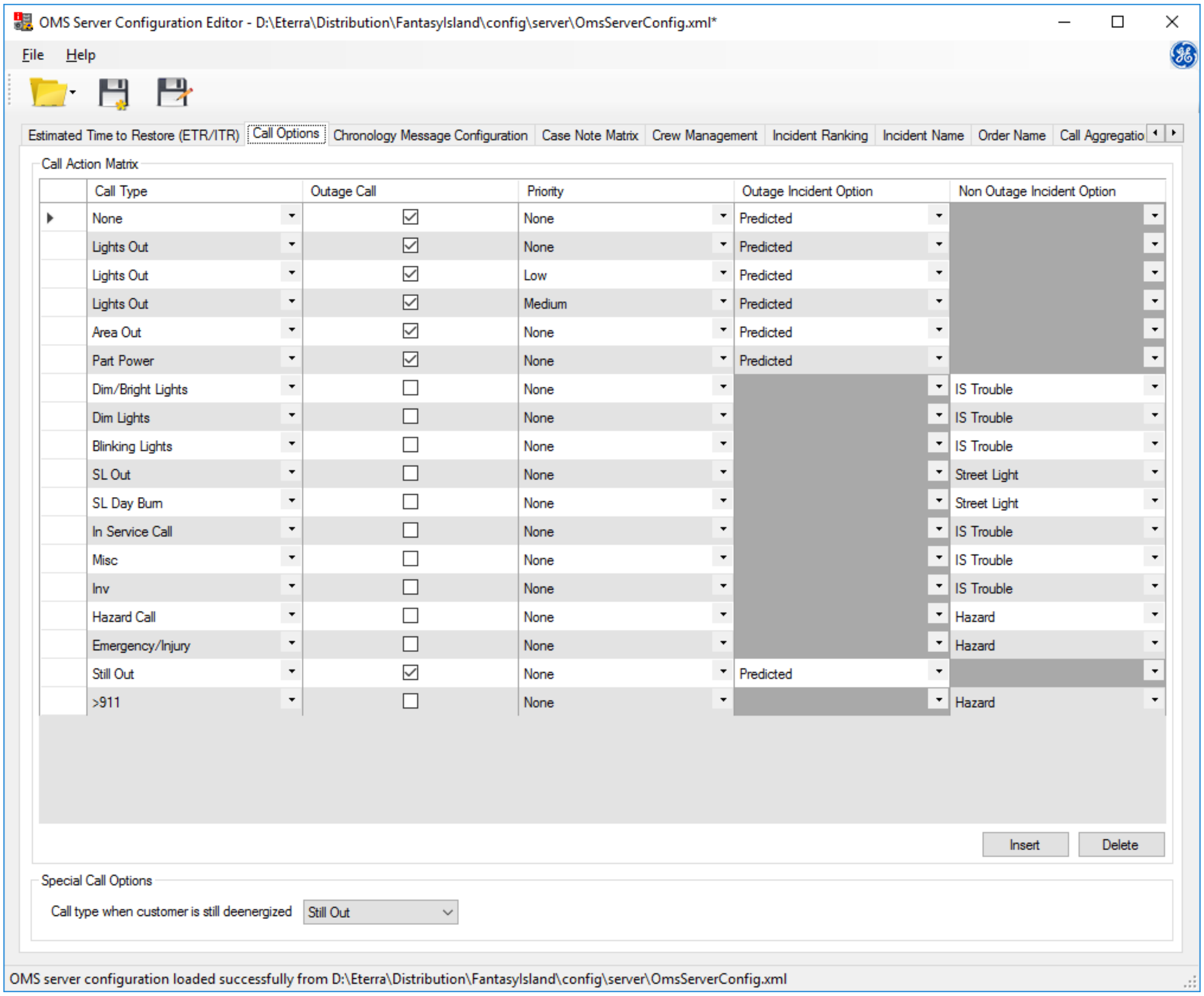


Figure 4. Call Options Tab

5.1. Call Action Matrix

The Call Actions table (Call Action Matrix) defines incident types that the system creates for specific customer or location call types.

When the system receives a call, it searches for an exact call type and call priority type match in the Call Action Matrix from beginning to end. If no exact match is found, the system searches again for the call type match only, regardless of the call priority type.

For customer calls from valid customer accounts, the system does the following:

- For an outage call (call type with the Outage Call option selected in the Call Action Matrix, for example, Lights Out), the system creates the specified outage incident.
- For a non-outage call (call type with the Outage Call option not selected in the Call Action Matrix, for example, Part Power), the system creates the specified outage incident.
- For a zero call priority type (None call priority type) unassociated call (call type that is not defined in the Call Action Matrix, for example, Unassociated), the system creates an unassociated call.
- For a non-zero call priority type (for example, Average call priority type) unassociated call, the system creates a non-outage isolated incident.

For customer calls from invalid or unlinked customer accounts, the system does the following:

- For a zero call priority type outage call, the system creates an isolated outage incident or an unassociated call (configurable).
- For a non-zero call priority type outage call, the system creates an isolated outage incident or a non-outage isolated incident (configurable).
- For a non-outage call, the system creates the specified non-outage incident.
- For a zero call priority type unassociated call, the system creates an unassociated call.
- For a non-zero call priority type unassociated call from an invalid or unlinked customer account, the system creates a non-outage isolated incident.

For location calls, the system does the following:

- For a zero call priority type outage or unassociated call, the system creates an unassociated call.
- For a non-zero call priority type outage or unassociated call, the system creates a non-outage isolated incident.
- For a non-outage call, the system creates the specified non-outage incident.

For calls from 911 customers, the system does the following:

- For a non-outage call, the system creates the specified non-outage incident.
- For an outage or non-priority unassociated call, the system does not create an incident. The call is only listed in the 911 Call Summary tab of the OMS display plug-in.
- For a non-zero call priority type unassociated call, the system creates a non-outage isolated incident.

When creating an incident from a 911 call or an unassociated call, the system does the following:

- For a 911 call or an unassociated call originated from an outage call type, the system creates an isolated incident.
- For a 911 call or an unassociated call originated from an unassociated call type, the system creates a non-outage isolated incident.

To define actions that the system takes for a call type, insert a row in the Call Action Matrix, select the call type, and specify its outage type, call priority type, and the system's action.

Each row in the matrix has the following fields:

- **Call Type:** Specifies a call type. For all call types that are not specified in the Call Action Matrix, the system creates unassociated calls.
- **Outage Call:** Specifies whether this is an outage call (selected) or a non-outage call (cleared). Outage calls are calls that report customer de-energization (for example, Lights Out). Non-outage calls are calls that do not reflect an outage, but rather an observation such as a blinking light or a wire down at a certain location.
- **Priority:** Contains different call priority types from the coded translation configuration.
- **Outage Incident Option:** Sets the system's action for customer outage calls with the specified type and call priority type. The system can create a single customer predicted or predicted stable incident. This field is enabled only when the Outage Call checkbox is selected.

> **Note**
>
> Only an automatic outage trip of a protective SCADA device can create a momentary incident. Momentary outage call types that are received during a momentary incident are added to the incident. Momentary outage calls that are not part of a momentary incident create unassociated incidents.

- **Non-Outage Incident Option:** Sets the system's action for non-outage calls with the specified type and call priority type. The system can create a non-outage incident of a specific type (for example, street light, hazard, or wire down). This field is enabled only when the Outage Call checkbox is cleared.

To add, delete, or move rows (Call Action rules), use the following buttons:

- **Insert:** To insert a row.
- **Delete:** To delete a row.

You can add or modify call types, call priority types, and non-outage incident types in the matrix using the CallType, Priority_Code, and NonOutageType coded translation items. For more information, refer to the *Coded Translation Editor User Guide*.

For more information about defining call priority type for different call conditions, see Call Prioritization Tab.

5.2. Special Call Options

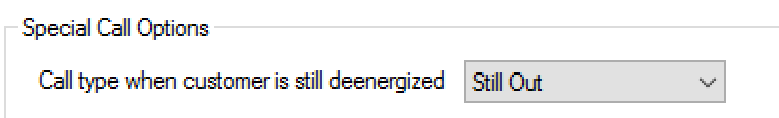


Figure 5. Special Call Options

Use the Special Call Options group to specify the call type indicating that a customer remains de-energized after restoration (Still Out). Calls of this type are highlighted in red in the Affected Customers and Customers that Called for an Incident panes on the Incident Detail display.

In addition, when a customer remains de-energized after restoration and creates a call of the specified type (for example, Still Out), a new single customer incident, non-outage incident, or unassociated call is created with the Is Callback flag set.

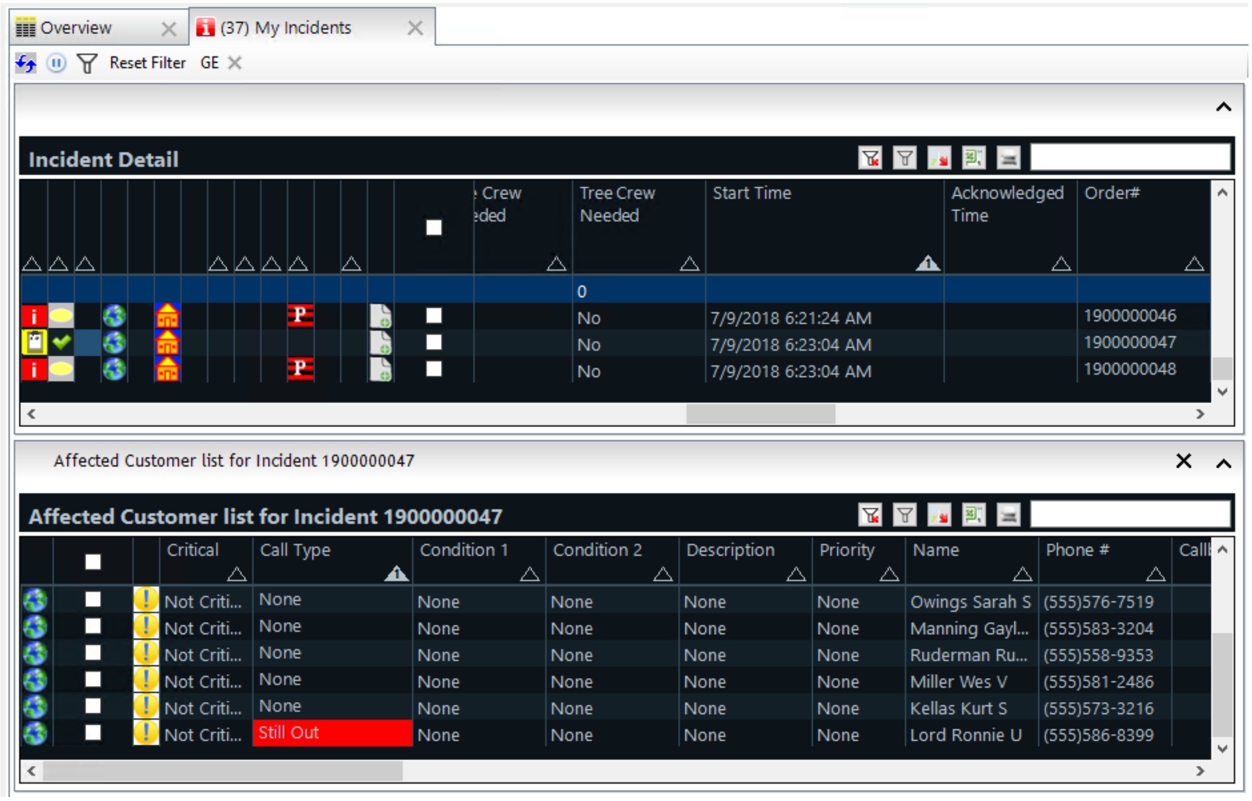


Figure 6. Incident Detail Display

6. Chronology Message Configuration Tab

The Chronology Message Configuration tab allows you to configure messages that appear in the Chronology tab of the Update Incident and Order Worksheet displays for certain event types.

The information that can be captured in a message is limited by the allowable query fields, which are listed for the given message type in the Chronology Message Types and Arguments List table.

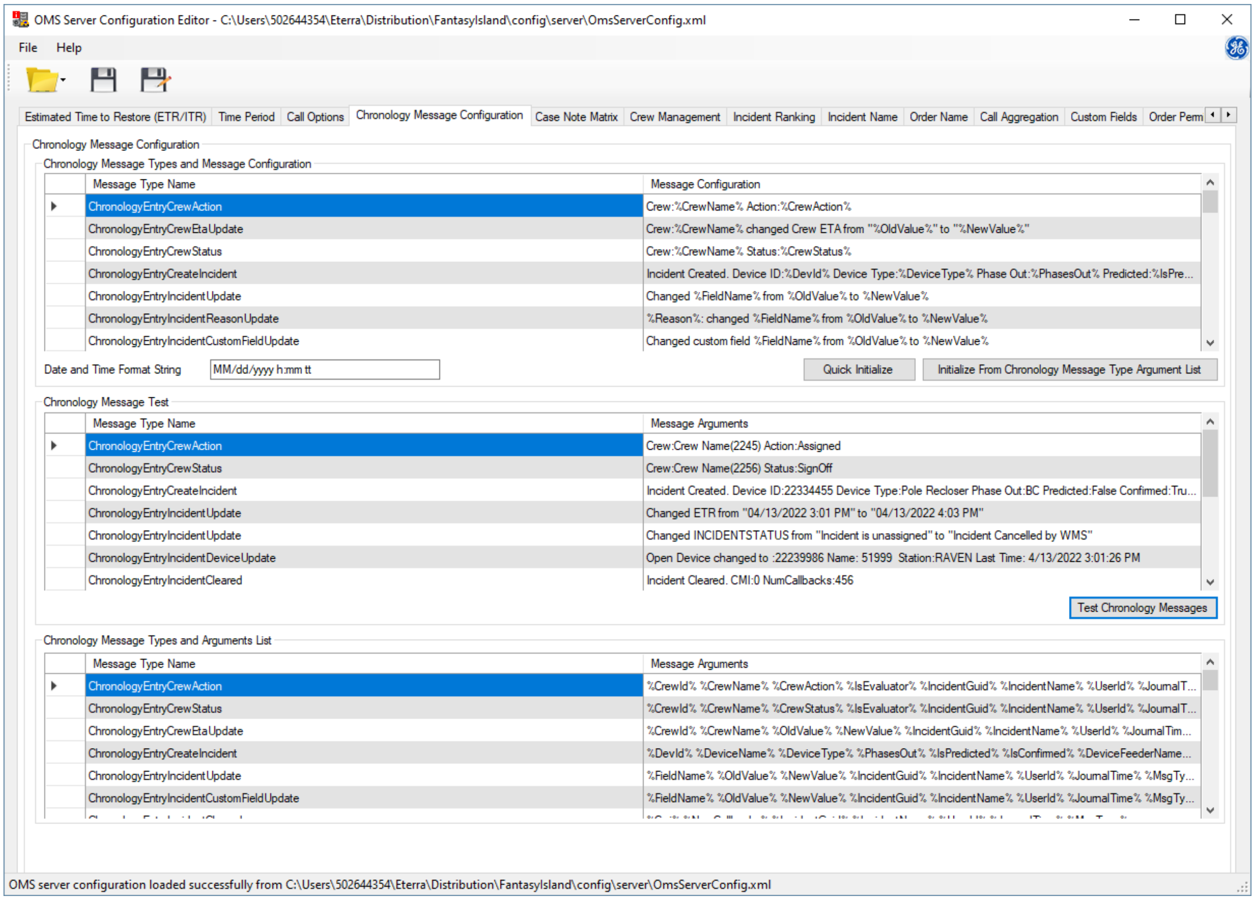


Figure 7. Chronology Message Configuration Tab

To configure a chronology message:

1. Click the Quick Initialize button to see the default chronology messages for each message type in the Chronology Message Types and the Message Configuration Table.
2. Click the Test Chronology Messages button, and review the human-readable translations in the Chronology Message Test table.
3. If you need to modify the default messages, do one of the following:
    - Copy the needed arguments from the Chronology Message Types and Arguments List table into the corresponding message type fields in the Chronology Message Types and Message Configuration table, and edit the messages as needed.

      Or:

    - Click the Initialize from Chronology Message Type ArgumentList button to copy all the arguments from the chronology message types and arguments list table into the Chronology Message Types and Message Configuration table, and edit the messages as needed.
4. Click the Test Chronology Messages button, to test the changes and review the human-readable translations in the Chronology Message Test table.
5. Save the changes via the File/Save menu.

6.1. Chronology Message Types and Message Configuration Table

The Chronology Message Types and Message Configuration table shown in the figure below allows you to define a human-readable message for every message type to appear in the chronology log.

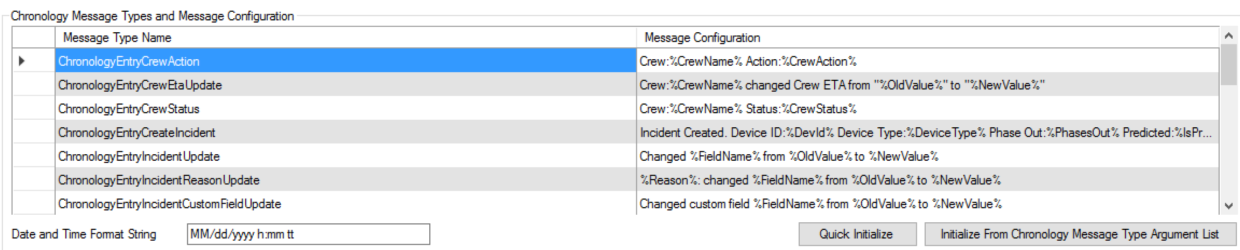


Figure 8. Chronology Message Types and Message Configuration

The following options appear below the table:

- **Date and Time Format String:** Sets the date and time format in the chronology log.
- **Quick Initialize:** From source code, reads examples of messages for each message type and displays them in the Message Configuration column.
- **Initialize from Chronology Message Type Argument List:** Copies all message arguments from the Chronology Message Types and Arguments List table at the bottom of the tab to the Chronology Message Types and Message Configuration table.

6.2. Chronology Message Test

The Chronology Message Test table displays the list of arguments from the Chronology Message Types and Message Configuration table in human-readable format.

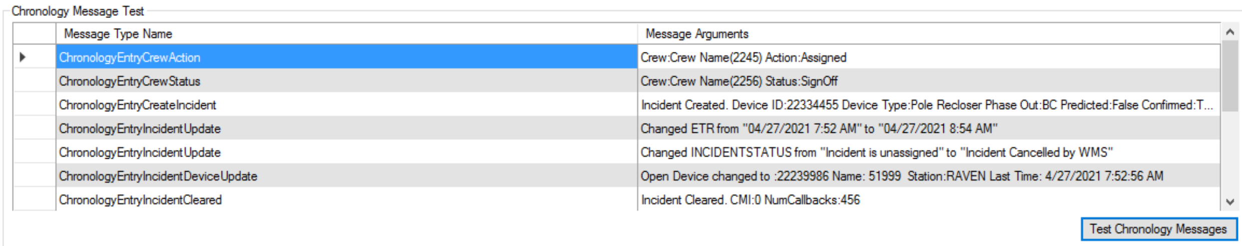


Figure 9. Chronology Message Test

- **Test Chronology Messages:** Tests the list of arguments from the Chronology Message Types and Message Configuration table, and displays the arguments in human-readable format.

6.3. Chronology Message Types and Arguments List Table

The chronology message types and arguments list table contains all message arguments available for each chronology message type. It serves as the analyst's reference.

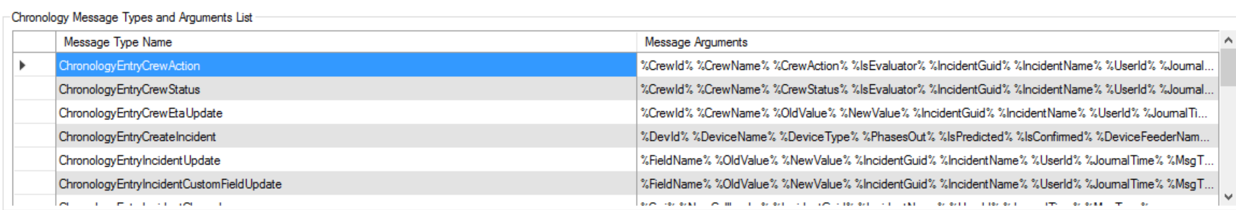


Figure 10. Chronology Message Types and Arguments List

The table below shows available query fields for all message types.

Table 2. Available Query Fields for All Message Types

Message Type	Available Query Fields
Crews	
ChronologyEntryCrewAction	CrewId, CrewName, CrewAction, IncidentGuid, IncidentName, UserId, JournalTime, and MsgType
ChronologyEntryCrewStatus	CrewId, CrewName, CrewStatus, IncidentGuid, IncidentName, UserId, JournalTime, and MsgType
ChronologyEntryCrewEtaUpdate	CrewId, CrewName, IncidentGuid, IncidentName, UserId, OldValue, NewValue, JournalTime, and MsgType
Incidents	
ChronologyEntryCreateIncident	DevId, DeviceType, PhasesOut, IsPredicted, IsConfirmed, DeviceFeederName, DeviceServiceCenterId, ETR, ITR, IncidentGuid, IncidentName, UserId, JournalTime, and MsgType
ChronologyEntryIncidentUpdate	FieldName, OldValue, NewValue, IncidentGuid, IncidentName, UserId, JournalTime, and MsgType

Message Type	Available Query Fields
ChronologyEntry IncidentCustomFieldUpdate	FieldName, OldValue, NewValue, IncidentGuid, IncidentName, UserId, JournalTime, and MsgType
ChronologyEntry IncidentReasonUpdate	Reason, FieldName, IncidentGuid, IncidentName, UserId, OldValue, and NewValue
ChronologyEntryIncidentCleared	Cmi, NumCallbacks, IncidentGuid, IncidentName, UserId, JournalTime, and MsgType
ChronologyEntry IncidentDeviceUpdate	OpenDeviceId, OpenDeviceName, OpenDeviceStation, LastChangeTime, IncidentGuid, IncidentName, UserId, JournalTime, and MsgType
ChronologyEntry CreatePlannedOutage	NumCustomers, Id, DevId, OpCenter, Station, Feeder, IncidentGuid, IncidentName, UserId, JournalTime, and MsgType
ChronologyEntry UpdatePlannedOutage	FieldName, OldValue, NewValue, Id, DevId, OpCenter, Station, Feeder, IncidentGuid, IncidentName, UserId, JournalTime, and MsgType
ChronologyEntry DeletePlannedOutage	FieldName, OldValue, NewValue, Id, DevId, OpCenter, Station, Feeder, IncidentGuid, IncidentName, UserId, JournalTime, and MsgType
ChronologyEntrySwitchOrderStep	Id, DevId, OpCenter, Station, Feeder, IncidentGuid, IncidentName, UserId, JournalTime, and MsgType
ChronologyEntry NtpSwitchOrderStep	DeviceName, DeviceId, StationName, RestoredCustomers, CustomersOut, IncidentGuid, IncidentName, UserId, JournalTime, OperationTime, and MsgType
ChronologyEntryRestorationStep	CustomerCount, DeenergizedTime, EnergizedTime, DeviceName, DeviceId, FeederName, StationName, OpCenterId, IncidentGuid, IncidentName, UserId, JournalTime, and MsgType
ChronologyEntry CreateSplitOrMergeIncident	Action, Inc1Guid, Inc1DeviceId, Inc1DeviceType, Inc1OrderNum, Inc2Guid, Inc2DeviceId, Inc2DeviceType, Inc2OrderNum, IncidentGuid, IncidentName, UserId, JournalTime, and MsgType
ChronologyEntryAssociateCall	CallId, IncidentGuid, IncidentName, UserId, JournalTime, and MsgType
ChronologyEntryIncidentCancelled	UserId and MsgType
ChronologyEntry IncidentUndoCancelled	UserId and MsgType
ChronologyEntry RestorationStepCancelled	UserId, StepNum, and MsgType
ChronologyEntry RestorationStepUndoCancelled	UserId, StepNum, and MsgType
ChronologyEntryRestorationStepUpdate	UserId, StepNum, FieldName, OldValue, NewValue, and MsgType
ChronologyEntryIncidentLog	UserId, Message, and MsgType
Customer Calls	
ChronologyEntryCallDetails	AccountNumber, CustomerName, Address, PhoneNumber, CallTime, CallType, Condition_1, Condition_2, Description, Priority, and DrivingDirections
Area Settings	
AreaNoteChronologyAction	UpdatedField and NewValue
Orders	
ChronologyEntryOrderCreated	OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType

Message Type	Available Query Fields
ChronologyEntryOrderHasFollowup	FollowupOrderName, FollowupOrderType, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntryOrderIsFollowup	OriginatingOrderName, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntry OrderHasPrerequisite	PrerequisiteOrderName, PrerequisiteOrderType, OrderType, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntry OrderIsPrerequisite	OriginatingOrderName, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntry RelatedOrderCreated	RelatedOrderName, RelatedOrderType, OrderType, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntry OrderCreatedAsRelated	OriginatingOrderName, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntry OrderStatusChanged	OldStatus, NewStatus, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntry OrderCrewAssigned	CrewId, CrewName, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntry OrderCrewUnassigned	CrewId, CrewName, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntry OrderCrewStatusChanged	CrewId, CrewName, OldCrewStatus, NewCrewStatus, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntryOrderTaskAdded	TaskType, Facility, FacilityType, FacilityName, CrewId, CrewName, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntry OrderTaskStatusChanged	TaskType, Facility, FacilityType, FacilityName, CrewId, CrewName, OldTaskStatus, NewTaskStatus OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntryOrderTaskAssigne d	TaskType, Facility, FacilityType, FacilityName, CrewId, CrewName, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntry OrderTaskUnassigned	TaskType, Facility, FacilityType, FacilityName, CrewId, CrewName, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntry OrderCrewStatusTimeChanged	CrewId, CrewName, StatusName,OldTime, NewTime, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntry OrderOrganizationChanged	OldOrganizationName, NewOrganizationName, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntryOrderOwnerChang ed	OldOwnerNameOrUnowned, NewOrderNameOrUnowned, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntry OrderCustomFieldChanged	CustomFieldName, OldValue, NewValue, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntryOrderOwnerSet	NewOwnerName, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType
ChronologyEntryOrderOwnerUnset	OldOwnerName, OrderGuid, OrderName, IncidentGuid, IncidentName, UserId, JournalTime, MsgType

7. Case Note Matrix Tab

The Case Note Matrix tab allows you to define rules to apply case notes based on the incident working status, the reason for the outage, and a manual override.

A "case note" is information about an incident that is sent along with the CaseNote ETR back to customers that may call. Case notes are tied to a transformer, or to a "service point" (the point on the electrical model where the customer's service connects, which is typically the low side of a distribution transformer). A service point is also called an "electrical address".

You can define numeric codes, names, descriptions, and bitmaps of case notes, working statuses, and causes using the coded translation editor tool. For more information, refer to the *Coded Translation Editor User Guide*.

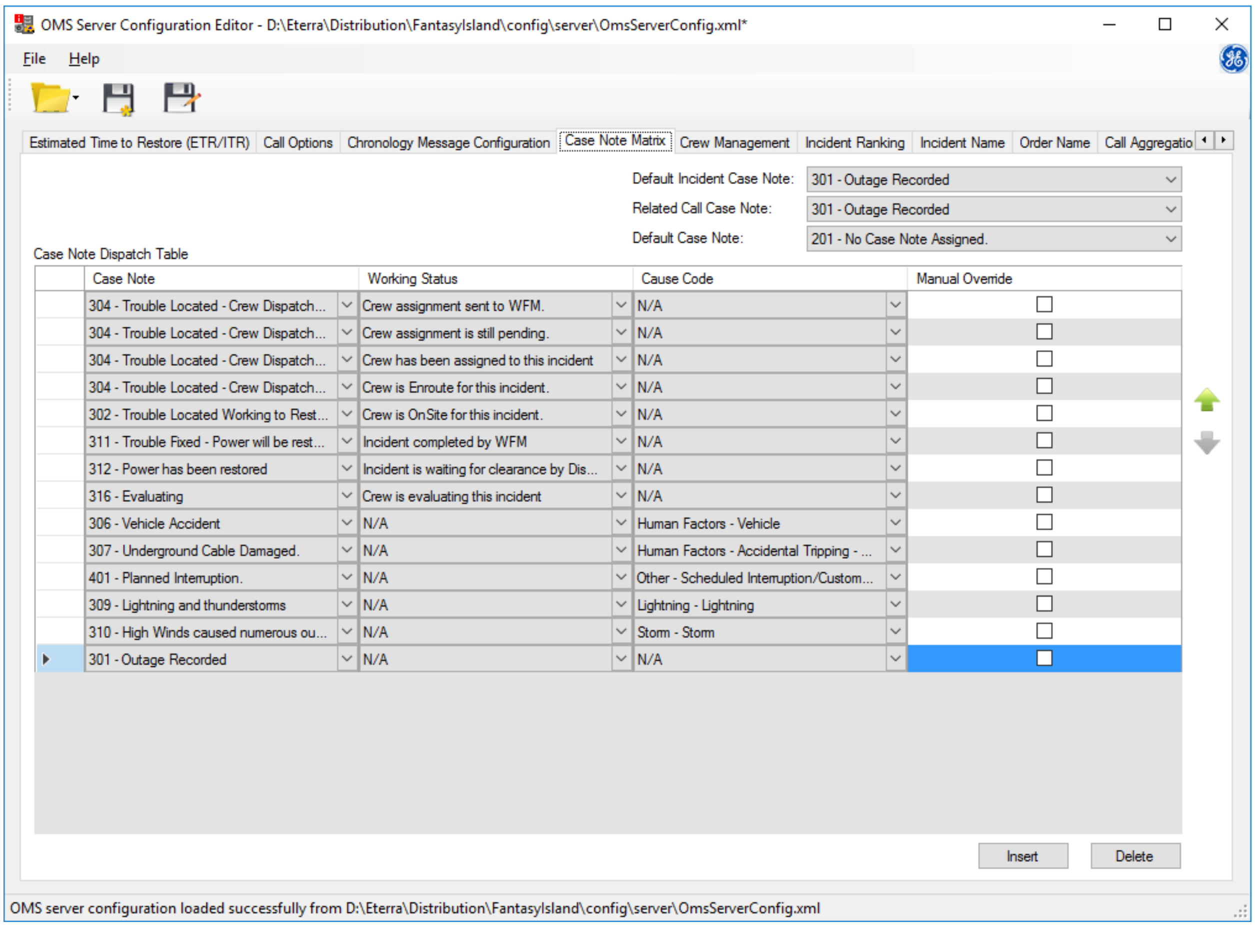


Figure 11. Case Note Matrix Tab

To add, delete, or move rows, use the following buttons:

- **Insert:** To insert a row.
- **Delete:** To delete a row.
- To move a row up.

- To move a row down.

You can also copy, paste, insert, delete, and move rows using the shortcut menu.

7.1. Case Note Dispatch Table

You can define case notes to be automatically applied to incidents using the case note dispatch table. The system evaluates each entry in the matrix every time the incident changes, and returns the first matching entry. This is referred to as the "highest priority case note".

The following fields are shown on the Case Note Matrix tab:

- **Default Incident Case Note:** The case note to use if there is an incident, but no matching results are found from the case note dispatch table.
- **Related Call Case Note:** The case note to use if there is another call on the electrical address, but no incident has been found yet (for example, a single call has been received but no incident has been created).
- **Default Case Note:** The default case note to use.

The case note dispatch table is an ordered list, with the following fields populated from the coded translations configuration file:

- **Case Note:** The case note's code and description (from the CaseNote coded translation entry).
- **Working Status:** The incident's working status (from the IncidentWorkingStatus coded translation entry).
- **Cause Code:** The cause (and sub-cause) for an incident (from the Reason coded translation entry).
- **Manual Override:** Flag to allow manually overriding the case note.

> **Note**
>
> When upgrading from **e-terra***distribution* 3.7 and earlier, rename the SchOutCustNotified short name of the Reason coded translation entry to SCHD to support case note updates for planned outages.

7.2. Order of Case Note Precedence

The overall order of precedence of case notes is:

- Case note dispatch table
- Default incident case note
- Related call case note
- Default case note

7.3. Translation of Case Note Codes

You can define case note numeric codes and other parameters using the coded translation editor tool. For more information, refer to the *Coded Translation Editor User Guide*.

- **201:** No Case Note Assigned (default message)
- **301:** Outage recorded – crew en route
- **302:** Trouble located – crew working to restore power
- **303:** Outage recorded – crew assigned not dispatched
- **304:** Trouble located – crew dispatched
- **305:** Large number of outages – crews not available
- **306:** Vehicle accident
- **307:** Underground cable damaged
- **308:** A tree has fallen on lines
- **309:** Lightning and thunderstorms
- **310:** High winds caused numerous outages
- **311:** Trouble fixed – power will be restored soon
- **312:** Power has been restored
- **401:** Planned interruption

8. Crew Management Tab

The Crew Management tab allows you to configure the name for automated and non-automated crews that are generated by static crew models from employees and temporary crews that can be tracked (including mobile). You can also specify crew statuses that prevent automatically overriding an incident's ETR/ITR by an area setting's ETR.

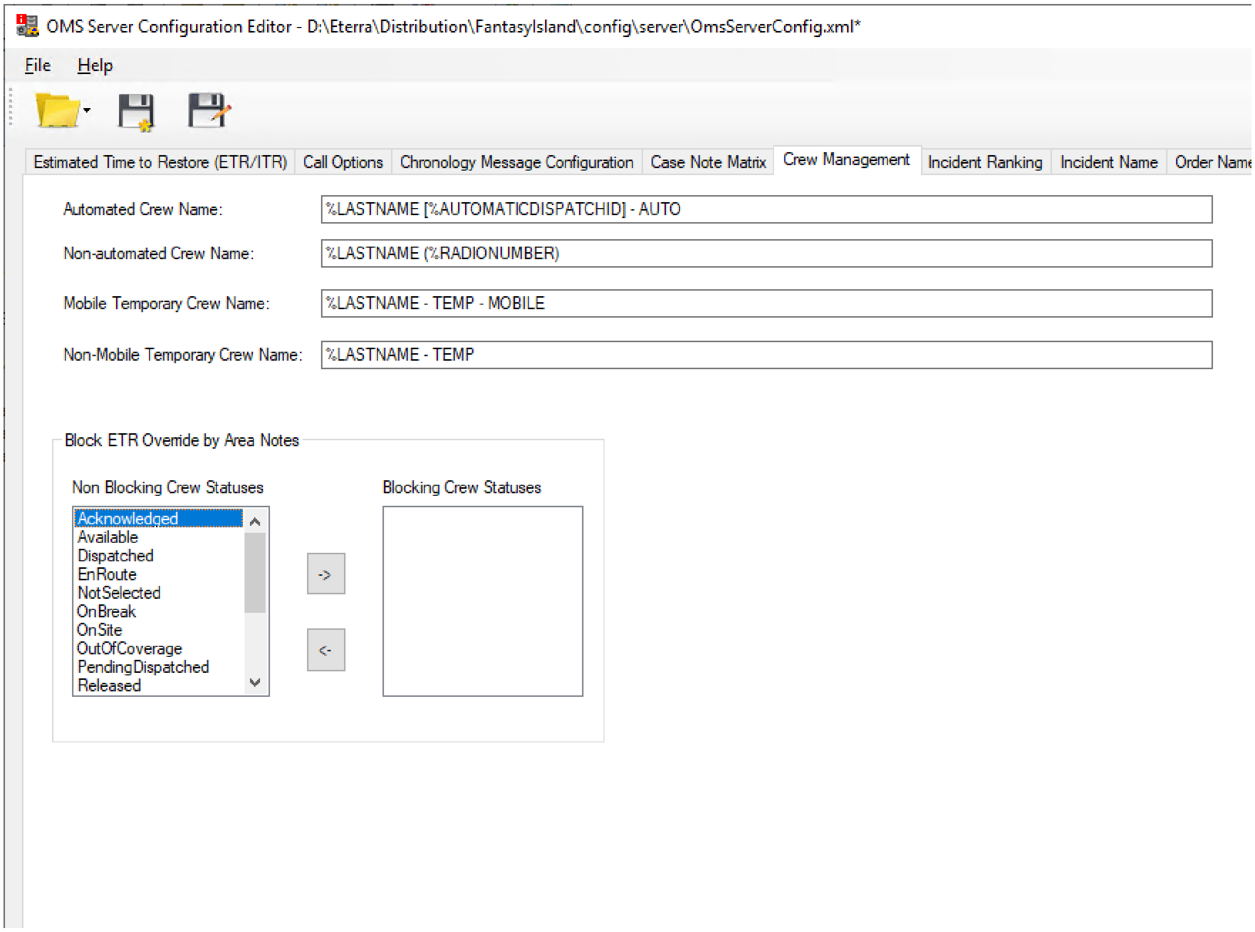


Figure 12. Crew Management Tab

8.1. Customizing the Crew Name

To customize the crew name, you can add the following fields:

- **%Name:** Primary employee's first name.
- **%Id:** Unique identification number for the primary employee.
- **%LastName:** Primary employee's last name.
- **%PhoneNumber:** Primary employee's phone number.
- **%RadioNumber:** Primary employee's radio number.

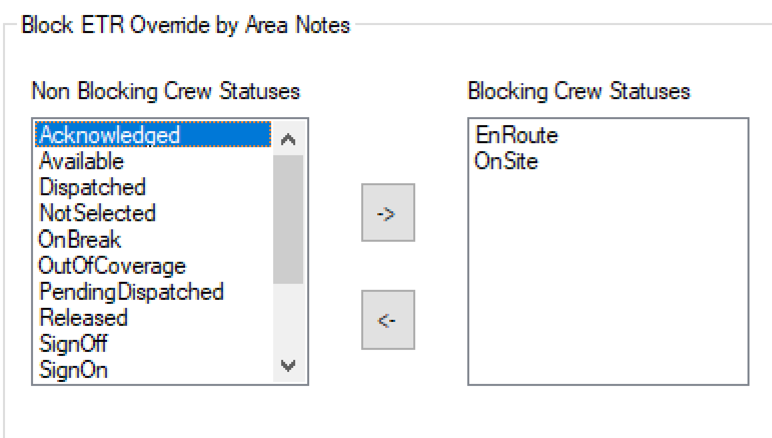
- **%AutomaticDispatchId:** Unique identification number from the Work Force Management system to track the primary employee's X, Y coordinates and communicate the location to the OMS system.
- **%CrewArea:** Primary employee's area polygon name.
- **%CrewAreaId:** Primary employee's area polygon ID.
- **%EmployeeTypeId:** Employee type identification number.
- **%EmployeeTypeName:** Employee type description.
- **%Dept:** Primary employee's department name.
- **%DeptId:** Primary employee's department ID.
- **%CrewDivision:** Name of the division the primary employee belongs to.

8.2. Blocking ETR/ITR Recalculation due to Area Setting ETR

You can disable overriding an incident's ETR/ITR by the area setting's ETR for incidents with dispatched crews.

To specify crew statuses that block overriding an incident's ETR/ITR by the area setting's ETR:

- Move the crew statuses from the Non-Blocking Crew Statuses list to the Blocking Crew Statuses list.



9. Incident Ranking Tab

The Incident Ranking tab allows you to configure a custom ranking for certain incident conditions. When an incident is created, the system calculates all weight values based on the incident's conditions and shows the total weight in the Ranking column on the Incident Detail grid of the OMS display plug-in.

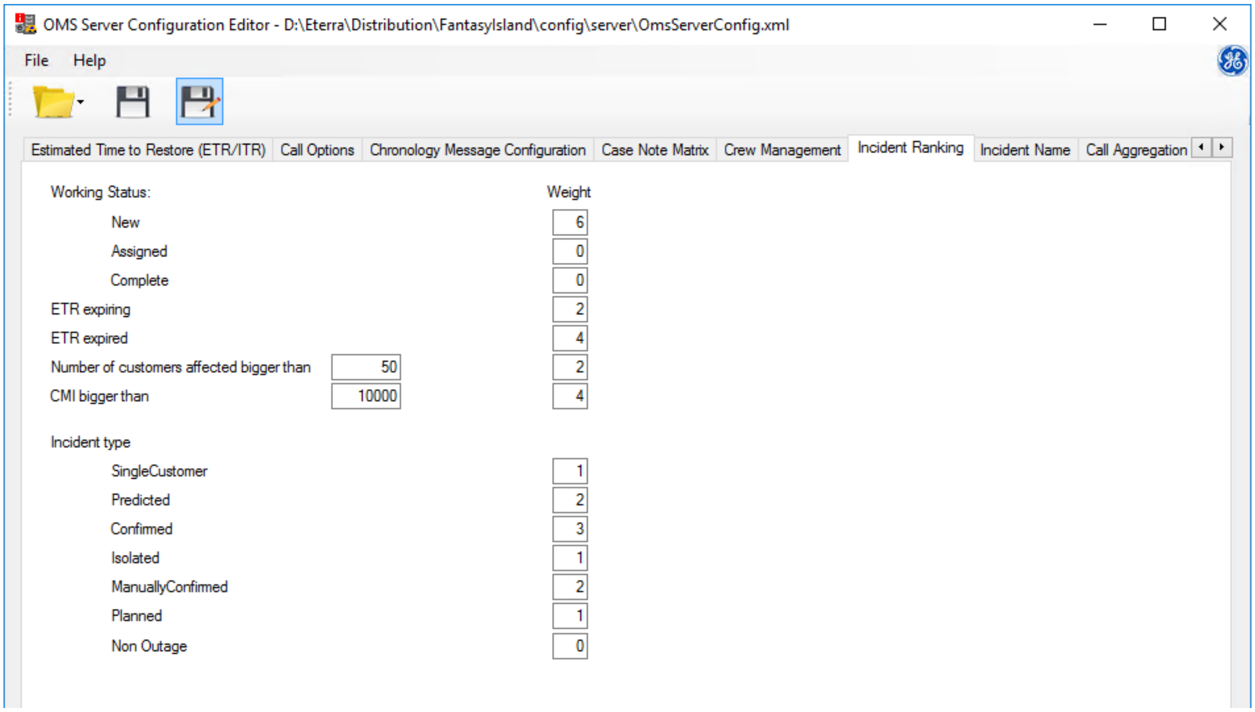


Figure 13. Incident Ranking Tab

To configure the ranking, add any number to the Weight field for each condition in the left column.

10. Incident Name Tab

The Incident Name tab allows you to customize the incident name format. You can choose to include incident details, such as sequential numbers in the year, service center, and date. The new format applies only to new incidents.

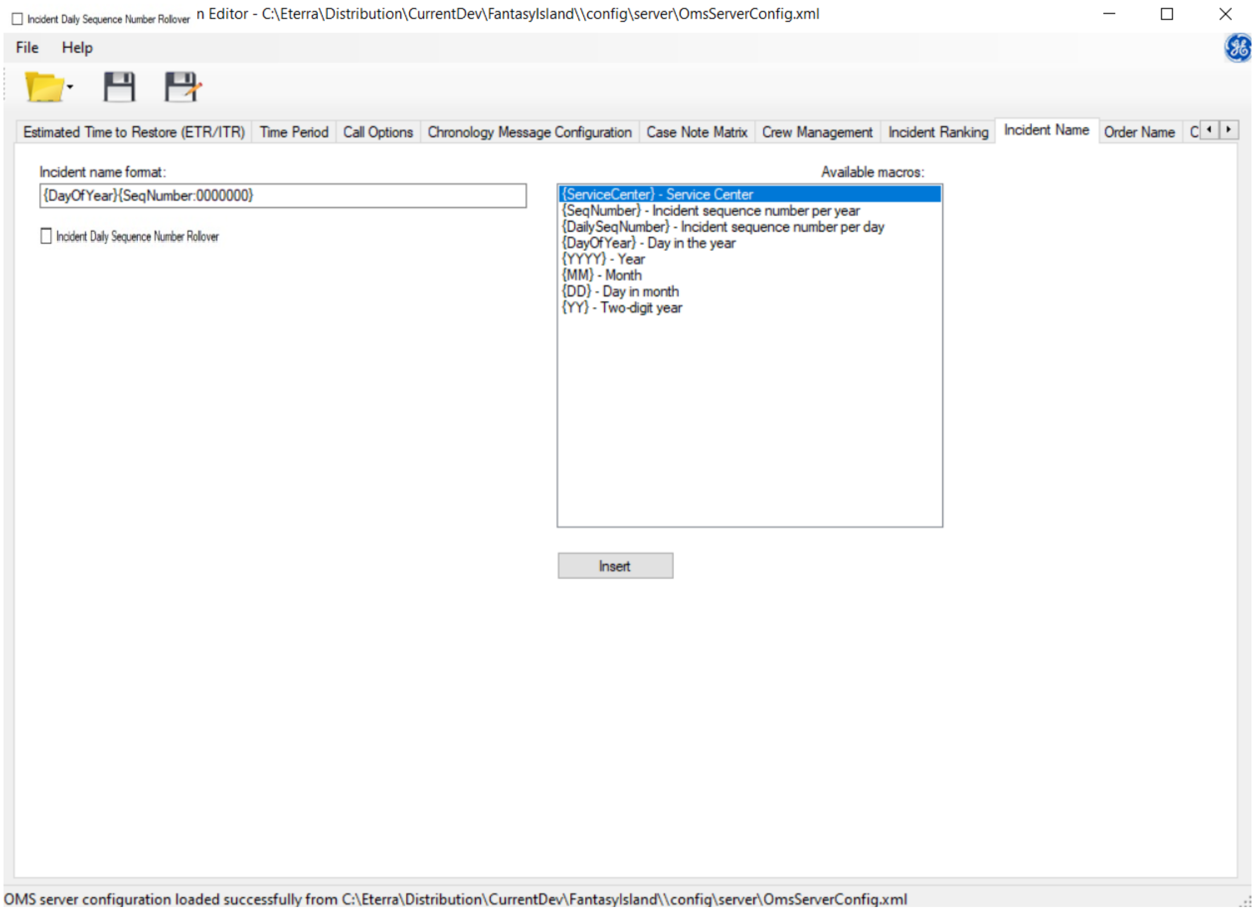


Figure 14. Incident Name Tab

The Incident Name tab includes the following fields and controls:

- **Incident Name Format:** Defines the incident name format that is applied in the Outage Management System. You can use any combination of plan text values and macro fields.

> **Note**
>
> The incident name supports only the following special characters: colon (:), dash (-), underscore (_), and backslash (\). Other special characters are not supported. Right-to-left text is not supported (such as Arabic or Hebrew).

- **Incident Daily Sequence Number Rollover:** The **Incident Daily Sequence Number Rollover** determines how the incident numbers are managed when the daily count reaches the maximum limit.

- When the checkbox is **unchecked**, the sequence continues beyond the limit.
- When the checkbox is **checked**, the digits of the sequence reset to zero.

- **Available Macros:** Includes a list of available macro fields that can be used to specify the incident name format.
  - **{ServiceCenter}:** Service center name.
  - **{SeqNumber:0000000}:** Sequential number of the incident in the current year. Zeroes configure the total number of digits in the incident name. Zeroes are replaced with the corresponding digits in the incident number, and unreplaced zeroes remain in the number. For example, 0000321.
  - **{DailySeqNumber:00000}:** Sequential number of the incident in the current day.

    > **Note**
    >
    > To differentiate among incidents that occur on different days, using the DailySeqNumber macro with the DayOfYear or DD macro is recommended.

  - **{DayOfYear}:** The ordinal number of a calendar day of the year. For example, the day number of February 20th is 50.
  - **{YYYY}:** Year.
  - **{MM}:** Month.
  - **{DD}:** Day.
  - **{YY}:** Two-digit year.
- **Insert:** Click Insert to place the selected macro in the Incident Name Format field. You can also double-click a macro in the Available Macros list to insert a macro in the Incident Name Format field.

Consider the following examples:

- With the default name format, {DayOfYear}{SeqNumber:0000000}, the name of the third incident in the system that appeared on April 1 is 09100000003.

- With the name format set to {ServiceCenter}-{YYYY}{MM}{DD}-{SeqNumber:000}, the name of the thirty-seventh incident in the system that appeared on April 24, 2018 is 3LIONSSC-20180424-037.

11. Order Name Tab

> **Note**
>
> This feature requires integration with the GE's Work Model system.

The Order Name tab allows you to customize the order name format. You can choose to include order details, such as sequential numbers in the year, service center, and date. The new format applies only to new orders.

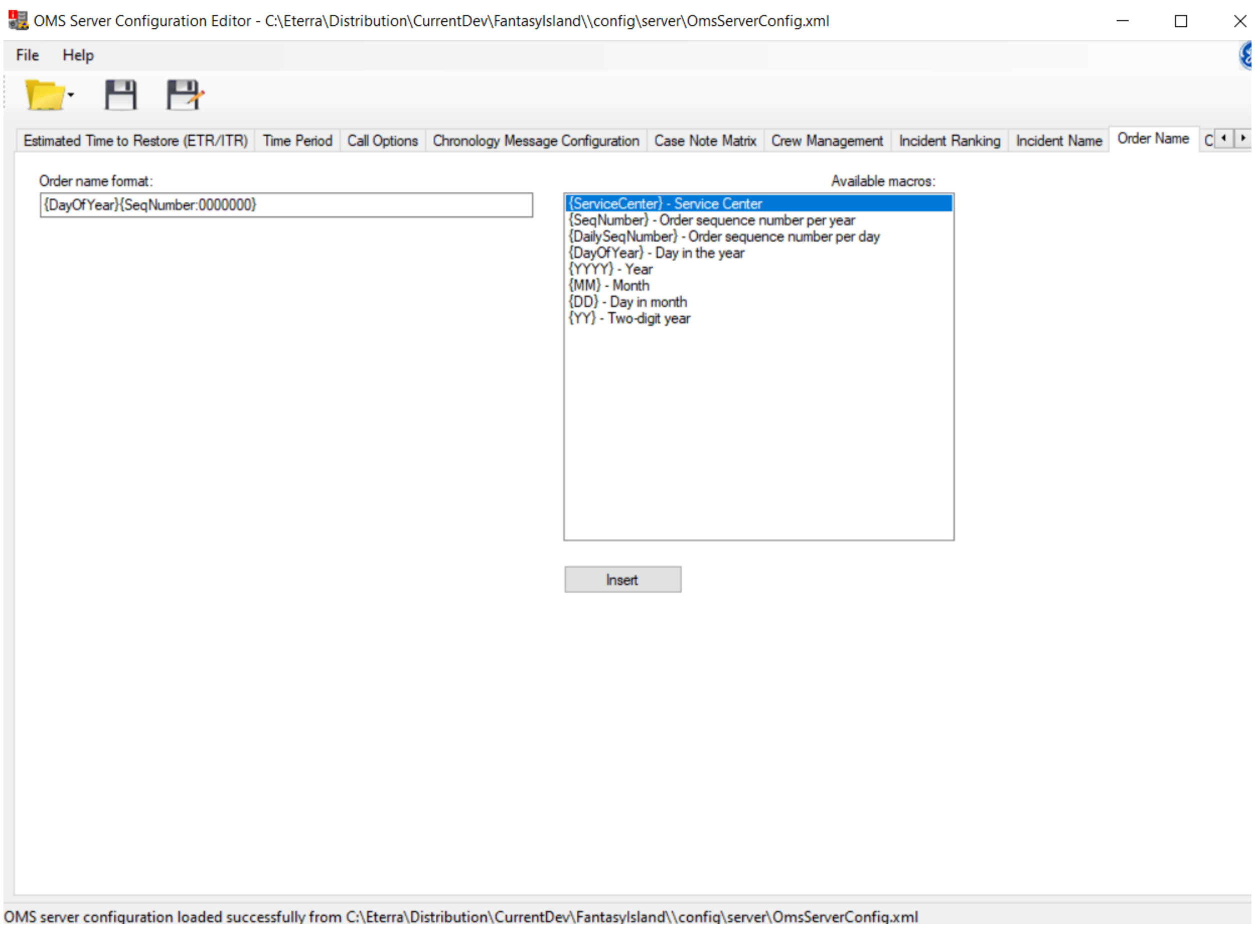


Figure 15. Order Name Tab

The Order Name tab includes the following fields and controls:

- **Order Name Format:** Defines the order name format that is applied in the Outage Management System. You can use any combination of plan text values and macro fields.

  > **Note**
  >
  > The order name supports only the following special characters: colon (:), dash (-), underscore (_), and backslash (\). Other special characters are not supported. Right-to-left text is not supported (such as Arabic or Hebrew).

- **Available Macros:** Includes a list of available macro fields that can be used to specify the order name format.
  - **{ServiceCenter}:** Service center name.
  - **{SeqNumber:0000000}:** Sequential number of the order in the current year. Zeroes configure the total number of digits in the order name. Zeroes are replaced with the corresponding digits in the order number, and unreplaced zeroes remain in the number. For example, 0000321.
  - **{DailySeqNumber:00000}:** Sequential number of the order in the current day.

    > **Note**
    >
    > To differentiate among orders that occur on different days, using the DailySeqNumber macro with the DayOfYear or DD macro is recommended.

  - **{DayOfYear}:** The ordinal number of a calendar day of the year. For example, the day number of February 20th is 50.
  - **{YYYY}:** Year.
  - **{MM}:** Month.
  - **{DD}:** Day.
  - **{YY}:** Two-digit year.
- **Insert:** Click Insert to place the selected macro in the Order Name Format field. You can also double-click a macro in the Available Macros list to insert a macro in the Order Name Format field.

For example, with the name format set to {DayOfYear}{SeqNumber:0000000}, the name of the third order in the system that appeared on April 1 is 09100000003.

12. Call Aggregation Tab

> **Note**
>
> The Call Aggregation feature applies only to Outages and does not support to Non-Outage scenarios.

The OMS server uses call aggregation rules to analyze the Call Type, Condition1, Condition2, and Description fields entered in the Add/Update Customer Call display and populate call aggregation report fields in OMS displays. This helps to better understand and analyze the extent of damage and provide a business intelligence tool to estimate power restoration efforts in a more efficient manner.

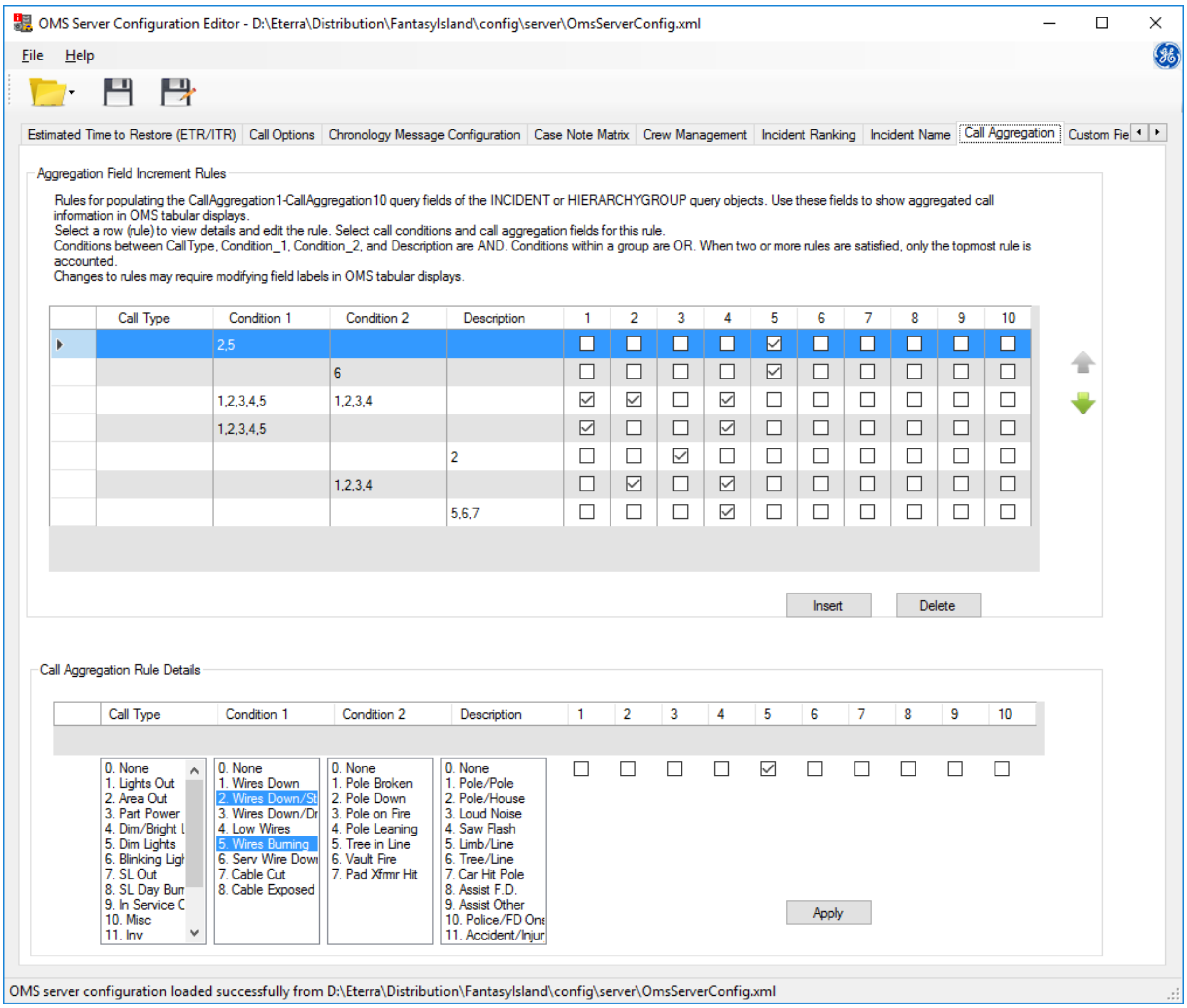


Figure 16. Call Aggregation Tab

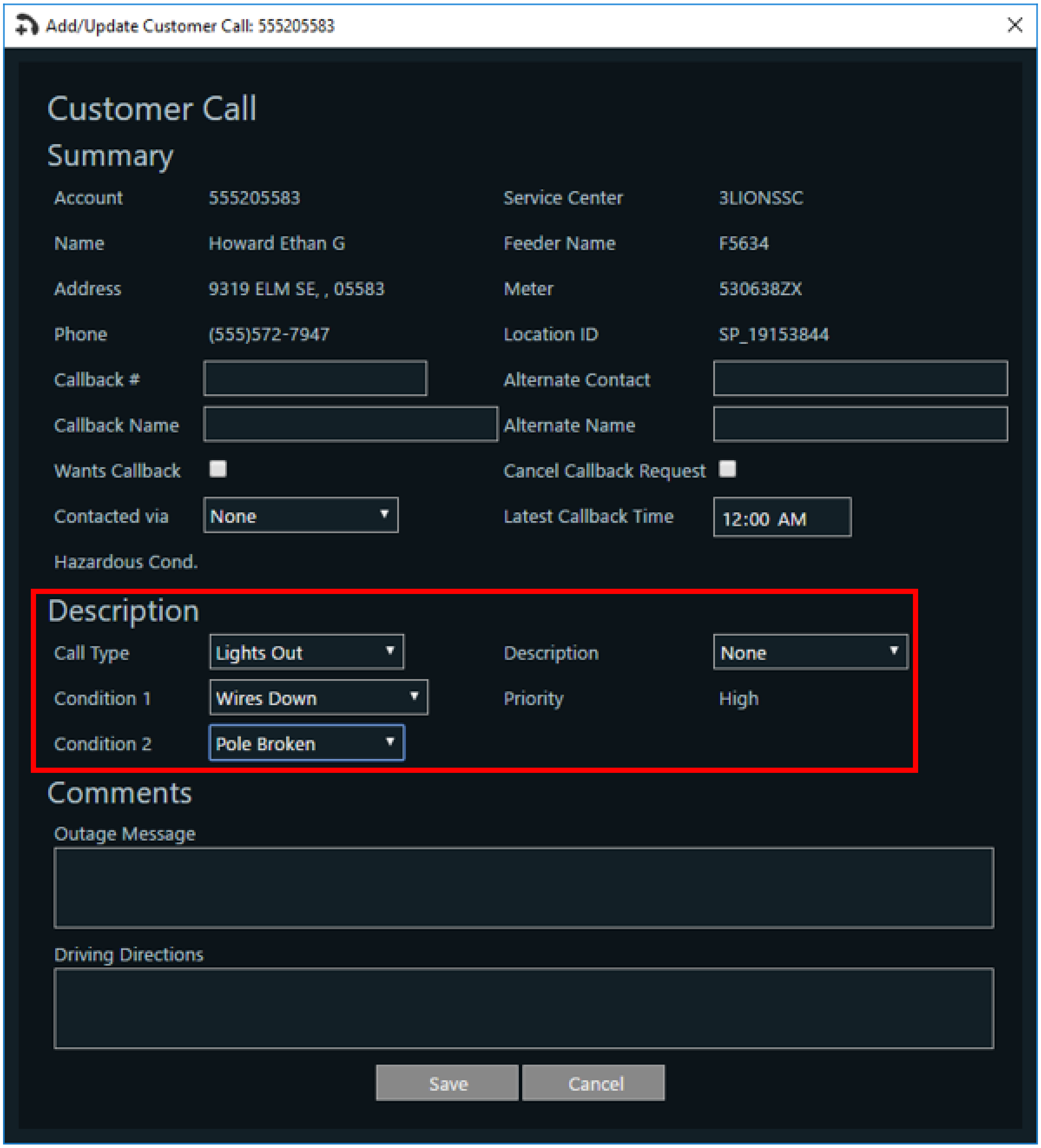


Figure 17. Add/Update Customer Call Display - Call Details

Call aggregation rules define how the CallAggregation1 - CallAggregation10 query field values are populated based on customer call details. These query fields can then be used in OMS displays to show summaries for wires down, poles down, services down, high priority, extreme priority, and other information from customer calls.

For a customer call, a satisfied call aggregation rule increases the counters in the corresponding enabled CallAggregation1 - CallAggregation10 fields by one.

By default, call aggregation query fields are used as follows:

- **CallAggregation1:** Wire Down.
- **CallAggregation2:** Poles Down.
- **CallAggregation3:** Services.
- **CallAggregation4:** High.
- **CallAggregation5:** Extreme.
- **CallAggregation6:** Not used.
- **CallAggregation7:** Not used.
- **CallAggregation8:** Not used.
- **CallAggregation9:** Not used.

Call aggregation report fields can be output using INCIDENT or HIERARCHYGROUP object CallAggregation query fields. The call aggregation results are shown on the Incident Detail, Overview, Company Summary, State Summary, County Summary, City Summary, Service Center Summary, Station Summary, and Feeder Summary displays.

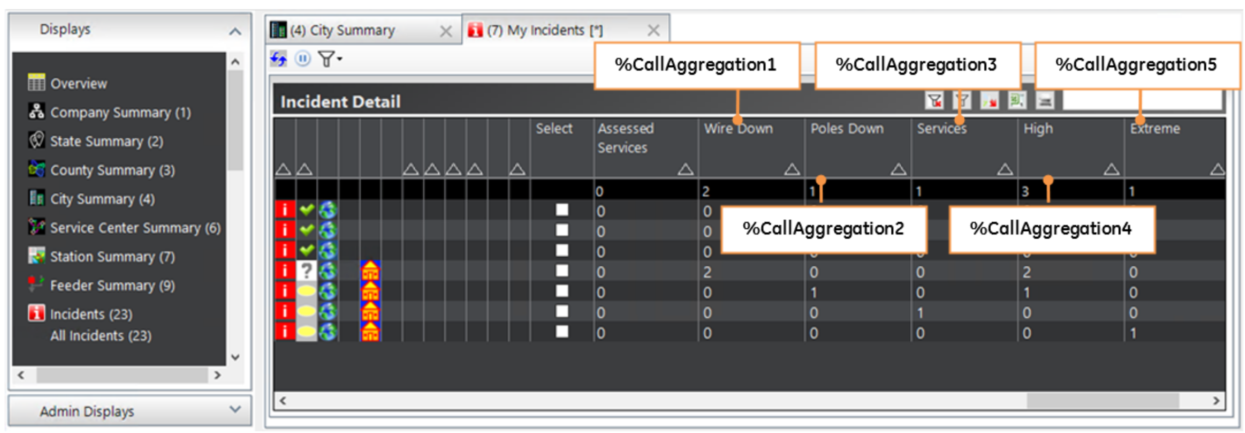


Figure 18. Call Aggregation Fields in the Incident Detail Display

12.1. Aggregation Field Increment Rules

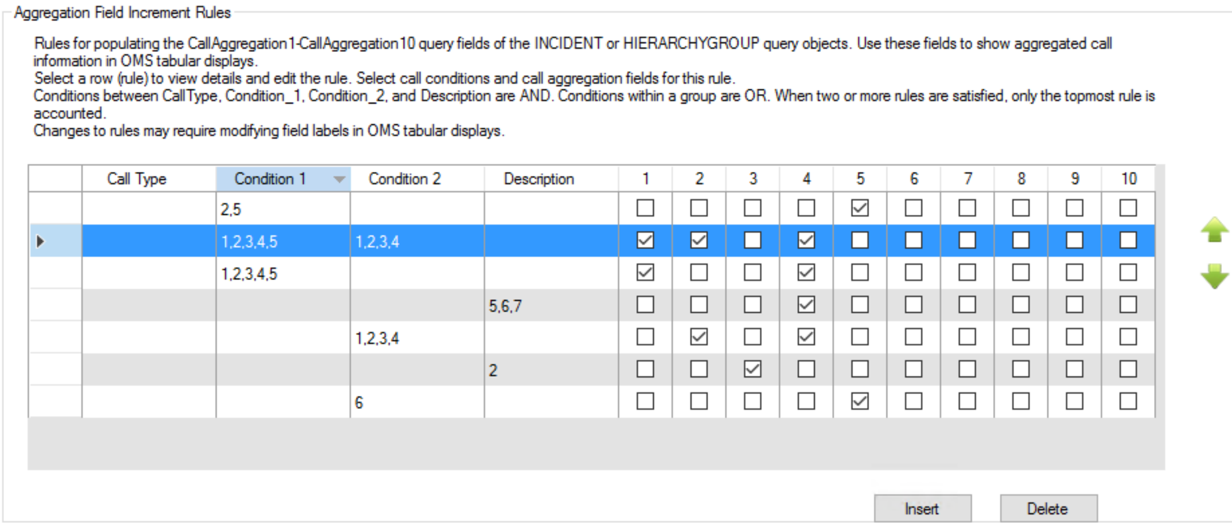


Figure 19. Aggregation Field Increment Rules Pane

The Aggregation Field Increment Rules pane shows defined call aggregation rules. Each rule includes the following fields:

- **Call Type:** Lists all Call Type items that are selected for this rule.
- **Condition 1:** Lists all Condition 1 items that are selected for this rule.
- **Condition 2:** Lists all Condition 2 items that are selected for this rule.
- **Description:** Lists all Description items that are selected for this rule.
- **Call Aggregation Fields 1-10:** Shows call aggregation fields (CallAggregation1 - CallAggregation10) that are enabled for this rule. Counters for the enabled call aggregation fields are incremented each time the rule is satisfied.

  Call aggregation fields can be included in OMS displays to produce a call aggregation report.

> **Note**
>
> Conditions within a group follow the OR condition. For example, when the Wires Down and Low Wires values are defined in the Condition 1 group, any of the defined values needs to be present in a call to satisfy the OR condition. Conditions between groups follow the AND condition. For example, when the Wires Down and Low Wires values are defined in the Condition 1 group and the Pole Broken and Pole Down values are defined in the Condition 2 group, then any combination of Condition 1 AND Condition 2 values (such as Wire Down & Pole Broken or Low Wire & Pole Down) needs to be present in a call to satisfy the AND condition.

> **Note**
>
> Only one call aggregation rule can be satisfied for a customer call. When two or more rules are satisfied, only the topmost rule is accounted. Therefore, it is necessary to carefully sort the rule

entries to get the desired behavior. When a rule is satisfied, multiple call aggregation fields may be incremented.

To add, delete, or move rows, use the following buttons:

- **Insert:** To insert a row
- **Delete:** To delete a row
- To move a row up
- To move a row down

12.2. Call Aggregation Rule Details

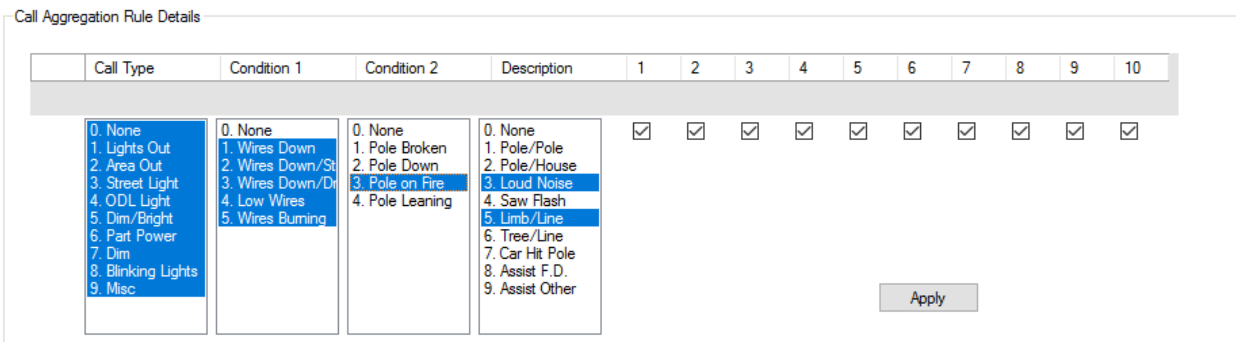


Figure 20. Call Aggregation Rule Details Pane

Use the Call Aggregation Rule Details pane to configure rules selected in the Aggregation Field Increment Rules pane.

- **Call Type:** Contains call types, which are defined by the CallType coded translation item. Use this field to specify the call types that satisfy the rule.
- **Condition 1:** Contains the first trouble call conditions, which are defined by the Condition_1 coded translation item. Use this field to specify the first trouble call conditions that satisfy the rule.
- **Condition 2:** Contains the second trouble call conditions, which are defined by the Condition_2 coded translation item. Use this field to specify the second trouble call conditions that satisfy the rule.
- **Description:** Contains the trouble call descriptions, which are defined by the Description coded translation item. Use this field to provide descriptions that satisfy the rule.
- **Call Aggregation Fields 1-10:** Use these fields to enable call aggregation fields for the rule. The counters of enabled call aggregation fields are incremented each time the rule is satisfied.
- **Apply:** Applies the current rule configuration.

Call condition field items are configurable via the Coded Translation Editor. For more information, refer to the *Coded Translation Editor User Guide*.

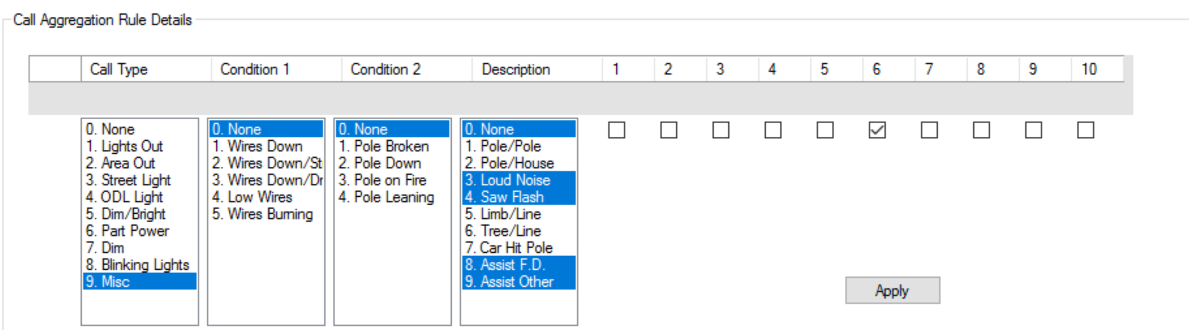
12.3. Adding and Modifying Rules

To view and/or modify the rules:

1. Select or add a row in the Aggregation Field Increment Rules pane. The details appear in the Call Aggregation Rule Details pane.
2. In the Call Aggregation Rule Details pane, specify the required call conditions by selecting or deselecting items in the following fields:
    - Call Type
    - Condition 1
    - Condition 2
    - Description

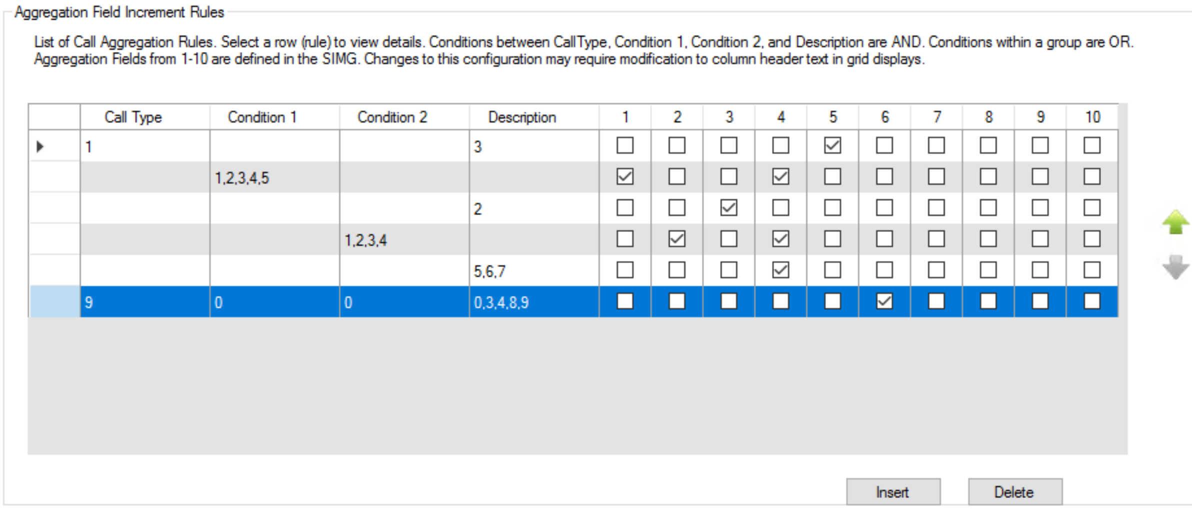
   You can select up to 10 call condition field items (from 0 to 9) for each call condition field. Conditions within a call condition field are OR. Conditions between the Call Type, Condition 1, Condition 2, and Description fields are AND.
3. In the Call Aggregation Rule Details pane, select the required call aggregation fields.

   For example, select the CallAggregation6 field for all miscellaneous calls that are not related to wire or pole down conditions.



4. Click Apply.

   The selected field item numbers and the enabled call aggregation fields for the selected conditions appear in the corresponding fields for the rule in the Aggregation Field Increment Rules pane.


Aggregation Field Increment Rules
List of Call Aggregation Rules. Select a row (rule) to view details. Conditions between CallType, Condition 1, Condition 2, and Description are AND. Conditions within a group are OR. Aggregation Fields from 1-10 are defined in the SIMG. Changes to this configuration may require modification to column header text in grid displays.
Call Type
Condition 1
Condition 2
Description
1
2
3
4
5
6
7
8
9
10
1
3
1,2,3,4,5
2
1,2,3,4
5,6,7
9
0
0
0,3,4,8,9
Insert
Delete

13. Custom Fields Tab

Use the Custom Fields tab to define custom fields for an incident, order, damage assessment, crew, area setting, or planned outage. These fields can be managed similarly to common fields. In addition, you can modify these fields the Incident's Custom Fields, Damage Assessment Location Custom Fields, and Order Worksheet pop-up displays.

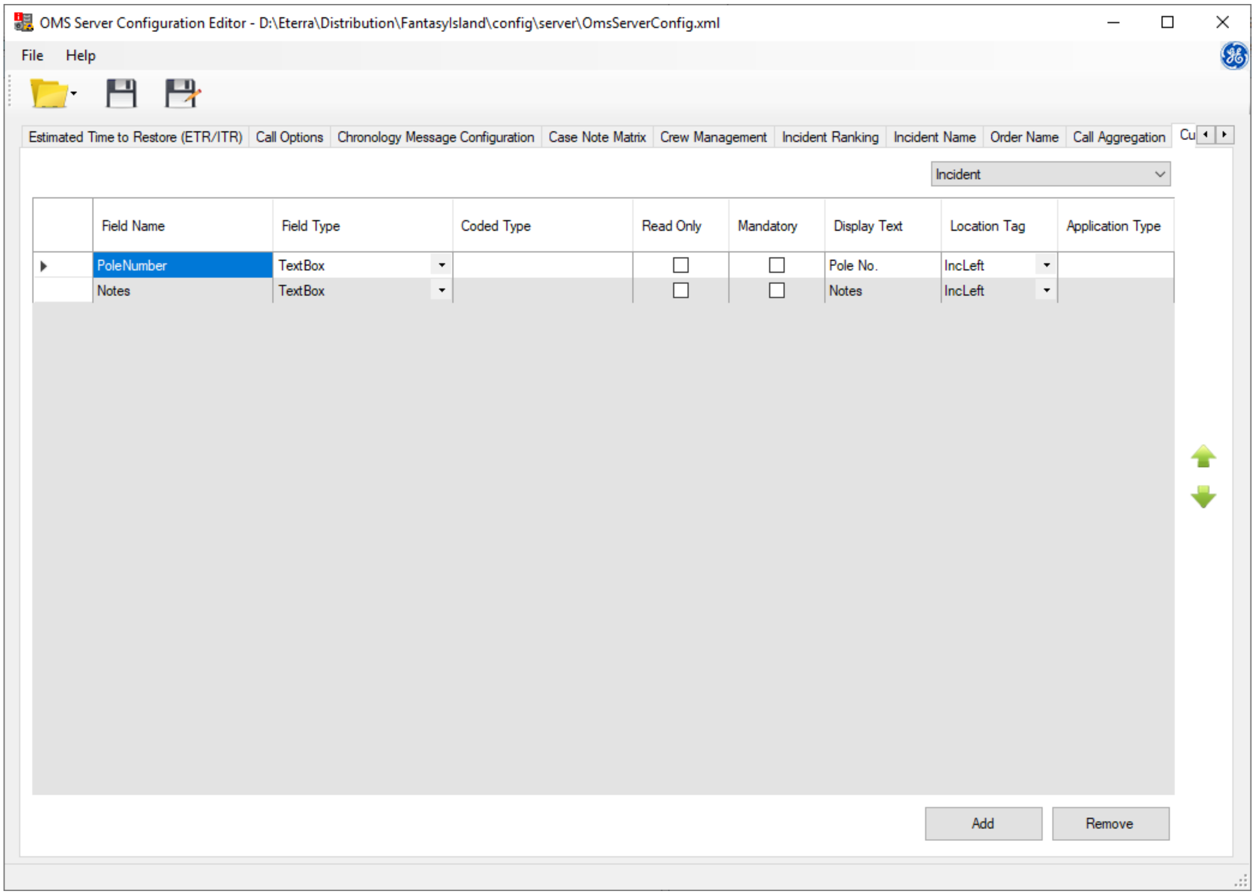


Figure 21. Custom Fields Tab

The Custom Fields tab includes the following fields and controls:

- **Object Type:** Specifies the object type to which the defined custom fields belong to (Incident, Order, DamageAssessment, Crew, AreaSettings, or PlannedOutage). This field is present in the upper right corner of the display.
- **Field Name:** The internal name of a custom field.

> **Note**
>
> The internal name cannot contain spaces and special characters.

- **Field Type:** The type of custom field. Supported types are text field, check box, drop-down menu (combo box), date and time field, and integer (numeric) field.

Note for Timestamp Fields: For custom fields of type Timestamp, values must always be in UTC format when saved or updated via API.

- **Coded Type:** Coded translation entry that defines the drop-down menu items. Applies to fields of the drop-down menu type only.
- **Read Only:** If selected, the custom field cannot be edited on the user interface.
- **Mandatory:** If selected, the custom field must be entered before clearing an incident or closing an order.

> **Note**
>
> Only incident and order custom fields support the Mandatory option. If the Mandatory option is selected, the Location Tag field must be set to one of the pre-defined location tag values. You cannot select the Read Only and Mandatory options together.

> **Warning**
>
> Do not set the Mandatory option for custom fields on the Restoration tab of the Update Incident display. The Restoration tab is hidden for non-outage incidents. Therefore, you will not be able to enter the custom fields and clear non-outage incidents.

- **Display Text:** The label for a custom field. For incidents, this text appears as a label in the Incident's Custom Fields pop-up display.
- **Location Tag:** The place in the user interface where the associated field appears.

  For the AreaSettings, Incident, and Order object types, the Location Text combo box includes a list of pre-defined location tag values to show the custom field in a pre-defined position on the Update Area Settings, Incident Update, or Order Worksheet display. You can also manually enter the Location Tag value, if needed (for example, to show the custom field in a custom position on the display). For more details, see Showing Custom Fields in the Incident Update Display.
- **Application Type:** Comma-separated list of application types (for example, "SA,WFM,OMS"). To use a custom field in ADMS displays, the Application Type value must contain the Application Type value defined in OmsClientConfiguration.xml or be null.

> **Note**
>
> Each custom field must have a unique combination of Field Name, Application Type, and Location Tag values. (You can use multiple custom fields with the same field names but different application types.)

To add, delete, or move rows (order permission rules), use the following buttons:

- **Insert:** To insert a row.
- **Delete:** To delete a row.
-

- To move a row up.
- To move a row down.

You can also copy, paste, insert, delete, and move rows using the shortcut menu.

> **Note**
>
> Planned outage custom field values are communicated to the associated planned incident custom fields (field names must be the same).

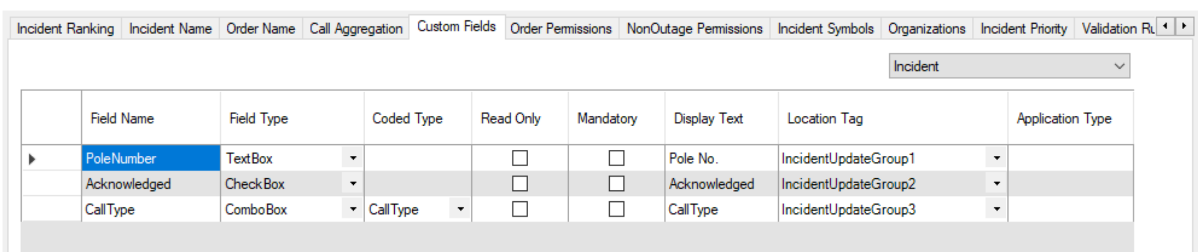
13.1. Showing Custom Fields in Grid Tabular Displays

Similar to the common incident and order fields, you can use the custom fields for grid tabular displays.

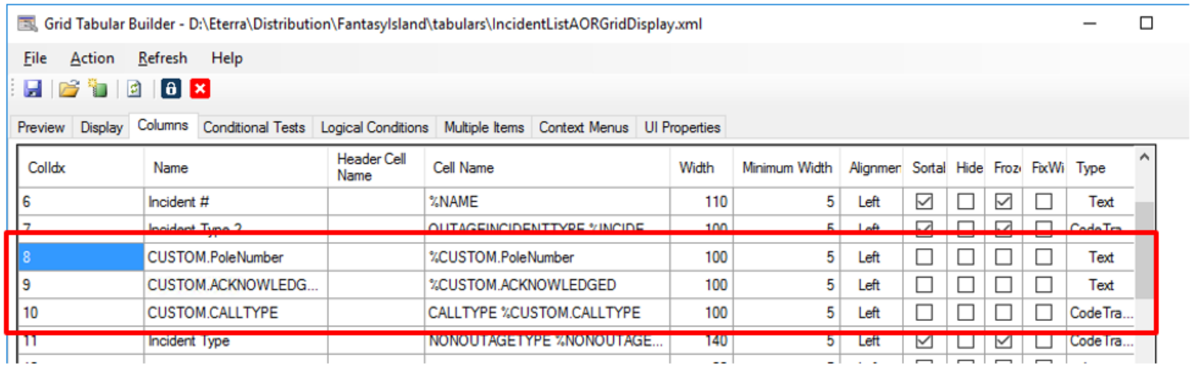
- When using on displays in the Standard OMS mode, the incident custom field query syntax is CUSTOM.<FIELDNAME> (for example, CUSTOM.POLENUMBER).
- When using on displays in the Enterprise OMS mode, the incident custom field query syntax is CUSTOM_INCIDENT.<FIELDNAME> (for example, CUSTOM_INCIDENT.POLENUMBER) and the order custom field query syntax is CUSTOM_ORDER.<FIELDNAME> (for example, CUSTOM_ORDER.LOCATION.

To show custom fields on the Incident Detail display:

1. Configure the custom fields on the Custom Fields tab of the OMS Server Configuration Editor tool.

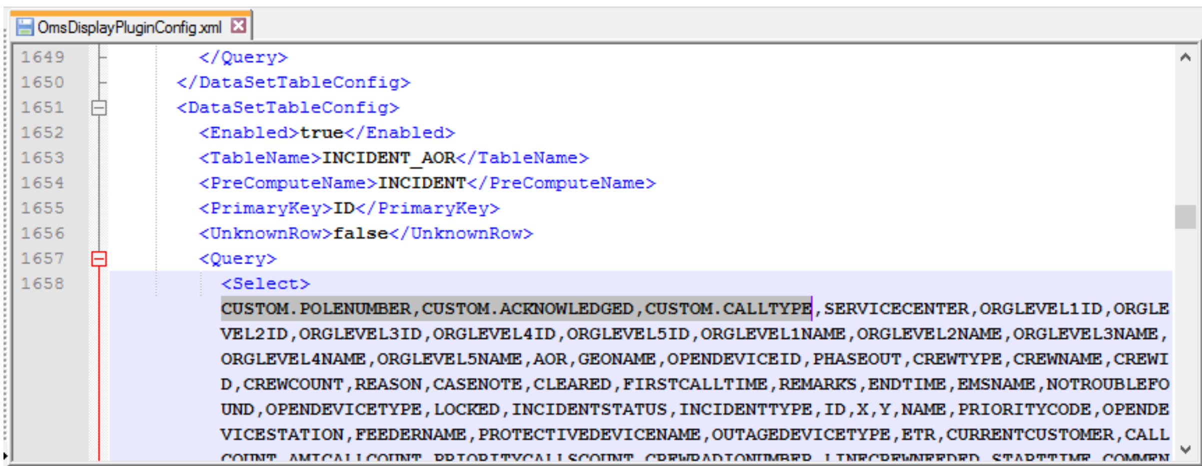


2. Add columns for custom fields to the IncidentListAORGridDisplay.xml using the Columns tab of the Grid Tabular Builder tool.



For more information about configuring the grid tabular displays, refer to the *Grid Tabular Builder User Guide*.

3. Add custom field queries to the INCIDENT_AOR table definition in the `OmsDisplayPluginConfig.xml` configuration file using the OMS Client Configuration Editor tool or a text editor.

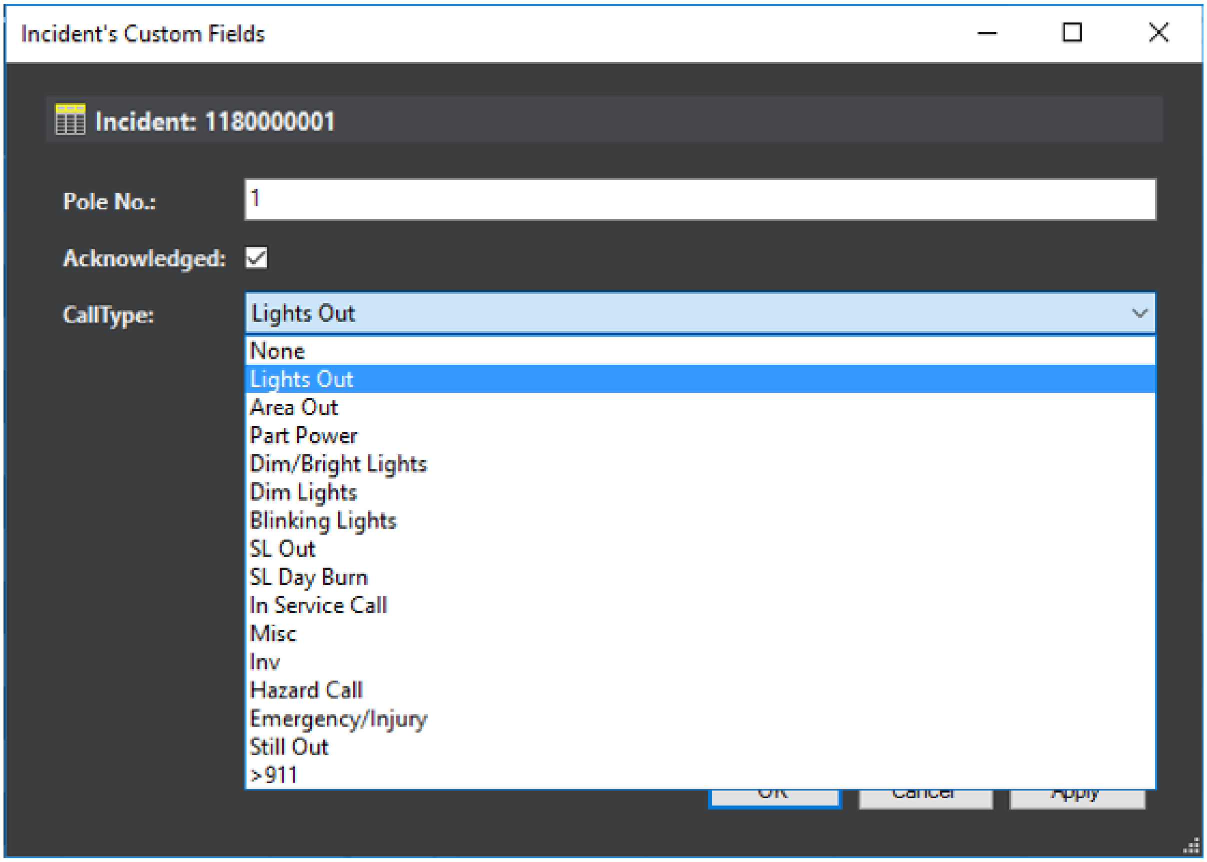


For more information about configuring the OMS Display Plugin, refer to the *OMS Client Configuration Editor User Guide*.

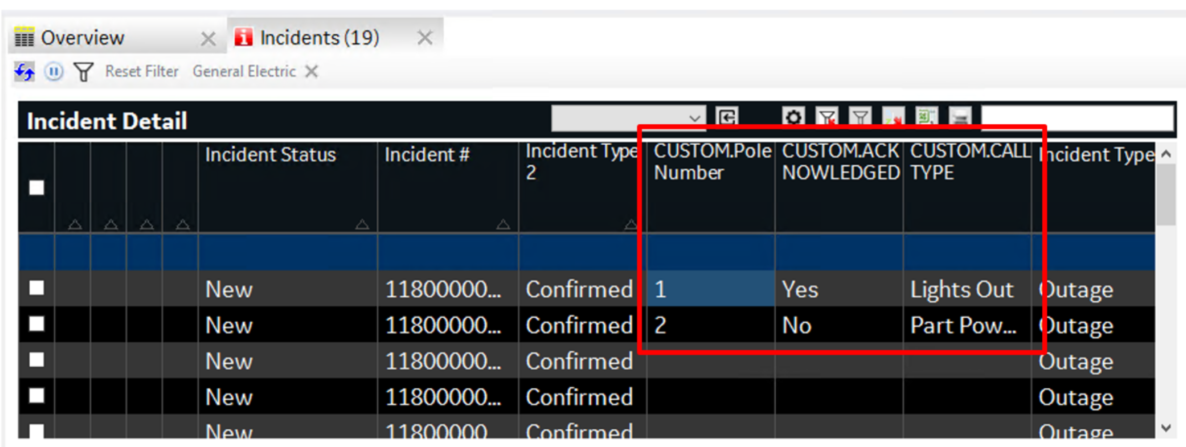
4. Restart the system.
5. Navigate to an incident in the Incident Detail display.
6. Select the Incident's Custom Fields option from the incident context menu.

   The Incident's Customs Fields window is displayed.

7. Enter custom field values for the incident.



8. Verify that the custom field values appear on the Incident Detail display.



The same logic applies when showing custom order fields on the Order List display.

13.2. Showing Custom Fields in the Incident Update Display

You can use the custom fields for the Incident Update display.

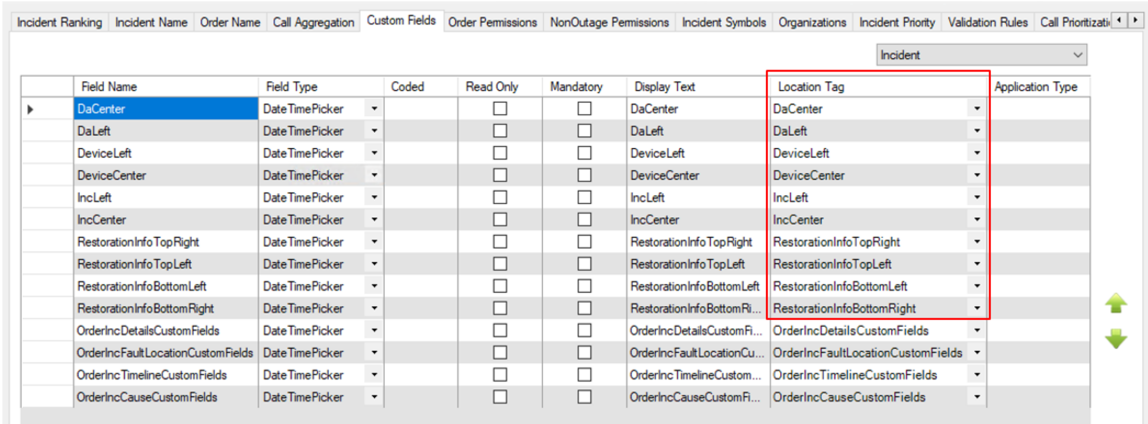
The Update Incident display has predefined placeholders for custom fields. You can also add custom fields at any position on the display by manually editing the associated HTML page.

To show custom fields in a predefined position on the Incident Update display:

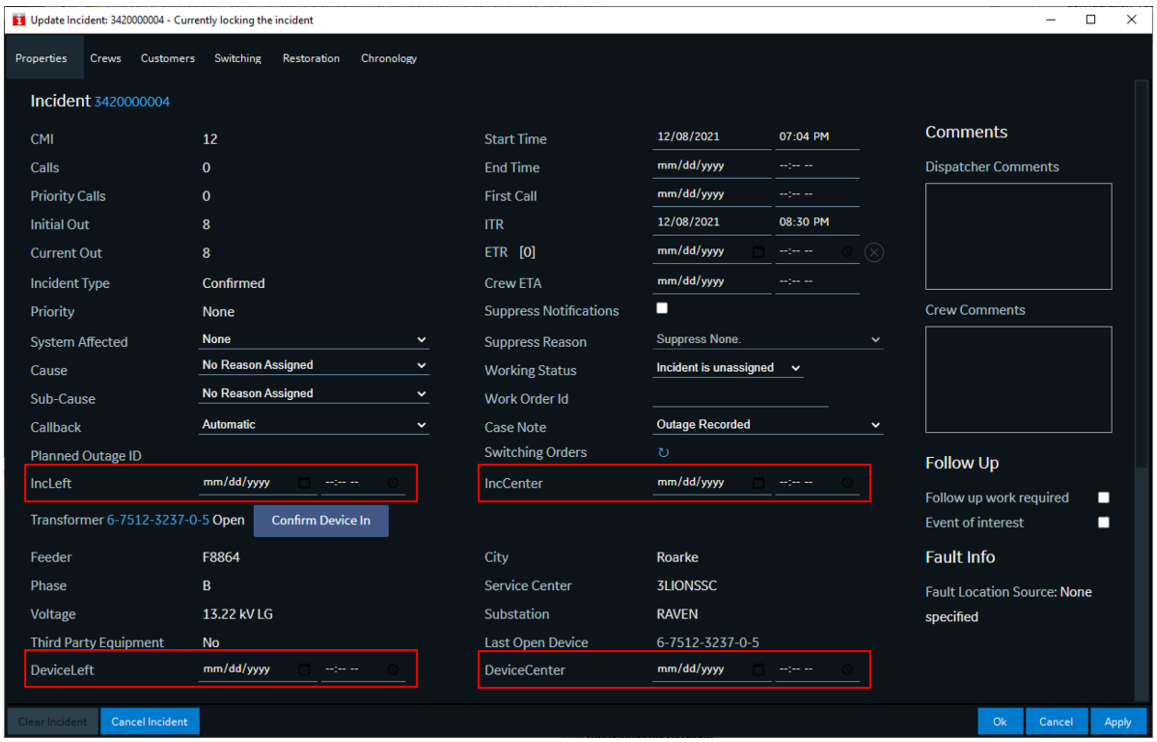
1. Configure Incident custom fields on the Custom Fields tab of the OMS Server Configuration Editor tool.
2. Use the following predefined Location Tag values:
    - **IncLeft:** Adds incident custom fields at the bottom of the Summary group on the Properties tab of Update Incident display (above the device information).
    - **IncCenter:** Adds incident custom fields at the bottom of the Incident Working Status group on the Properties tab of Update Incident display (above the service center information).
    - **DeviceLeft:** Adds incident custom fields at the bottom of the left Device group on the Properties tab of Update Incident display (above the damage assessment information).
    - **DeviceCenter:** Adds incident custom fields at the bottom of the right Device group on the Properties tab of Update Incident display.
    - **DaLeft:** Adds incident custom fields at the bottom of the left Damage Assessment group on the Properties tab of Update Incident display.
    - **DaCenter:** Adds incident custom fields at the bottom of the right Damage Assessment group on the Properties tab of Update Incident display.
    - **RestorationInfoTopLeft:** Adds incident custom fields at the bottom of the left Restoration Properties group on the Restoration tab of Update Incident display.
    - **RestorationInfoTopRight:** Adds incident custom fields at the bottom of the right Restoration Properties group on the Restoration tab of Update Incident display.
    - **RestorationInfoBottomLeft:** Adds incident custom fields under the Failed Equipment box on the Restoration tab of Update Incident display, in the left column.
    - **RestorationInfoBottomRight:** Adds incident custom fields under the Failed Equipment box on the Restoration tab of Update Incident display, in the right column.

    > **Note**
    >
    > The OrderIncDetailsCustomFields, OrderIncFaultLocationCustomFields, OrderIncTimelineCustomFields, and OrderIncCauseCustomFields are used to place incident custom fields on the Order Worksheet display.

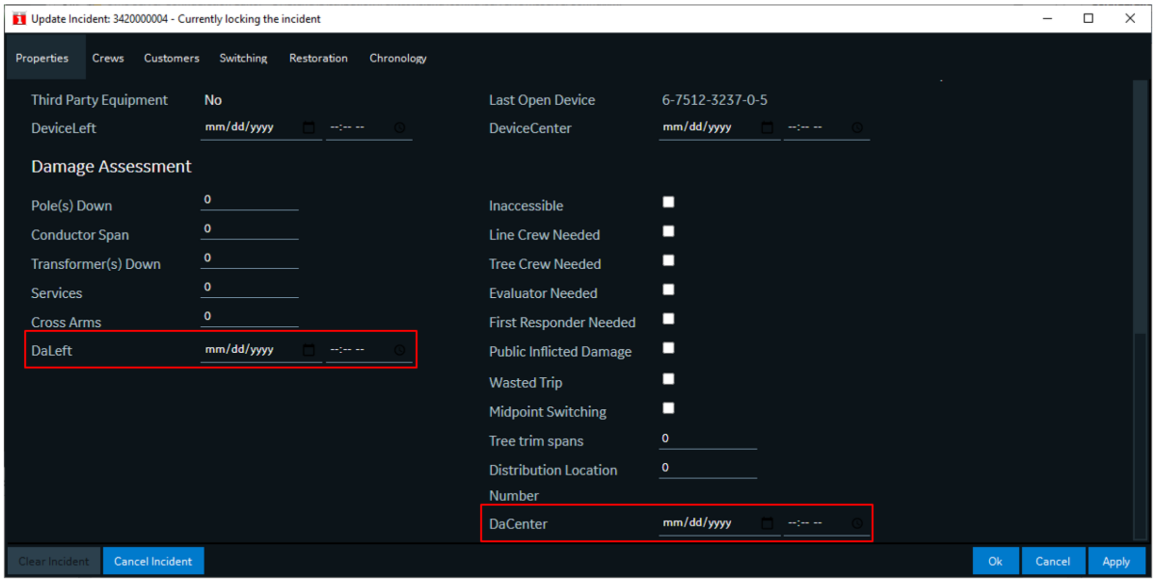
    For example:

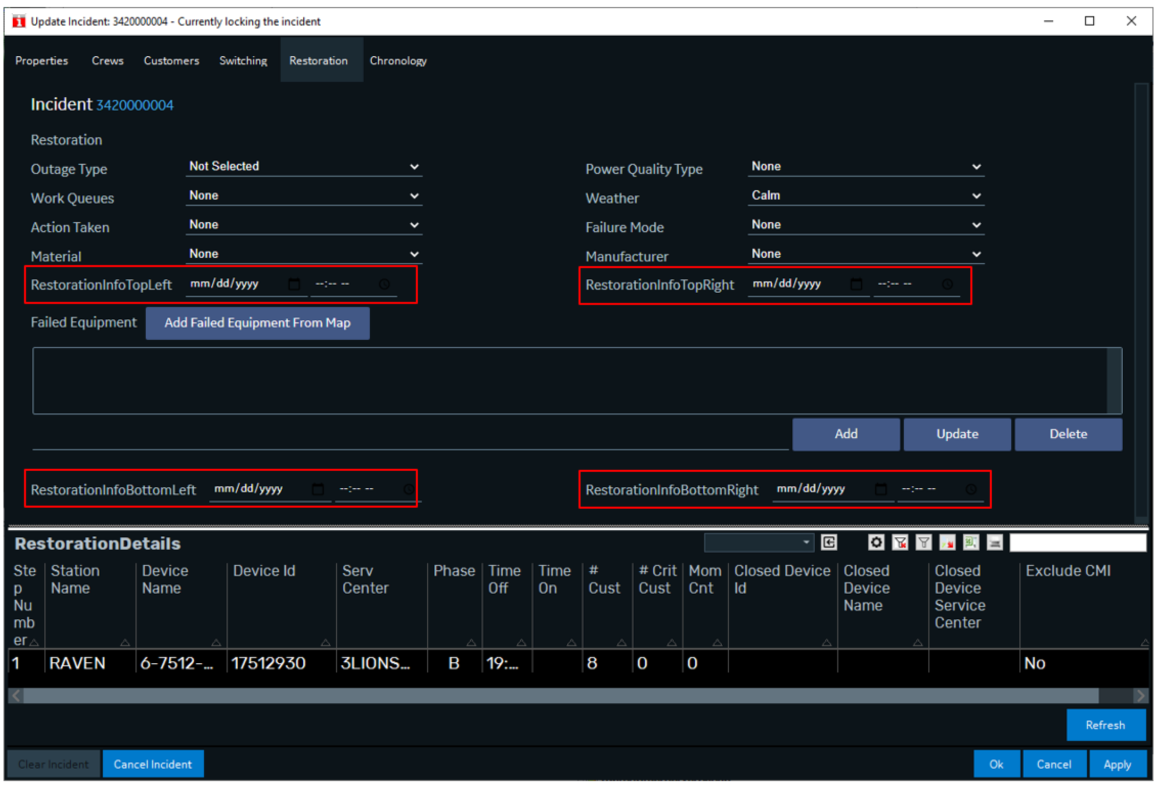


Incident Ranking
Incident Name
Order Name
Call Aggregation
Custom Fields
Order Permissions
NonOutage Permissions
Incident Symbols
Organizations
Incident Priority
Validation Rules
Call Prioritizati
Incident
Field Name
Field Type
Coded
Read Only
Mandatory
Display Text
Location Tag
Application Type
DaCenter
DaLeft
DeviceLeft
DeviceCenter
IncLeft
IncCenter
RestorationInfoTopRight
RestorationInfoTopLeft
RestorationInfoBottomLeft
RestorationInfoBottomRight
OrderIncDetailsCustomFields
OrderIncFaultLocationCustomFields
OrderIncTimelineCustomFields
OrderIncCauseCustomFields
DateTimePicker



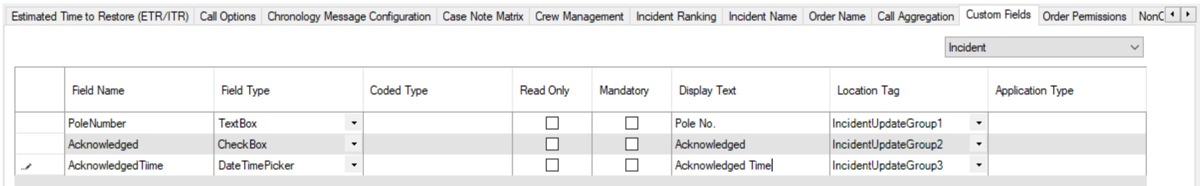
Update Incident: 3420000004 - Currently locking the incident
Properties
Crews
Customers
Switching
Restoration
Chronology
Incident 3420000004
CMI 12
Calls 0
Priority Calls 0
Initial Out 8
Current Out 8
Incident Type Confirmed
Priority None
System Affected None
Cause No Reason Assigned
Sub-Cause No Reason Assigned
Callback Automatic
Planned Outage ID
IncLeft mm/dd/yyyy
Transformer 6-7512-3237-0-5 Open
Confirm Device In
Feeder F8864
Phase B
Voltage 13.22 kV LG
Third Party Equipment No
DeviceLeft mm/dd/yyyy
Start Time 12/08/2021 07:04 PM
End Time mm/dd/yyyy
First Call mm/dd/yyyy
ITR 12/08/2021 08:30 PM
ETR [0] mm/dd/yyyy
Crew ETA mm/dd/yyyy
Suppress Notifications
Suppress Reason Suppress None.
Working Status Incident is unassigned
Work Order Id
Case Note Outage Recorded
Switching Orders
IncCenter mm/dd/yyyy
City Roarke
Service Center 3LIONSSC
Substation RAVEN
Last Open Device 6-7512-3237-0-5
DeviceCenter mm/dd/yyyy
Comments
Dispatcher Comments
Crew Comments
Follow Up
Follow up work required
Event of interest
Fault Info
Fault Location Source: None specified
Clear Incident
Cancel Incident
Ok
Cancel
Apply





To show custom fields in a custom position on the Incident Update display:

1. Configure the custom fields on the Custom Fields tab of the OMS Server Configuration Editor tool.



2. Open the UpdateIncidentPage.html file (stored in %IDMS_ROOT%\webpages\incident\ in a text editor.
3. Add the entries for incident custom fields under existing fields.

   The format is:

   ```
   <div class="grid-custom-fields-column-<ExistingColumnNumber>" id="<CustomFieldLocationTag>"></div>
   ```

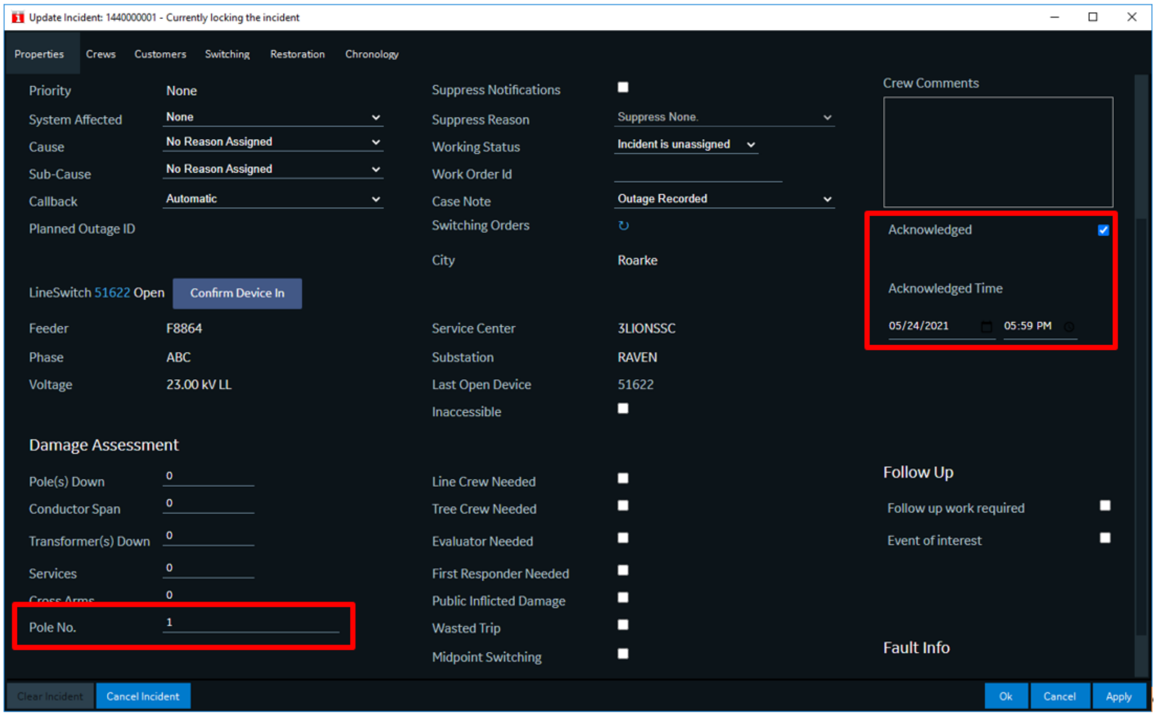
   Where:

   - grid-custom-fields-column-<ExistingColumnNumber>: Existing column number tag in the UpdateIncidentPage.html.
   - CustomFieldLocationTag: Location Tag value from the custom field configuration in OmsServerConfig.xml.

     For example:

     ```
     <label class="grid-label-column-1">Cross Arms</label>
     <input class="grid-value-column-1" id="numberOfCrossArms" type="number" min="0" data-property-
     name="NOOFCROSSARMS" onfocusout="onChange(this)" />
     <div class="grid-custom-fields-column-1" id="IncidentUpdateGroup1"></div>
     ...
     <label class="grid-label-column-3 grid-label-vertical grid-label-textarea">Crew
     Comments</label>
     <div class="grid-value-column-3 grid-value-vertical grid-value-vertical-4">
     <textarea data-property-name="CREWCOMMENTS" onfocusout="onChange(this)"></textarea>
     </div>
     <div class="grid-custom-fields-column-3" id="IncidentUpdateGroup2"></div>
     ```

     This adds all fields with the specified Location Tag values in the specified places in the Incident Update display.



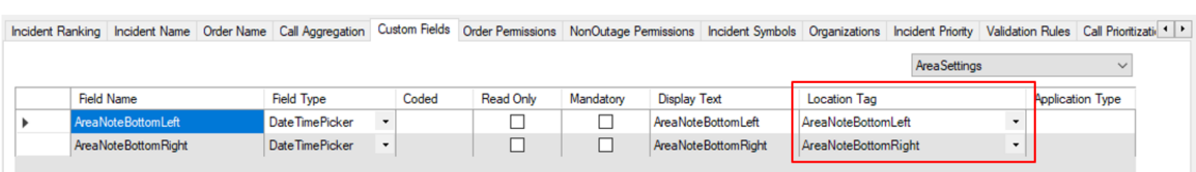
13.3. Showing Custom Fields in the Update Area Settings Display

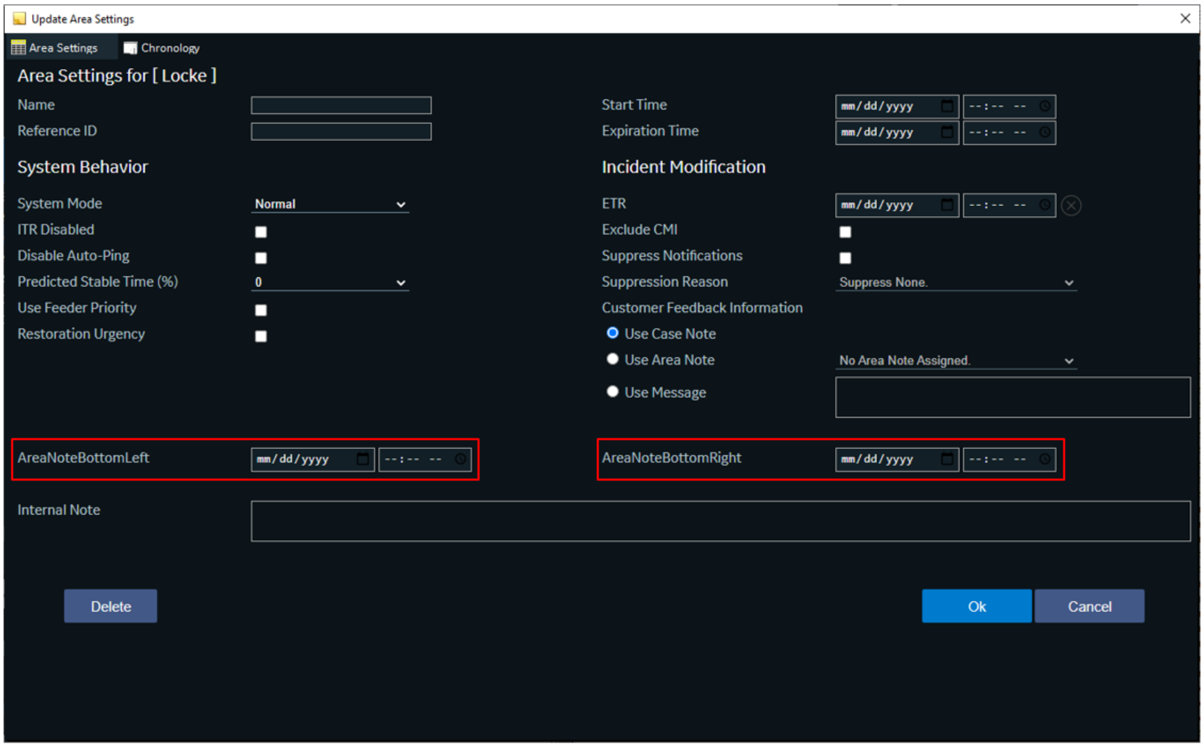
You can use custom fields on the Update Area Settings display, which has predefined placeholders for custom fields.

To show custom fields in a predefined position on the Incident Update display:

1. Configure the AreaSettings custom fields on the Custom Fields tab of the OMS Server Configuration Editor tool.
2. Use the following predefined Location Tag values:
    - **AreaNoteBottomLeft:** Adds area note custom fields above the Internal Note field on the Update Area Settings display, in the left column.
    - **AreaNoteBottomRight:** Adds area note custom fields above the Internal Note field on the Update Area Settings display, in the right column.

   For example:





13.4. Showing Custom Fields in the Order Worksheet Display

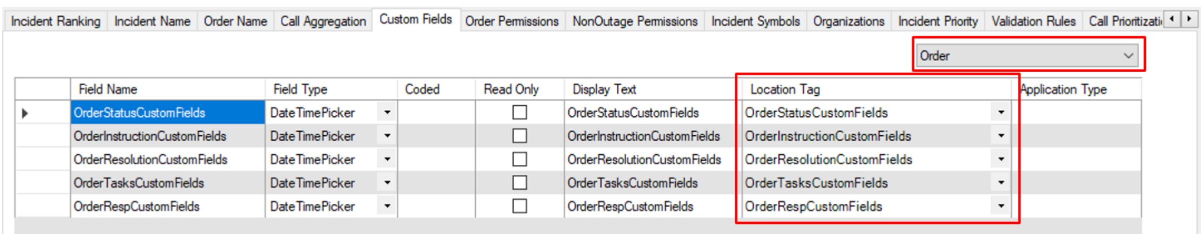
You can use custom fields on the Order Worksheet display, which has predefined placeholders for custom fields.

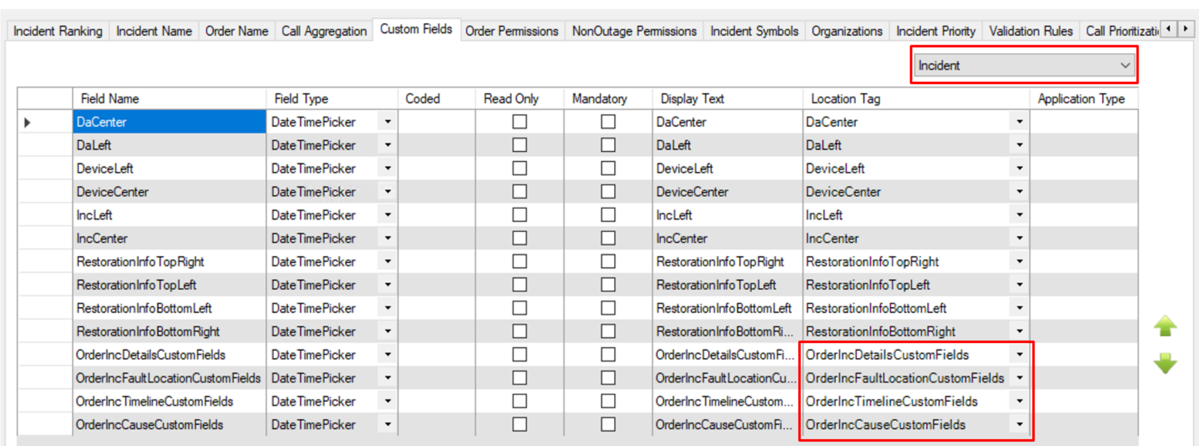
To show custom fields in a predefined position on the Incident Update display:

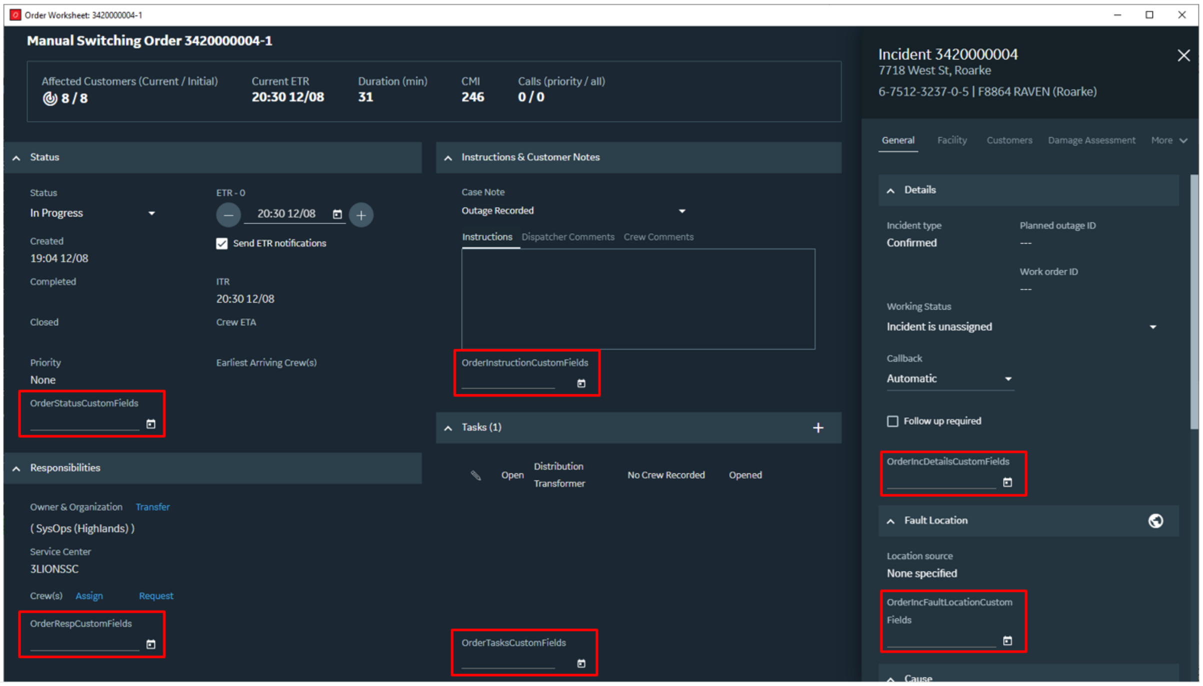
1. Configure the Order custom fields on the Custom Fields tab of the OMS Server Configuration Editor tool.
2. Use the following predefined Location Tag values for the Order custom fields:
   - **OrderStatusCustomFields:** Adds order custom fields at the bottom of the Status section on the main panel.
   - **OrderRespCustomFields:** Adds order custom fields at the bottom of the Responsibilities section on the main panel.
   - **OrderInstructionCustomFields:** Adds order custom fields at the bottom of the Instructions and Customer Notes section on the main panel.
   - **OrderTasksCustomFields:** Adds order custom fields at the bottom of the Tasks section on the main panel.
   - **OrderResolutionCustomFields:** Adds order custom fields at the bottom of the Resolution section on the main panel.

3. Configure the Incident custom fields on the Custom Fields tab of the OMS Server Configuration Editor tool.
4. Use the following predefined Location Tag values for the Incident custom fields:
    - **OrderIncDetailsCustomFields:** Adds incident custom fields at the bottom of the Details section on the General tab of the side panel.
    - **OrderIncFaultLocationCustomFields:** Adds incident custom fields at the bottom of the Fault Location section on the General tab of the side panel.
    - **OrderIncCauseCustomFields:** Adds incident custom fields at the bottom of the Cause section on the General tab of the side panel.
    - **OrderIncTimelineCustomFields:** Adds incident custom fields at the bottom of the Timeline section on the General tab of the side panel.

    For example:



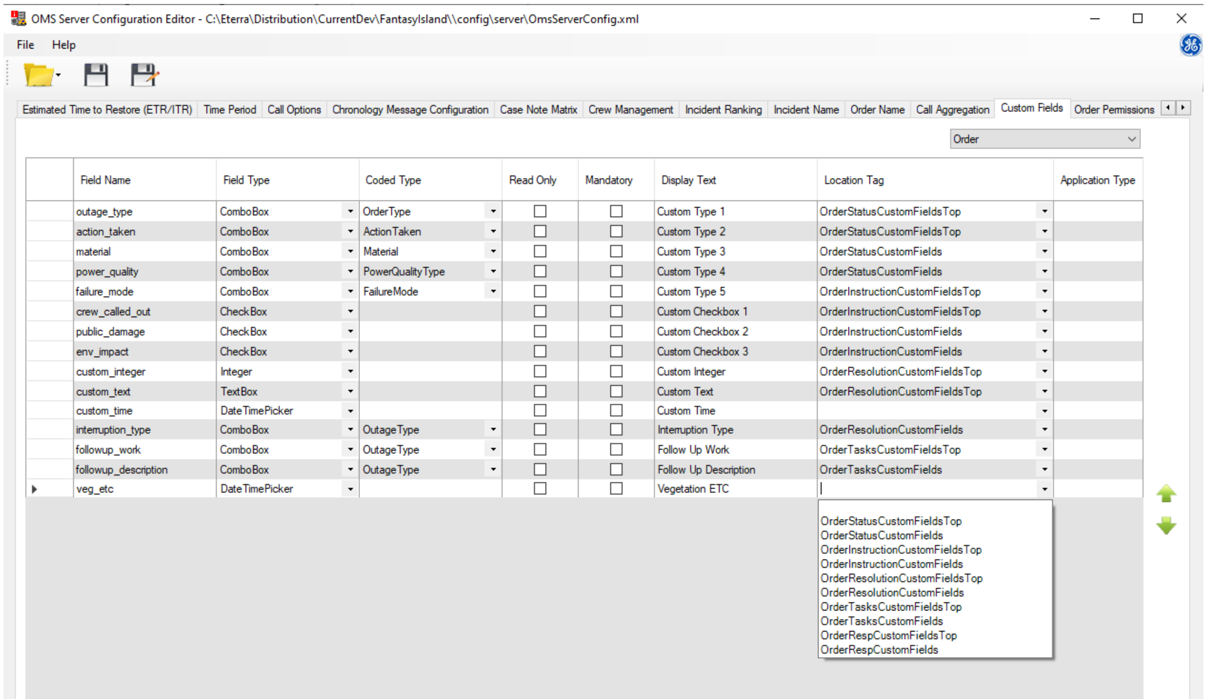


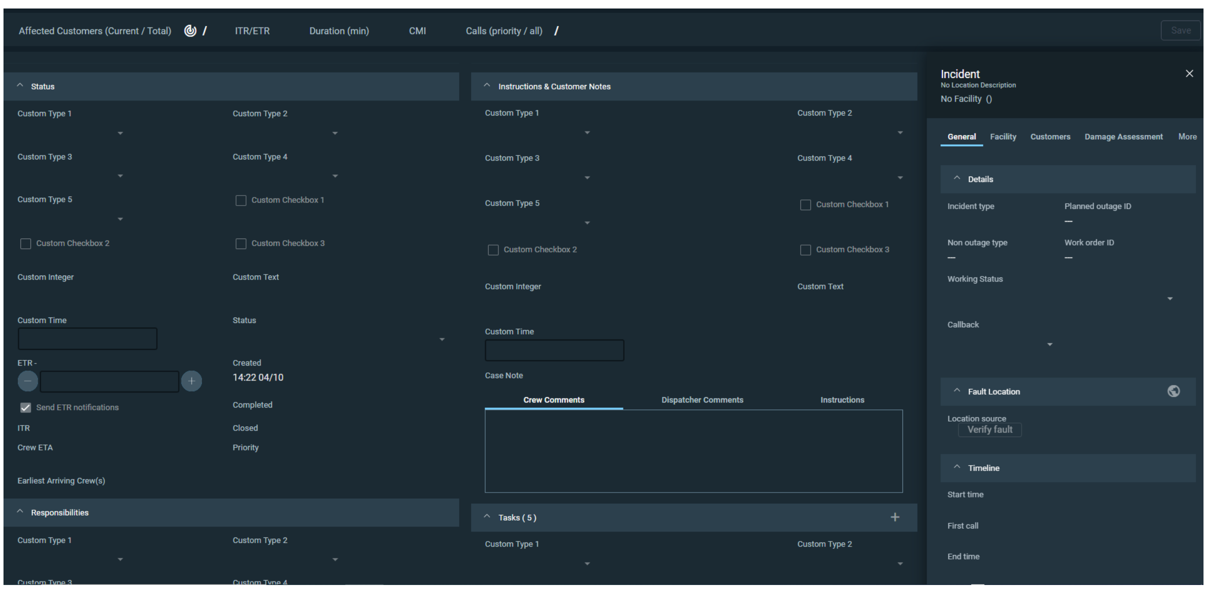


5. Use the following predefined Location Tag values for the Order custom fields:

    - **OrderStatusCustomFieldsTop:** Adds order custom fields at the top of the Status section on the main panel.
    - **OrderRespCustomFieldsTop:** Adds order custom fields at the top of the Responsibilities section on the main panel.
    - **OrderInstructionCustomFieldsTop:** Adds order custom fields at the top of the Instructions and Customer Notes section on the main panel.
    - **OrderTasksCustomFieldsTop:** Adds order custom fields at the top of the Tasks section on the main panel.
    - **OrderResolutionCustomFieldsTop:** Adds order custom fields at the top of the Resolution section on the main panel.

    For example:


OMS Server Configuration Editor - C:\Eterra\Distribution\CurrentDev\FantasyIsland\\config\server\OmsServerConfig.xml
File Help
Estimated Time to Restore (ETR/ITR) Time Period Call Options Chronology Message Configuration Case Note Matrix Crew Management Incident Ranking Incident Name Order Name Call Aggregation Custom Fields Order Permissions
Order
Field Name Field Type Coded Type Read Only Mandatory Display Text Location Tag Application Type
outage_type ComboBox OrderType Custom Type 1 OrderStatusCustomFieldsTop
action_taken ComboBox ActionTaken Custom Type 2 OrderStatusCustomFieldsTop
material ComboBox Material Custom Type 3 OrderStatusCustomFields
power_quality ComboBox PowerQualityType Custom Type 4 OrderStatusCustomFields
failure_mode ComboBox FailureMode Custom Type 5 OrderInstructionCustomFieldsTop
crew_called_out CheckBox Custom Checkbox 1 OrderInstructionCustomFieldsTop
public_damage CheckBox Custom Checkbox 2 OrderInstructionCustomFields
env_impact CheckBox Custom Checkbox 3 OrderInstructionCustomFields
custom_integer Integer Custom Integer OrderResolutionCustomFieldsTop
custom_text TextBox Custom Text OrderResolutionCustomFieldsTop
custom_time DateTimePicker Custom Time
interruption_type ComboBox OutageType Interruption Type OrderResolutionCustomFields
followup_work ComboBox OutageType Follow Up Work OrderTasksCustomFieldsTop
followup_description ComboBox OutageType Follow Up Description OrderTasksCustomFields
veg_etc DateTimePicker Vegetation ETC
OrderStatusCustomFieldsTop
OrderStatusCustomFields
OrderInstructionCustomFieldsTop
OrderInstructionCustomFields
OrderResolutionCustomFieldsTop
OrderResolutionCustomFields
OrderTasksCustomFieldsTop
OrderTasksCustomFields
OrderRespCustomFieldsTop
OrderRespCustomFields


Affected Customers (Current / Total) ITR/ETR Duration (min) CMI Calls (priority / all)
Status
Custom Type 1
Custom Type 2
Custom Type 3
Custom Type 4
Custom Type 5
Custom Checkbox 1
Custom Checkbox 2
Custom Checkbox 3
Custom Integer
Custom Text
Custom Time
Status
ETR -
Created
14:22 04/10
Send ETR notifications
Completed
ITR
Closed
Crew ETA
Priority
Earliest Arriving Crew(s)
Responsibilities
Instructions & Customer Notes
Case Note
Crew Comments
Dispatcher Comments
Instructions
Tasks (5)
Incident
No Location Description
No Facility ()
General
Facility
Customers
Damage Assessment
More
Details
Incident type
Planned outage ID
Non outage type
Work order ID
Working Status
Callback
Fault Location
Location source
Verify fault
Timeline
Start time
First call
End time

14. Order Permissions Tab

Use the Order Permissions tab to define functional permissions (Areas of Responsibility) in the PERMIT model that are required for updating specific types of follow-up work orders.

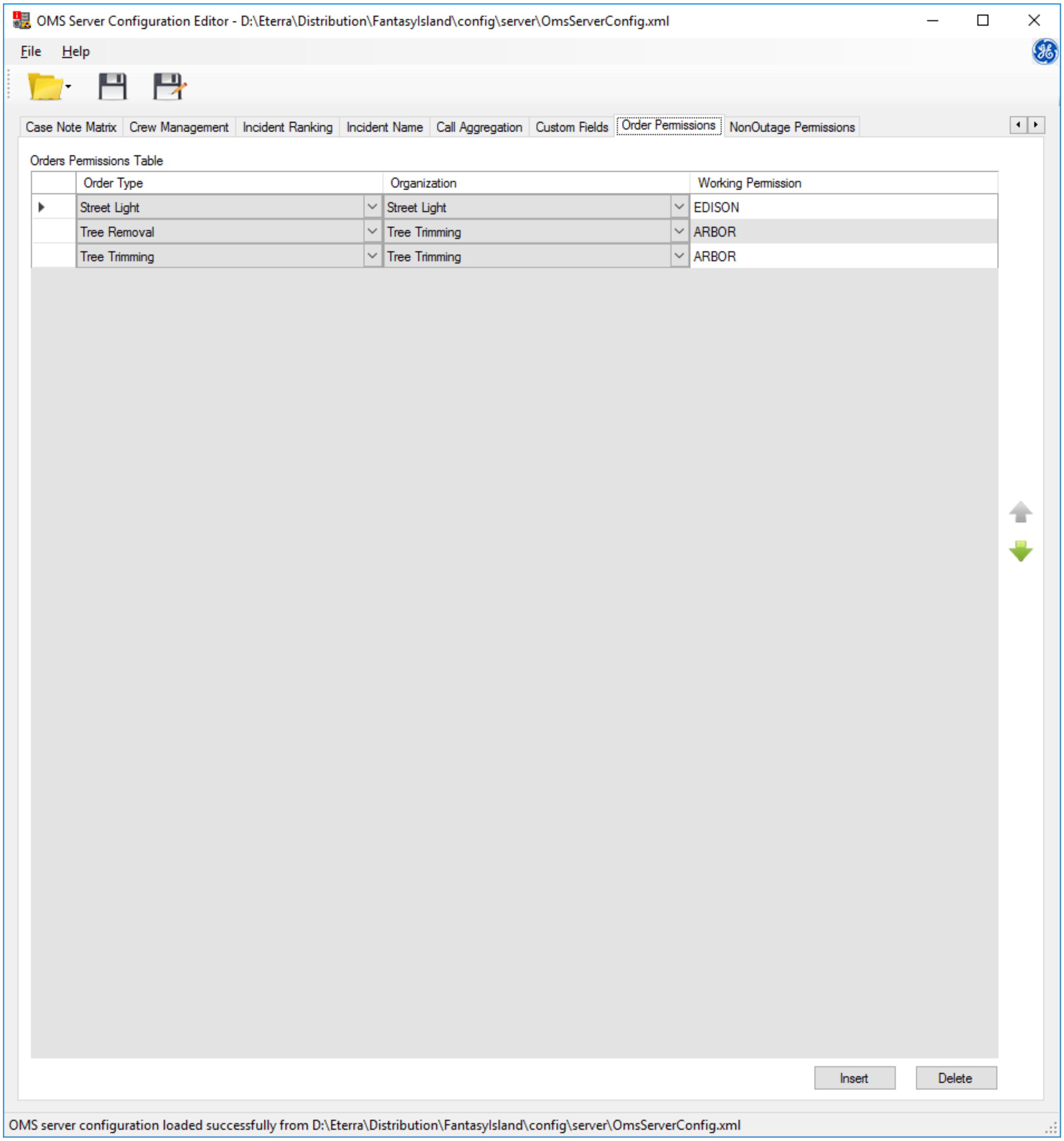


Figure 22. Order Permissions Tab

The Order Permissions tab includes the following fields and controls:

- **Order Type:** The follow-up work order type.

- **Organization:** The follow-up work order organization.
- **Working Permission:** Specifies the name of the functional permission in PERMIT that is required to update follow-up work orders of the specified type and organization.

> **Note**
>
> The Order Type and Organization values are taken from the coded translation configuration file.

For example, the first rule in the above image specifies that only users with the EDISON functional permission in PERMIT can update follow-up work orders with the Street Light type and Street Light organization.

To add, delete, or move rows (order permission rules), use the following buttons:

- **Insert:** To insert a row.
- **Delete:** To delete a row.
- To move a row up.
- To move a row down.

You can also copy, paste, insert, delete, and move rows using the shortcut menu.

15. NonOutage Permissions Tab

Use the NonOutage Permissions tab to define functional permissions (Areas of Responsibility) in the PERMIT model that are required for updating or clearing specific types of non-outage incidents.

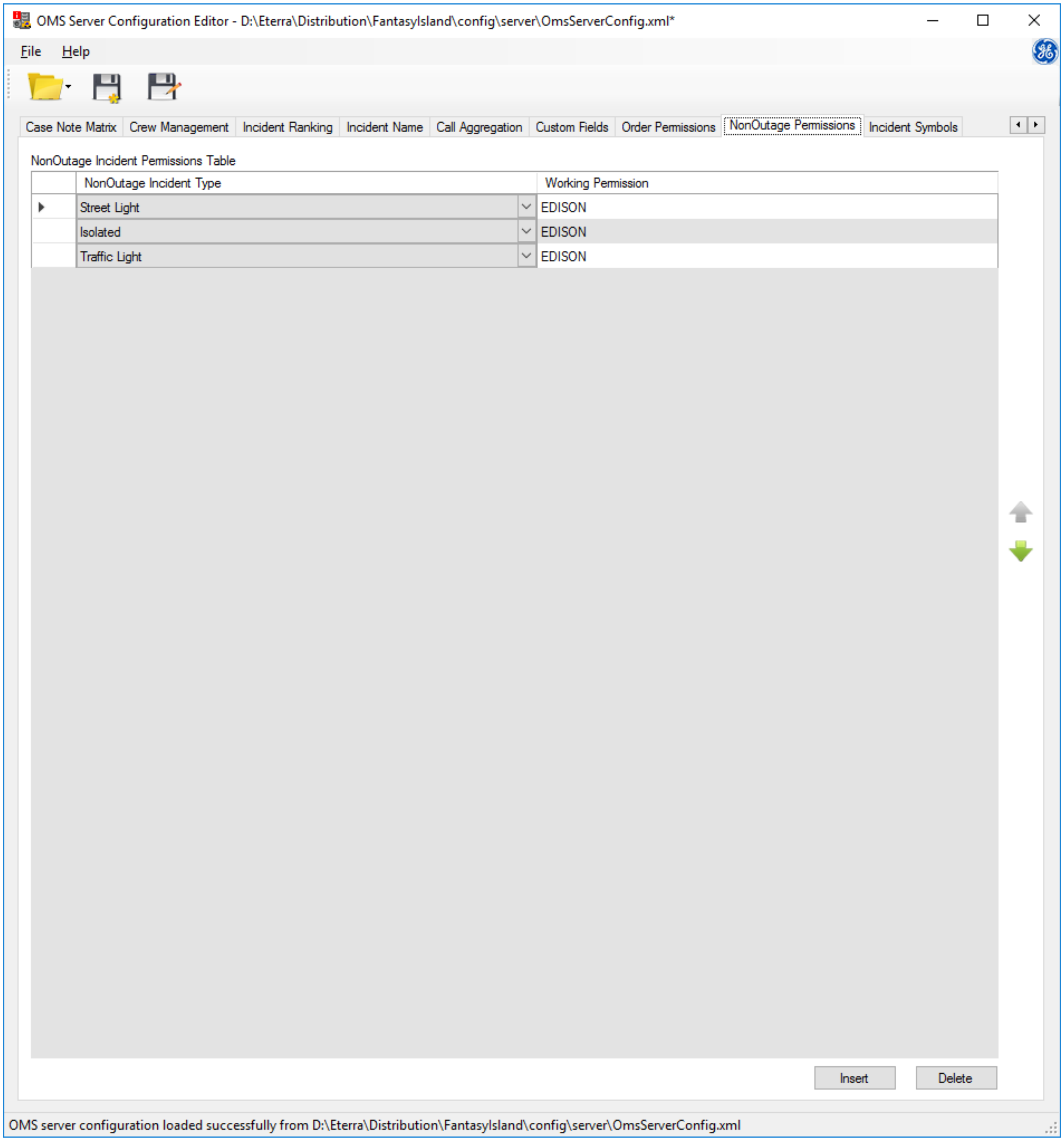


Figure 23. NonOutage Permissions Tab

The NonOutage Permissions tab includes the following fields and controls:

- **NonOutage Incident Type:** The non-outage incident type.

- **Working Permission:** Specifies the name of the functional permission in PERMIT that is required to update or clear non-outage incidents.

> **Note**
>
> NonOutage Incident Type values are taken from the coded translation configuration file.

For example, the first rule in the above image specifies that only users with the EDISON functional permission in PERMIT can update or clear non-outage incidents of the Street Light type.

To add, delete, or move rows (non-outage permission rules), use the following buttons:

- **Insert:** To insert a row
- **Delete:** To delete a row
- To move a row up
- To move a row down

You can also copy, paste, insert, delete, and move rows using the shortcut menu.

16. Incident Symbols Tab

Use the Incident Symbols tab to define incident icons on the geographic network view for different incident categories, types, devices, and working statuses. In addition, you can use the IncidentSymbol query field, as text or as a bitmap, in grid tabular and attribute displays.

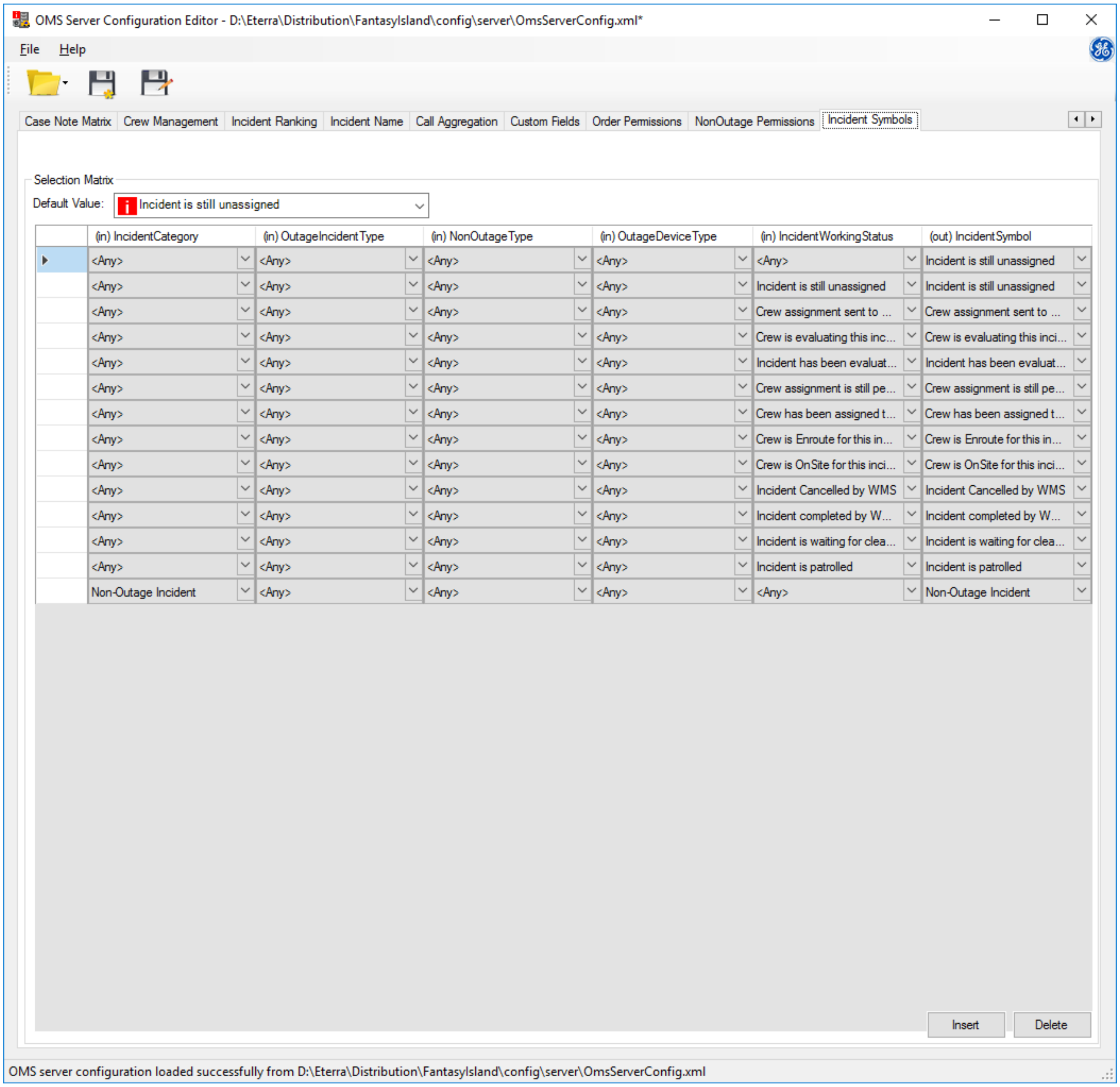


Figure 24. Incident Symbols Tab

The Incident Symbols tab includes the following controls:

- **Default Value:** Shows the default entry of the IncidentSymbol coded translation item that is applied when no incident symbol selection rule is satisfied.
- **Insert:** Adds a row for a new incident symbol selection rule.

- **Delete:** Removes the selected rule.

The Selection Matrix table includes the following input (in) and output (out) fields:

- **IncidentCategory:** Includes values of the IncidentCategory coded translation item to specify the incident's category (outage incident or a non-outage incident).
- **OutageIncidentType:** Includes values of the OutageIncidentType coded translation item to specify the incident type (for example, confirmed or non-outage).
- **NonOutageType:** Includes values of the NonOutageType coded translation item to specify the non-outage incident type (for example, street light or isolated).
- **OutageDeviceType:** Includes values of the OutageDeviceType coded translation item to specify the incident device type (for example, fuse or sectionalizer).
- **IncidentWorkingStatus:** Includes values of the IncidentWorkingStatus coded translation item to specify the incident working status.
- **IncidentSymbol:** Includes values of the IncidentSymbol coded translation item to specify the resulting incident symbol.

For example, the below image shows a non-outage incident icon on the geographic network view.

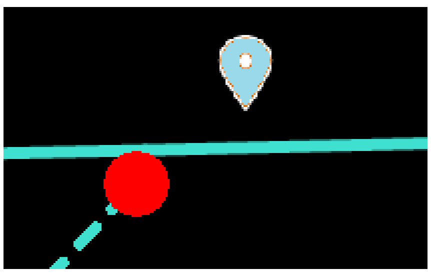


Figure 25. Non-Outage Incident on the Geographic View

Determining the symbol to use for a particular incident is based on the left-to-right evaluation of the attribute rules defined by the input (in) columns. Starting from the left, the rows that exactly match the incident's attribute value (the <Any> tag is not a match) for the column being evaluated are kept and those that don't are discarded. If the incident's attribute value doesn't exactly match any of the column criteria being evaluated, rows with the <Any> tag in that column are kept and others are discarded.

For example, when selecting the symbol for a Non-Outage Incident it matches the first column "(in) Incident Category" so, it will keep the last row and discard all other rows since they do not match the field. It will then continue with the second column and so on, and since the rest of the columns for the last row are <Any> it will match any values for the fields and finally select the symbol "Non-Outage Incident".

On the other hand when selecting the symbol for an Outage Incident, it will try to match the first column "(in) Incident Category" and since none of the rows match it will keep all the rows with the <Any> tag, discard the last row with value "Non-Outage Incident" and continue with the second column and since the next columns all contain the <Any> tag it will keep them and select symbol based on column "(in) IncidentWorkingStatus".

If at the end of selection none of the rows match, the "Default Value" symbol is used.

17. Order Symbols Tab

Use the Order Symbols tab to define order icons on the geographic network view for different order categories, types, devices, and working statuses. In addition, you can use the OrderSymbol query field, as text or as a bitmap, in grid tabular and attribute displays.

> **Note**
>
> To show order symbols instead of incident symbols on the network view, the Show Orders Instead Of Incidents option must be enabled in the Net Dynamics Configuration Editor.

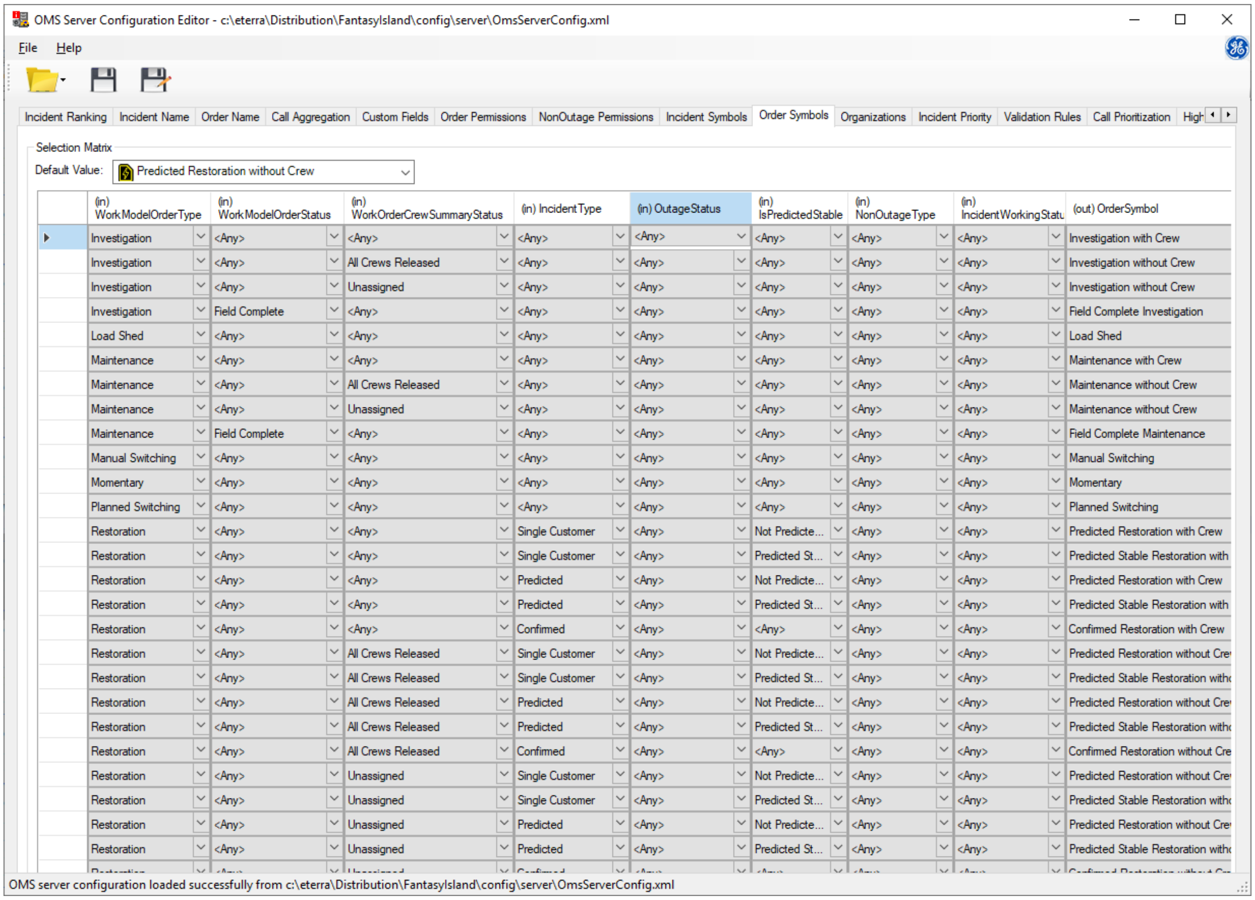


Figure 26. Order Symbols Tab

The Order Symbols tab includes the following controls:

- **Default Value:** Shows the default entry of the OrderSymbol coded translation item that is applied when no order symbol selection rule is satisfied.
- **Insert:** Adds a row for a new order symbol selection rule.
- **Delete:** Removes the selected rule.

The Selection Matrix table includes the following input (in) and output (out) fields:

- **WorkModelOrderType:** Includes values of the WorkModelOrderType coded translation item to specify the order type (for example, restoration or investigation).
- **WorkModelOrderStatus:** Includes values of the WorkModelOrderStatus coded translation item to specify the order status (for example, field complete).
- **WorkOrderCrewSummary:** Includes values of the WorkOrderCrewSummary coded translation item to specify the crew status (for example, unassigned).
- **IncidentType:** Includes values of the IncidentType coded translation item to specify the incident type (for example, confirmed or non-outage).
- **OutageStatus:** Includes values of the OutageStatus coded translation item to specify the outage incident status (for example, predicted or confirmed).
- **IsPredictedStable:** Includes values to specify the predicted stable status of the associated incident (Predicted Stable or Not Predicted Stable).
- **NonOutageType:** Includes values of the NonOutageType coded translation item to specify the non-outage incident type (for example, street light or isolated).
- **IncidentWorkingStatus:** Includes values of the IncidentWorkingStatus coded translation item to specify the incident working status (for example, incident is unassigned).
- **OrderSymbol:** Includes values of the OrderSymbol coded translation item to specify the resulting order symbol.

For example, the below image shows an unassigned restoration order icon on the geographic network view.

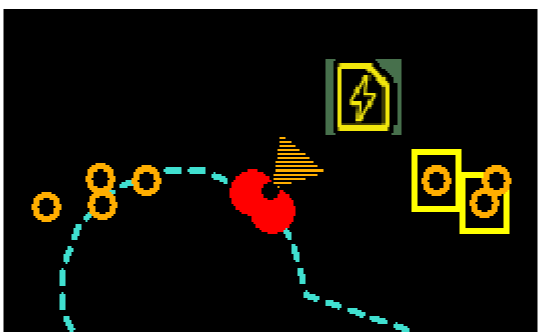
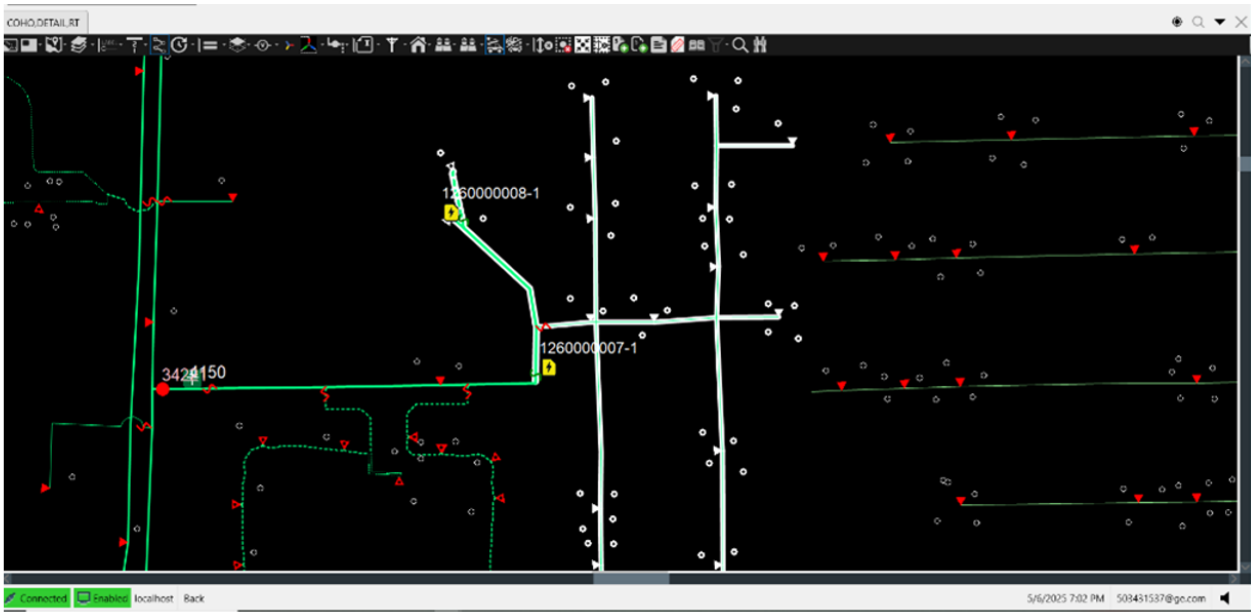


Figure 27. Unassigned Restoration Order on the Geographic View

The logic for the selection of a symbol to user for orders is similar to that used for incidents. It is based on the left-to-right evaluation of the attribute rules defined by the input (in) columns. Starting from the left, the rows that exactly match the order's attribute value (the <Any> tag is not a match) for the column being evaluated are kept and those that don't are discarded. If the order's attribute value doesn't exactly match any of the column criteria being evaluated, rows with the <Any> tag in that column are kept and others are discarded. Refer to Incident Symbol Tab for an example of the logic used for symbol selection.

For example, when a new restoration order (outage incident) is created, the system automatically matches the `WorkOrderCrewSummaryStatus` column with the value Unassigned. To ensure correct behavior, include a lookup entry with Unassigned explicitly specified. The <Any> tag does not account for this value.



The system determines the `IncidentType` based on the crew status Unassigned. It checks for incident types categorized under Unassigned. Because Manually Confirmed is not currently included in this category, the system defaults to a predefined value. If Manually Confirmed is added to the Unassigned category, the correct symbol is selected. The <Any> tag is used only when no specific entries exist for the Unassigned category. This behavior applies to all columns in the `OmsServerConfig` table.

18. Organizations Tab

> **Note**
>
> This feature requires that the WORKMODEL_WEBDB_ENABLED parameter is enabled (= 1) in the `NetServerOmsConfig.txt` configuration file. In other words, the Work Model function must be enabled.

> **Note**
>
> Before defining the tables in the Organizations Tab, define the WorkOrganizationTypes (organization types), WorkOrganizations (organizations), OutageIncidentType (outage incident types), and NonoutageType (non-outage incident types), CallType (call types), PlannedOutageType (planned incident subtypes), and ConfirmedIncidentSubtype (confirmed incident subtypes) coded translation items in the in Coded Translation Editor.

> **Important**
>
> WorkOrganizations coded translation item ShortName value length must be 8 characters or less, because WorkOrganizations correspond to AREA names in the Permit database.

Use the Organizations tab to map organizations to polygon groups and organization types and define which organization type is assigned to a specific incident (and order) type.

For example, you can configure that when an outage incident (and restoration order) occurs, this incident and order are assigned to a dispatch center organization type (for example, Dispatch). The system will then determine a specific organization that covers the outage area (for example, DSP_WST) based on the location of the incident's device of record.

Example 1

A single customer incident (Outage type) with the Any call type is created.

- The Organizations Matrix table specifies that these conditions match the Dispatch organization type.

Organizations Matrix

Incident Type	Outage?	Call Type	Organization Type
SingleCustomer	Outage	Any	Dispatch
Momentary	Outage	Any	Dispatch

- The Organization Types table specifies that the Dispatch organization type maps to the DISPATCH polygon group.

Organization Types

Organization Type	Polygon Group	Assign Organization
Dispatch	DISPATCH	
Planning		Planning

- The Organizations table specifies that the Dispatch organization type maps to the Dispatch

Organization, Dispatch (Beaches), Dispatch (Highlands), Dispatch (Plains), and Dispatch (Delta) organizations.

Organizations

Organization	Organization Type
Dispatch Organization	Dispatch
Dispatch (Beaches)	Dispatch
Dispatch (Highlands)	Dispatch
Dispatch (Plains)	Dispatch
Dispatch (Delta)	Dispatch
Planning	Planning

As a result, the DISPATCH polygon group is consulted to provide a specific organization name (for example, Dispatch (Delta) based on the location of the incident.

Example 2

A non-outage incident of the Hazard type with a Hazard call type is being created.

- The Organizations Matrix table specifies that these conditions match the Hazard Management organization type.

Organizations Matrix

Incident Type	Outage?	Call Type	Organization Type
Outage	Outage	Any	Dispatch
Hazard	Nonoutage	Hazard Call	Hazard Management

- The Organization Types table specifies that the Hazard Management organization type does not map to a polygon group. Instead, the Hazard Management organization type maps to the Hazard Management organization.

Organization Types

Organization Type	Polygon Group	Assign Organization
Dispatch	DISPATCH	
Planning		Planning
Storm Teams		Storm Team
Forestry	FOREST	
System Operations	SYSOPS	
Hazard Management		Hazard Management
Construction-Maintenance	CM	
Power Quality	PWRQUAL	
Data Quality		Data Quality
Unassociated Calls		Unassociated Calls

As a result, the Hazard Management organization is assigned.

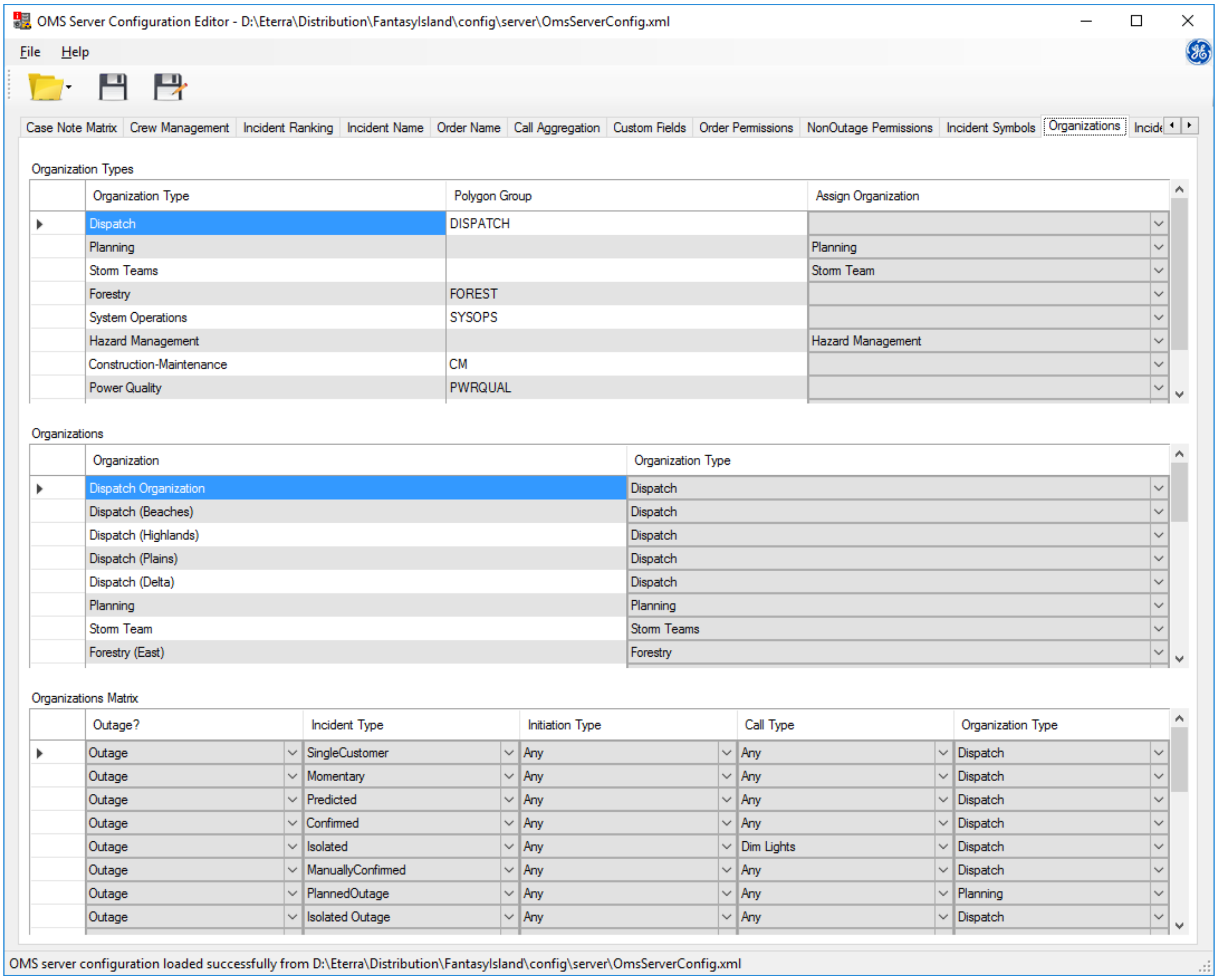


Figure 28. Work Organizations Tab

18.1. Organization Types

The Organization Types table defines mapping between organization types and company polygon groups (maps) or specifies which organization to assign for an organization type.

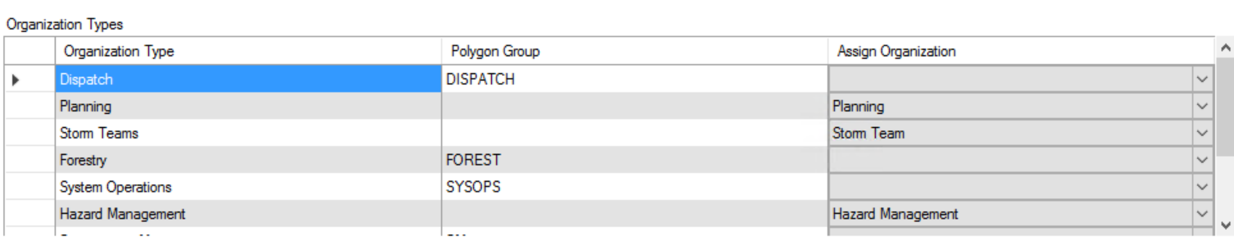


Figure 29. Organization Types Configuration

The Organization Types table includes the following fields:

- **Organization Type:** The organization type to which the mapping entry applies.

- **Polygon Group:** The polygon group to which the organization type is mapped. If specified, the incident and order are assigned to an organization based on a polygon group. If not specified, the Assign Organization must be specified.

  > **Note**
  >
  > Polygon groups are defined in the customer model using the Customer List Compiler tool.

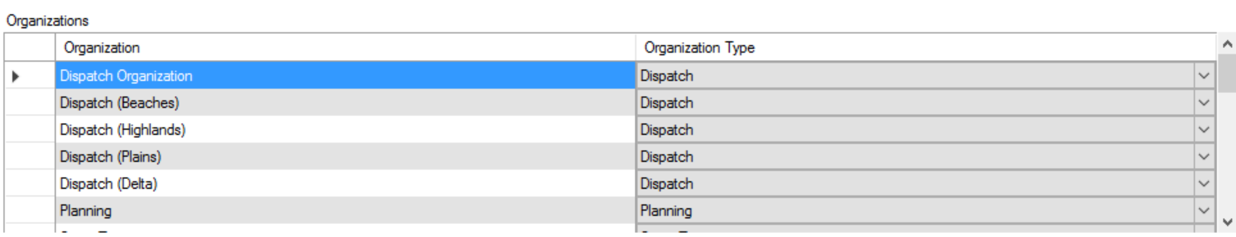
- **Assign Organization:** The organization to which the organization type is mapped. If specified, the incident and order are assigned to this specific organization.

  If both the Polygon Group and Assign Organization fields are specified, the system first tries to determine an organization based on the polygon group. If the system cannot resolve an organization based the polygon group, the system uses the Assign Organization name.

  If the system fails to determine an organization based on the Organization tables, the default organization defined using the DEFAULT_ORDER_ORGANIZATION_NAME parameter in the `NetServerOmsConfig.txt` configuration file is used.

18.2. Organizations

The Organizations table defines mapping between organization types and organizations.


Organizations

Organization	Organization Type
Dispatch Organization	Dispatch
Dispatch (Beaches)	Dispatch
Dispatch (Highlands)	Dispatch
Dispatch (Plains)	Dispatch
Dispatch (Delta)	Dispatch
Planning	Planning

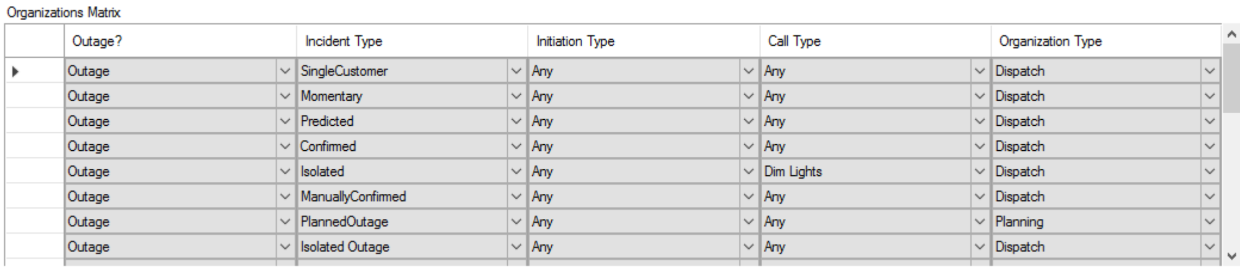
Figure 30. Organizations Configuration

The Organizations table includes the following fields:

- **Organization:** The organization name. Organization names correspond to AREA names in the Permit database.
- **Organization Type:** The organization type.

18.3. Organizations Matrix

The Organizations Matrix table defines organization types that are assigned for specific incident and call types. For example, you can specify that a Construction-Maintenance organization is assigned to a non-outage incident of a StreetLight type that originated from a customer call of an SL Out type.


Organizations Matrix

	Outage?	Incident Type	Initiation Type	Call Type	Organization Type
	Outage	SingleCustomer	Any	Any	Dispatch
	Outage	Momentary	Any	Any	Dispatch
	Outage	Predicted	Any	Any	Dispatch
	Outage	Confirmed	Any	Any	Dispatch
	Outage	Isolated	Any	Dim Lights	Dispatch
	Outage	ManuallyConfirmed	Any	Any	Dispatch
	Outage	PlannedOutage	Any	Any	Planning
	Outage	Isolated Outage	Any	Any	Dispatch

Figure 31. Organizations Matrix Configuration

The Organizations Matrix table includes the following fields:

- **Outage:** The outage or non-outage incident type.
- **Incident Type:** The incident type.
- **Initiation Type:** The initiation type, such as an incident subtype (for example, Commanded) or planned incident type (for example, Emergent).
- **Call Type:** The call type.
- **Organization Type:** The organization type that is mapped to the specified incident and call types.

19. Incident Priority Tab

The Incident Priority tab allows you to define priority settings for different incident conditions, such as customer types, area setting details, and outage duration. The system calculates all weight values based on the incident’s conditions and uses the Priority Index value to sort incidents by priority on the Incident Detail grid of the OMS display plug-in.

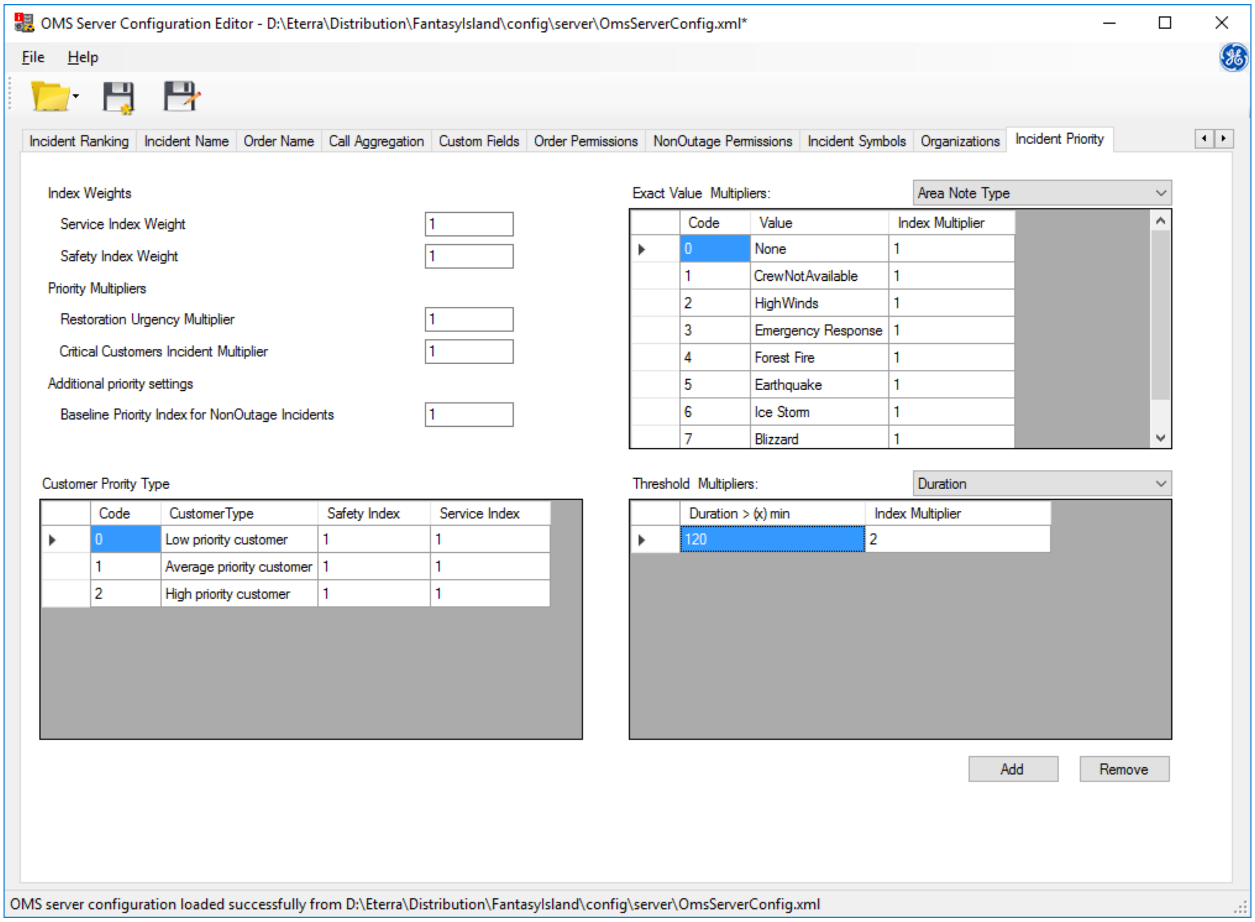


Figure 32. Incident Priority Tab

The Index Priority Weights Type group includes the following fields:

- **Safety Index Weight:** The weight for the safety index. The safety index for the incident is calculated as a sum of safety indexes for all affected customers. The safety index weight is used for calculating the baseline incident priority index.
- **Service Index Weight:** The weight for the service (customer impact) index. The service index for the incident is calculated as a sum of service indexes for all affected customers. The service index weight is used for calculating the baseline incident priority index.
- **Restoration Urgency Multiplier:** The multiplier for restoration urgency. Used for calculating the adjusted incident priority index.
- **Critical Customers Incident Multiplier:** The multiplier for affected critical customers. Used for

calculating the adjusted incident priority index.

- **Baseline Priority Index for Non-Outage Incidents:** The fixed baseline priority index for non-outage incidents.

The Customer Priority Type grid includes the following fields:

- **Code:** The customer priority type key value as defined using the CustomerPriorityType coded translation item.
- **Customer Type:** The customer priority type short name value as defined using the CustomerPriorityType coded translation item.
- **Safety Index:** The safety index for the specified customer priority type.
- **Service Index:** The service (customer impact) index for the specified customer priority type.

The Exact Value Multipliers grid includes the following fields:

- **Type:** The coded translation item for which the multiplier settings are specified (AreaNote, Priority_Code, IncidentType, or OutageDeviceType).
- **Code:** The key value of the associated coded translation item.
- **Code:** The short name value of the associated coded translation item.
- **Index Multiplier:** The multiplier for coded translation item type.

The Threshold Multipliers grid includes the following fields:

- **Type:** The threshold multiplier type for which the priority settings are specified (Duration, Customer Count, or CMI).
- **Threshold > X:** Specifies the threshold value.
- **Index Multiplier:** The multiplier for exceeding the threshold.

The Exact Value and Threshold multipliers are used for calculating the adjusted incident priority index.

19.1. Configuring the Incident Priority

To configure the incident priority calculation settings, do the following:

- Customize the following coded translation items: CustomerPriorityType, AreaNote, Priority_Code, IncidentType, and OutageDeviceType. For more details, refer to the *Coded Translation Editor User Guide*.
- Specify incident priority weights.
- Specify safety and service indexes for customer priority types.
- Specify multipliers for specific area setting, incident, priority, and outage device types.
- Specify the incident duration, customer count, and CMI thresholds and multipliers for exceeding these threshold values.

19.2. Calculating the Incident Priority

The incident priority indexes are calculated as follows:

- **Service Index:** Calculated as a sum of service indexes for all affected customers (service indexes are based on customer types).

$$\sum_{Customer} ServiceIndex_{CustomerType}$$

- **Safety Index:** Calculated as a sum of safety indexes for all affected customers (safety indexes are based on customer types).

$$\sum_{Customer} SafetyIndex_{CustomerType}$$

- **Baseline Priority Index:** For outage incidents, calculated as a sum of safety indexes multiplied by safety index weights and service indexes multiplied by service index weights. For non-outage incidents, the value is fixed and set to the Baseline Priority Index for Non-Outage Incidents value.

```
BaselineIncidentPriorityIndex = SafetyIndex*SafetyWeight + ServiceIndex*ServiceWeight
```

- **Adjusted Priority Index:** Calculated as a baseline priority index multiplied by the Restoration Urgency, Critical Customers, Exact Value, and Threshold multipliers (Area Setting Type, Outage Device Type, Duration, CMI, etc.).

```
AdjustedncidentPriorityIndex =
BaselineIncidentPriorityIndex*CMIMutiplier*CustomerCountMultiplier*AreaNoteTypeMutiplier*OutageDev
iceTypeMutiplier*DurationMultiplier*CallSeverityMultiplier*IncidentTypeMultiplier*RestorationUrgen
cyMultiplier (applied if AreaNote.RestorationUrgency = True)*CriticalCustomerIncidentMultiplier
```

- **Priority Index:** If the incident's feeder restoration priority (FRP) value is set to a value other than "-1" (occurs only for breaker incidents when the area setting's UseFRP flag is set to True), the priority index is set to Double.MaxValue (1.7976931348623157E+308) minus FRP so that incidents with lower FRP are ranked higher. If FRP = -1, the priority index is set to the adjusted priority index.

```
Incident.FRP = Feeder.FRP (if AreaNote.UseFrp = True and Incident.DeviceType = Breaker)

Incident.FRP = -1 (if AreaNote.UseFrp != True or Incident.DeviceType != Breaker)

Incident.PriorityIndex = (Int32.MaxValue) - Incident.FRP (if Incident.FRP != -1)

Incident.PriorityIndex = AdjustedncidentPriorityIndex (if Incident.FRP = -1)
```

The Service, Safety, Baseline Priority, Adjusted Priority, and Priority indexes are recalculated when an incident is created, the list of affected customers is changed, or the OMS server is restarted. The Adjusted Priority and Priority indexes are also recalculated at the time period specified by the PriorityRecalculationIntervalInSeconds parameter in `OmsServerConfig.xml` (by default, once a minute).

20. Validation Rules Tab

You can use the Validation Rules tab to set validation rules for incident clearance.

> **Note**
>
> The validation rules described in this section are applied only to outage incidents.

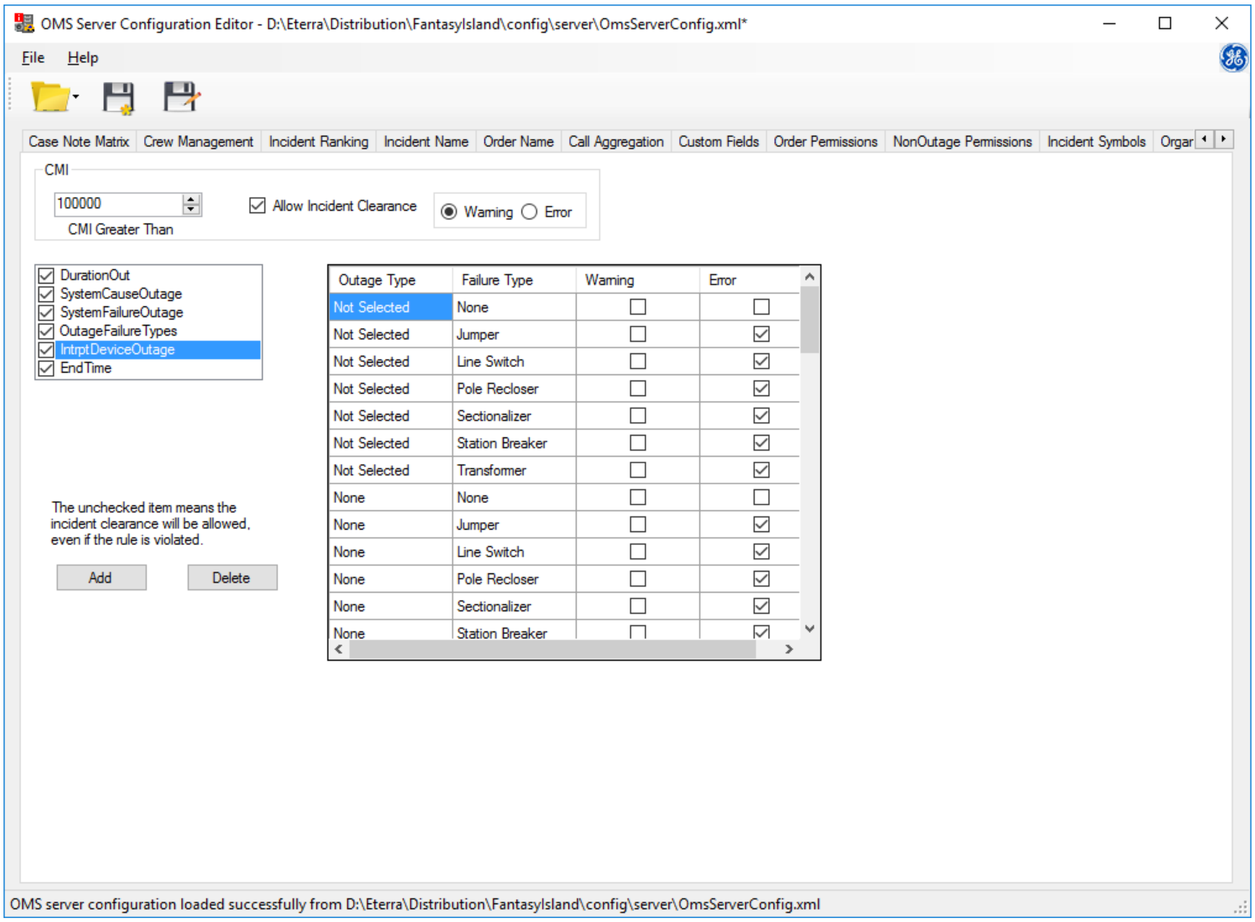


Figure 33. Validation Rules Tab

- **Customer Minutes Interrupted:** The CMI group sets the validation rules for an incident clearance when Customer Minutes Interrupted(CMI) exceeds a certain number.
    - **CMI Greater Than:** Sets the number of CMI. If this number is exceeded, a warning or an error message appears during the incident clearance.
    - **Allow Incident Clearance:** Specifies whether the incident clearance is allowed when the CMI number is exceeded.
    - **Warning/Error:** Specifies whether a warning or an error is issued during the incident clearance when the CMI number is exceeded.
- **Validation Rules:** The set of validation rules is available in the list box under the CMI group. When a rule is selected, the rule conditions appear in the table at the right side.

- **Enable/Disable:** When a rule is enabled (checked), the rule conditions are considered during the incident clearance and a warning or error appears if the rule is violated.
- **Add:** Shows the Add Validation Rule dialog.
- **Delete:** Deletes the selected validation rule.

> **Note**
>
> You cannot delete default validation rules. You can disable them, as appropriate.

20.1. Creating an Incident Clearance Validation Rule

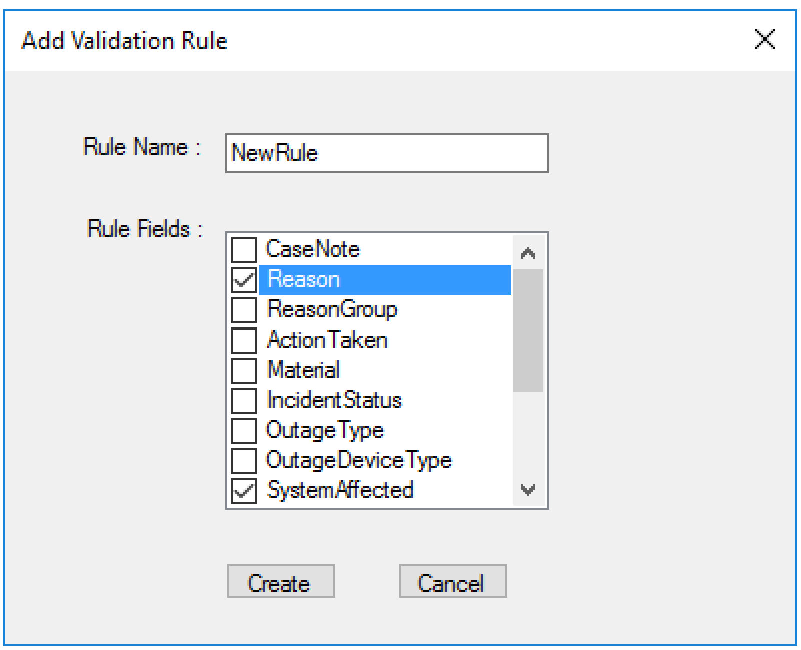
This section describes how to create a validation rule for an incident clearance. For example, you can define that the system shows an error message if the operator does not assign a cause of the outage for the overhead system or selects an invalid cause (for example, "dig in").

To create a validation rule:

1. Click the Add button.

   The Add Validation Rule form appears.

2. Enter the rule name.
3. Select the fields of the Update Incident pop-up to be validated. Up to four coded translation items (fields) can be selected.

4. Click Create.

   The validation rule is created for the selected fields. The rule appears at the bottom of the

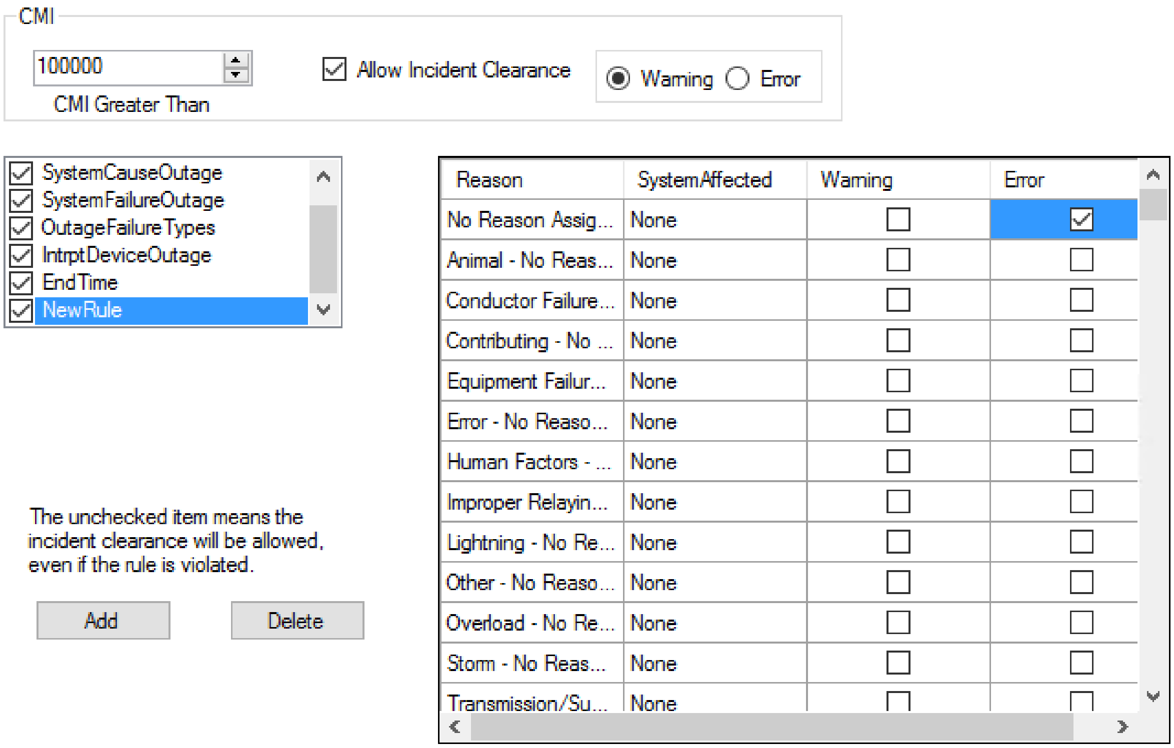
Validation Rules list.

5. Select the rule in the Validation Rules list.

   A table with the corresponding conditions appears at the right side of the screen.

6. Reorder any condition columns, as appropriate.

7. Set a Warning flag or an Error flag for the conditions you want to enforce during the incident clearance.



   This validation rule example defines that an error appears when No Reason Assigned is set in the Cause field (Reason) of the Update Incident display when None is selected in the System Affected field.

8. Save the changes via the File/Save menu.

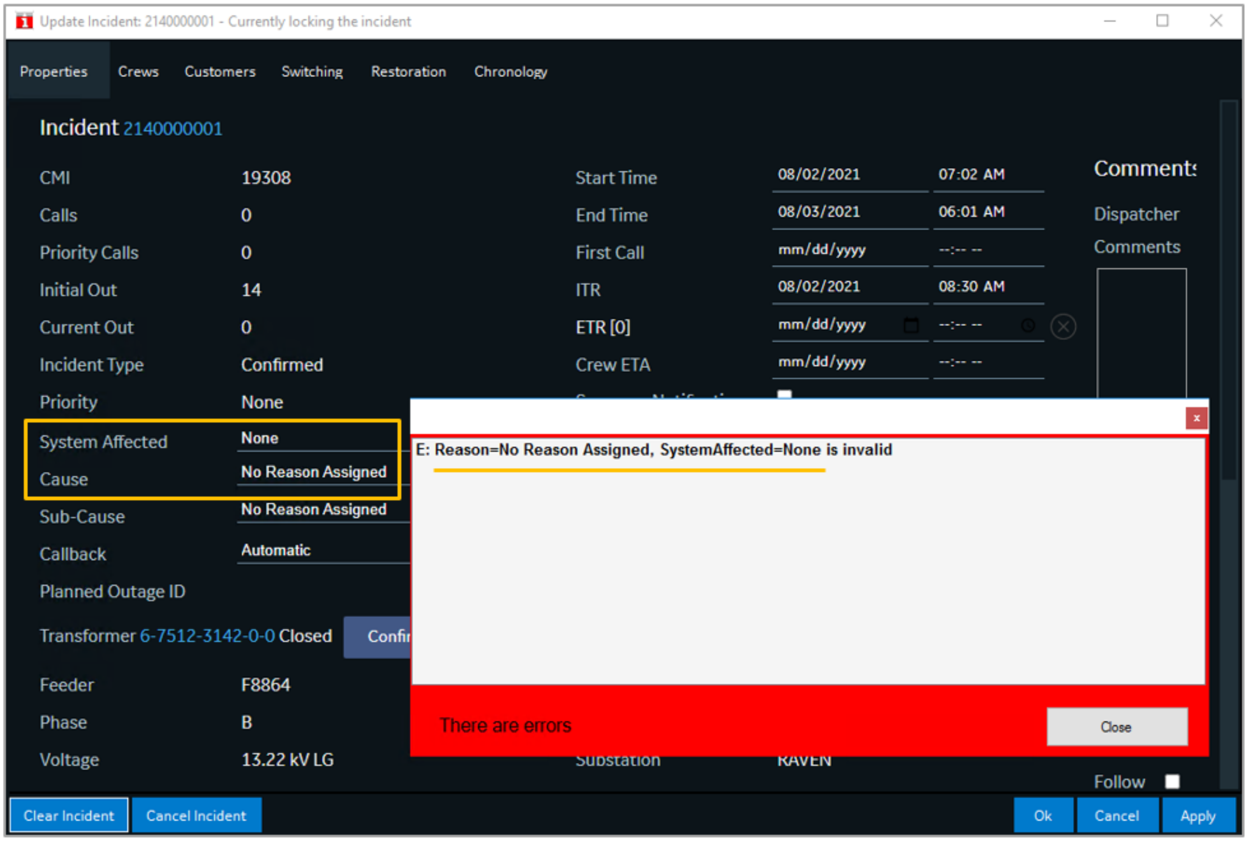


Figure 34. Incident Clearance Validation Example

21. Call Prioritization Tab

You can use the Call Prioritization tab to define which call priority type is applied to a call based on the call type, condition, and description parameters.

For a customer call, the system applies a call priority type according to the rule with the maximum number of matches of the Call Type, Condition 1, Condition 2, and Description fields, where the N/A value is equal to mismatch.

> **Note**
>
> Only one call priority type rule can be satisfied for a customer call. When two or more call priority type rules have the same number of matches, only the topmost rule is applied. Therefore, it is necessary to carefully sort the rule entries to get the desired behavior.

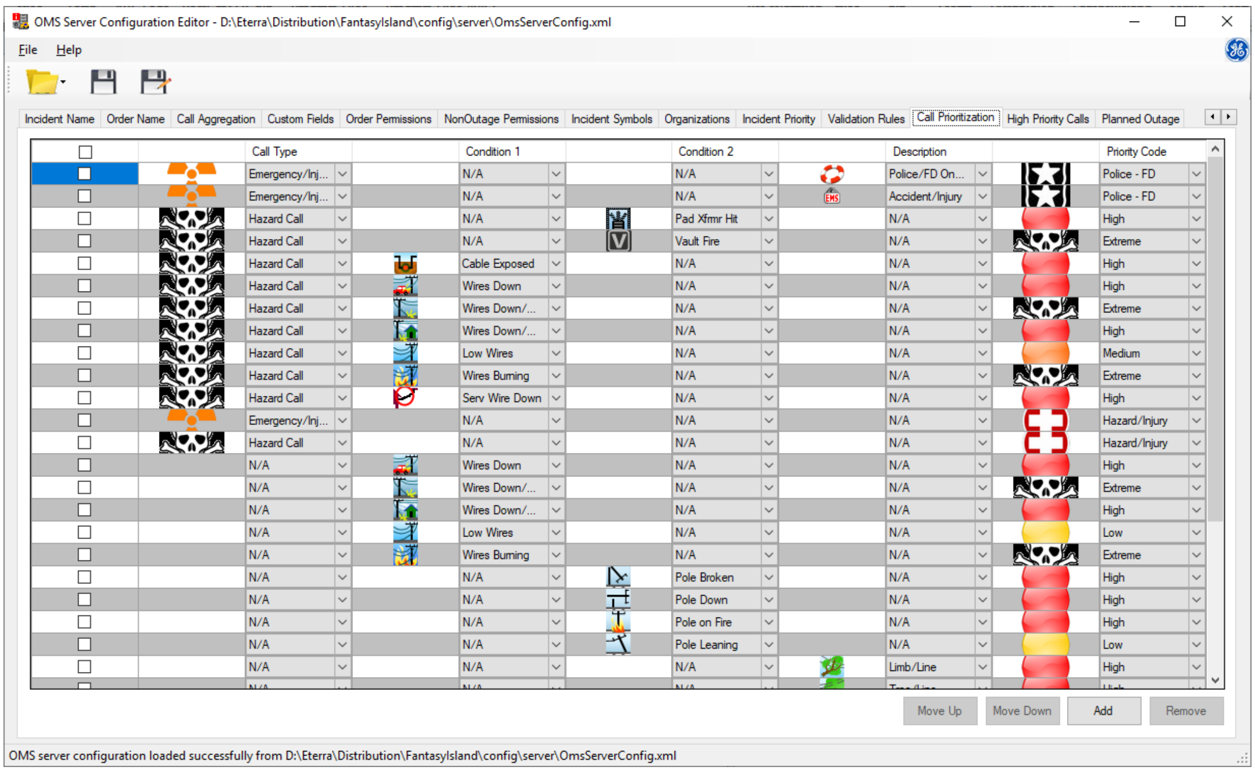


Figure 35. Call Prioritization Tab

The Call Prioritization tab includes the following fields:

- **Call Type:** Describes the type of problem (for example, lights out, area outage, and blinking lights).
- **Condition 1:** Describes the first trouble code condition (for example, wires down, wires burning, etc.).
- **Condition 2:** Describes the second trouble code condition from the unassociated call (for example, pole on fire, pole leaning, etc.).

- **Description:** Displays the trouble code description, if any (for example, pole/house, loud noise, etc.).
- **Priority:** Defines the call priority type to be assigned to the specified Call Type, Condition 1, Condition 2, and Description fields combination.
- **Move Up:** Move a row up.
- **Move Down:** Move a row down.
- **Add:** Add a new call priority type configuration.
- **Remove:** Remove a call priority type configuration.

21.1. Adding a New Call Prioritization Rule

To add a new call prioritization configuration:

1. Click Add.

   A new blank row is added.
2. Select as needed from the corresponding drop-down lists: Call Type, Description, Condition 1, and Condition 2 parameters.
3. Select the call priority type from the Priority drop-down list.
4. Navigate to File> Save to save the changes.

21.2. Removing a Call Prioritization Rule

To remove a call prioritization configuration:

1. Select the checkbox for the row to be deleted.
2. Click Remove.
3. Navigate to File> Save to save the changes.

22. High Priority Calls Tab

You can use the High Priority Calls tab to define which call priority types are considered a high priority call. For a high priority call, the PriorityCall flag is set to True and the Priority Call field is set to Yes in the OMS Display plugin.

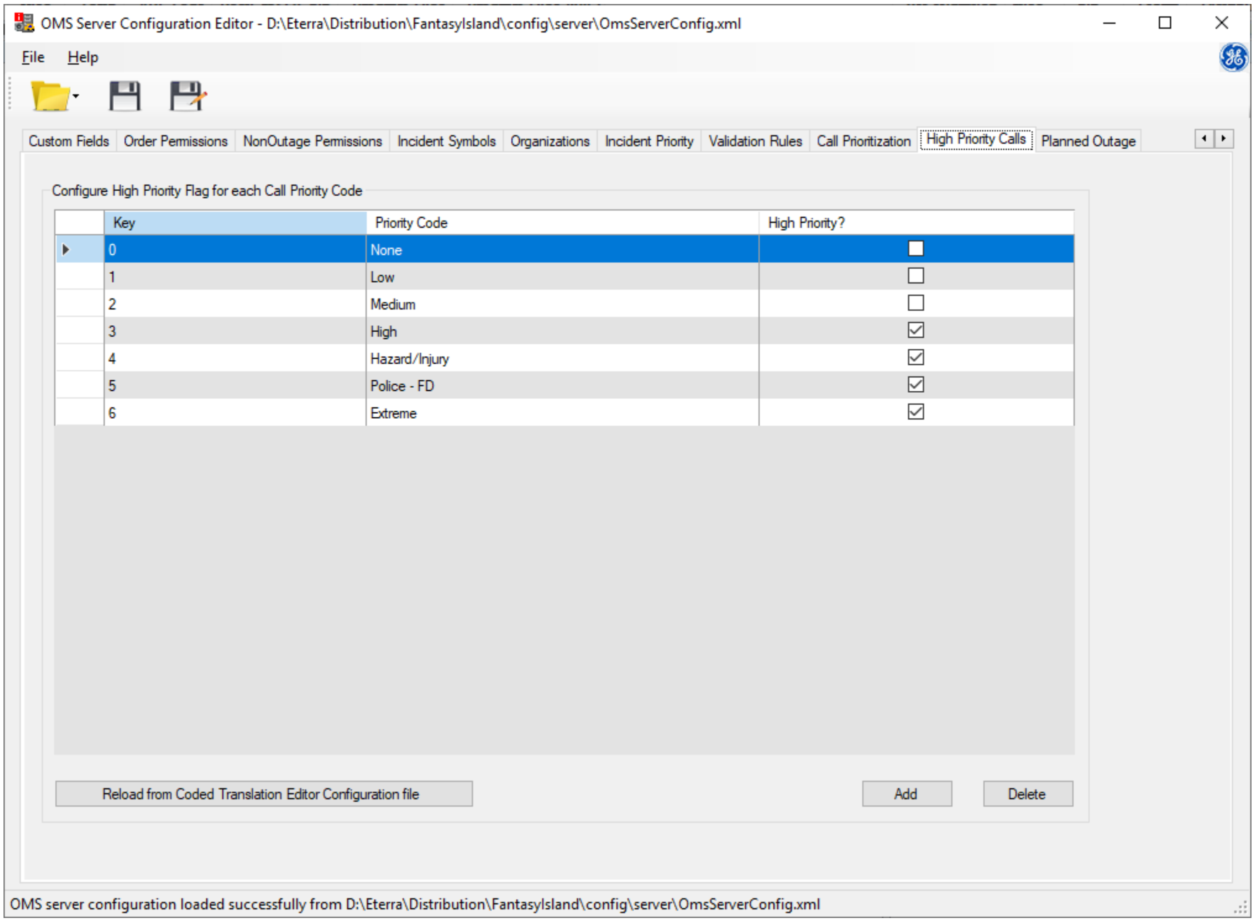


Figure 36. High Priority Calls Tab

The High Priority Calls tab includes the following fields:

- **Key:** The key value of the Priority_Code coded translation item's entry.
- **Priority Code:** The short name value of the Priority_Code coded translation item's entry.
- **High Priority?:** Defines whether the call with the associated Priority_Code value (call priority type) is considered a priority call.
- **Reload from Coded Translation Editor Configuration File:** Reloads the Priority_Code coded translation item entries from the `CodedTranslationConfig.xml` file.

23. Planned Outage Tab

You can use the Planned Outage tab to define planned outage notification, conversion, and retirement behavior.

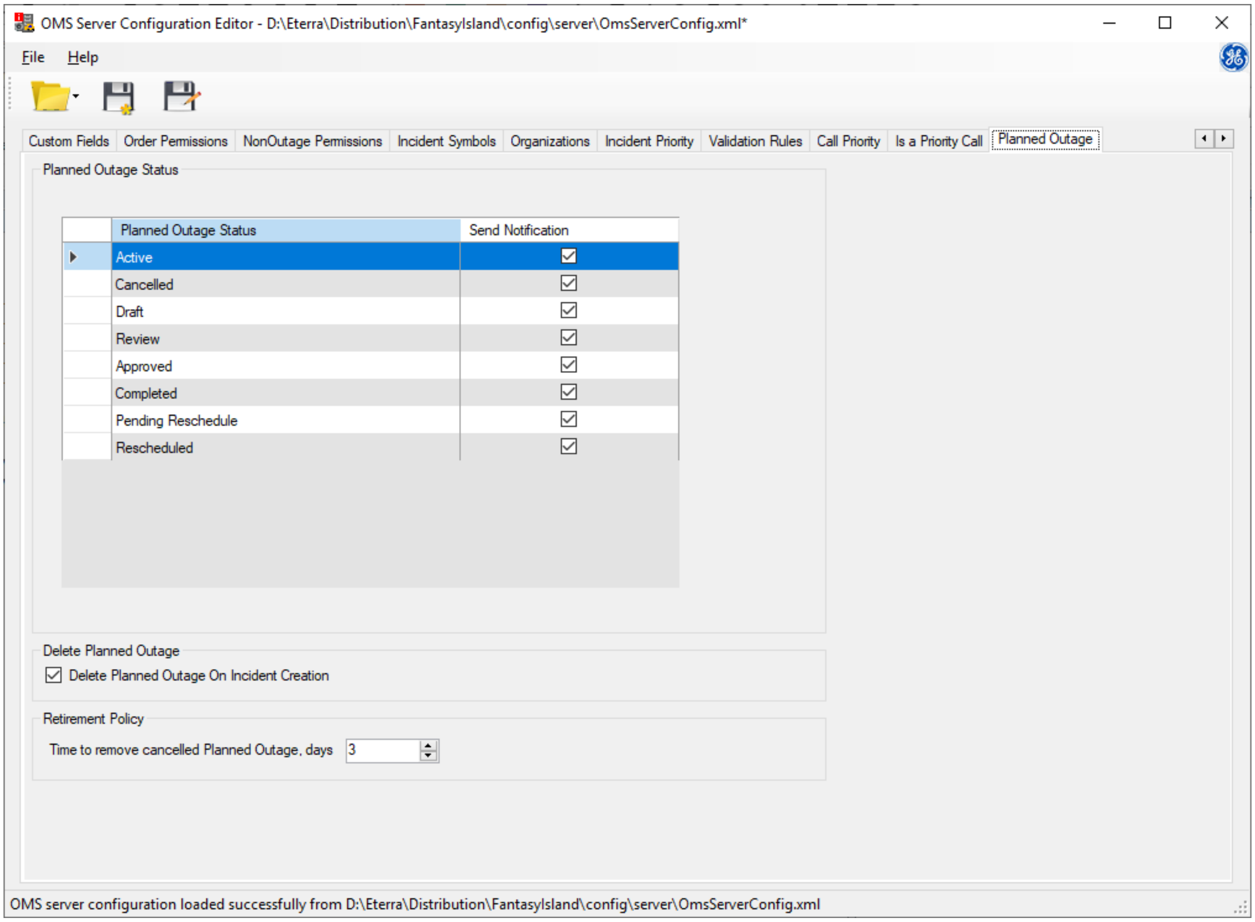


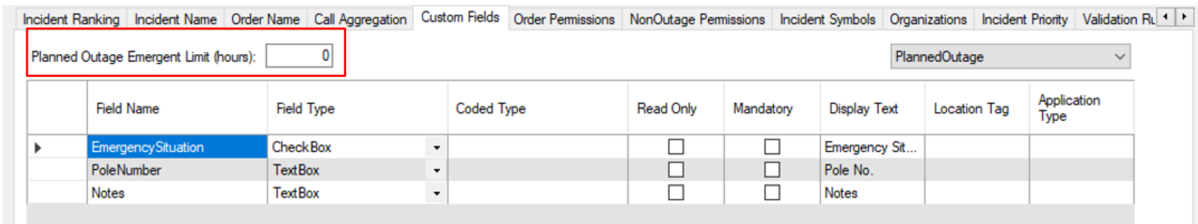
Figure 37. Planned Outage Tab

The Planned Outage tab includes the following fields:

- **Planned Outage Status Grid:** Configures which planned outage statuses result in customer notifications.
  - **Planned Outage Status:** The planned outage status, as defined by the PlannedOutageStatus coded translation item.
  - **Send Notification:** Defines whether a customer notification is sent for the associated planned outage status.
- **Delete Planned Outage on Incident Creation:** Defines whether a planned outage is deleted after the planned outage is converted into a planned incident.
- **Time to Remove Cancelled Planned Outage, Days:** The number of days to wait before deleting a cancelled planned outage.
- **Planned Outage Emergent Limit (Hours):** The maximum number of hours from the planned

outage creation time till the planned outage start time to consider a planned outage emergent. If set to 0, a planned outage is never considered emergent.

> **Note**
>
> This option currently appears on the Custom Fields tab when the PlannedOutage object is selected.



24. Summary Matrix Tab

You can use the Summary Matrix tab to define summary fields for the Overview displays.

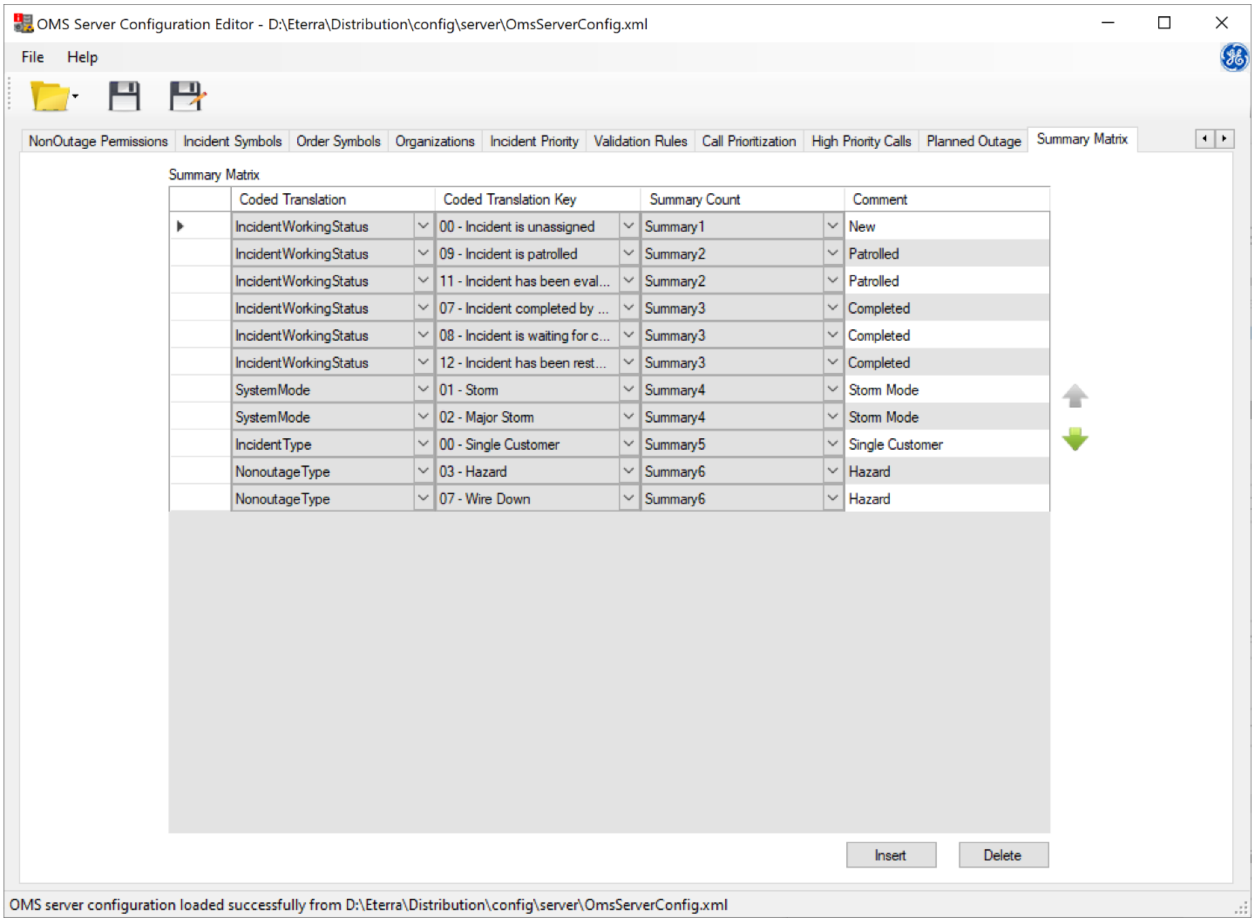


Figure 38. Summary Matrix Tab

These fields are available on the OMS Overview and grid tabular displays that show a summary of incidents, such as the Feeder, Station, and Service Center summaries.

The Summary Matrix tab includes the following fields:

- **Coded Translation:** These are coded translations related to an incident. There are four supported coded translation fields:
  - **Incident Working Status:** Uses the current working status of the incident.
  - **System Mode:** If there is an Area Note associated with the incident, this reflects the system mode of the Area Note.
  - **Incident Type:** This reflects the incident type. Some incident types are already calculated and are shown on the Overview and other summary displays.
  - **Non-outage Type:** For incidents that are not outage related, this reflects the type of the non-outage incident based on the customer call.

- **Coded Translation Key:** When a coded translation is selected, this column will change based on the Coded Translation defined in the first column.
- **Summary Count:** Select from SummaryCount1 through SummaryCount8.
- **Comment:** This field is used to document the intent of this row and is not used at run-time.

Rows may be inserted or deleted using the buttons on the tab, or by using the right-click menu on the row. Rows may be inserted in any order.

At run-time, the OMS server evaluates all rows in the summary matrix for every incident. If there is a match for the corresponding field of the incident, it increments the count of that summary field.

The field names to be used on the Overview and grid tabular summaries are SummaryCount1 through SummaryCount8.

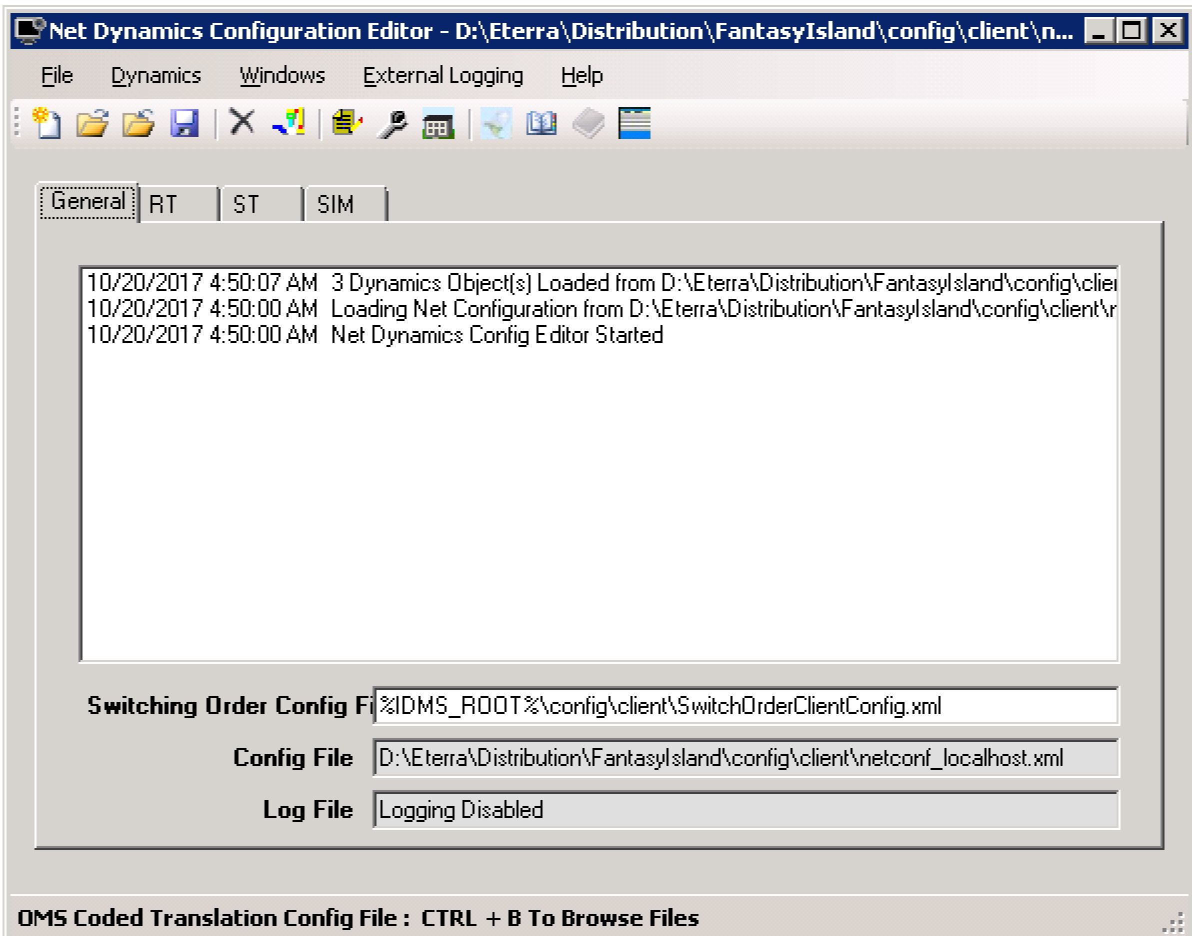
> **Note**
>
> If two Coded Translations are used for the same summary field, it is possible that the incident will be double counted.

25. Configuring the Dynamics Instance

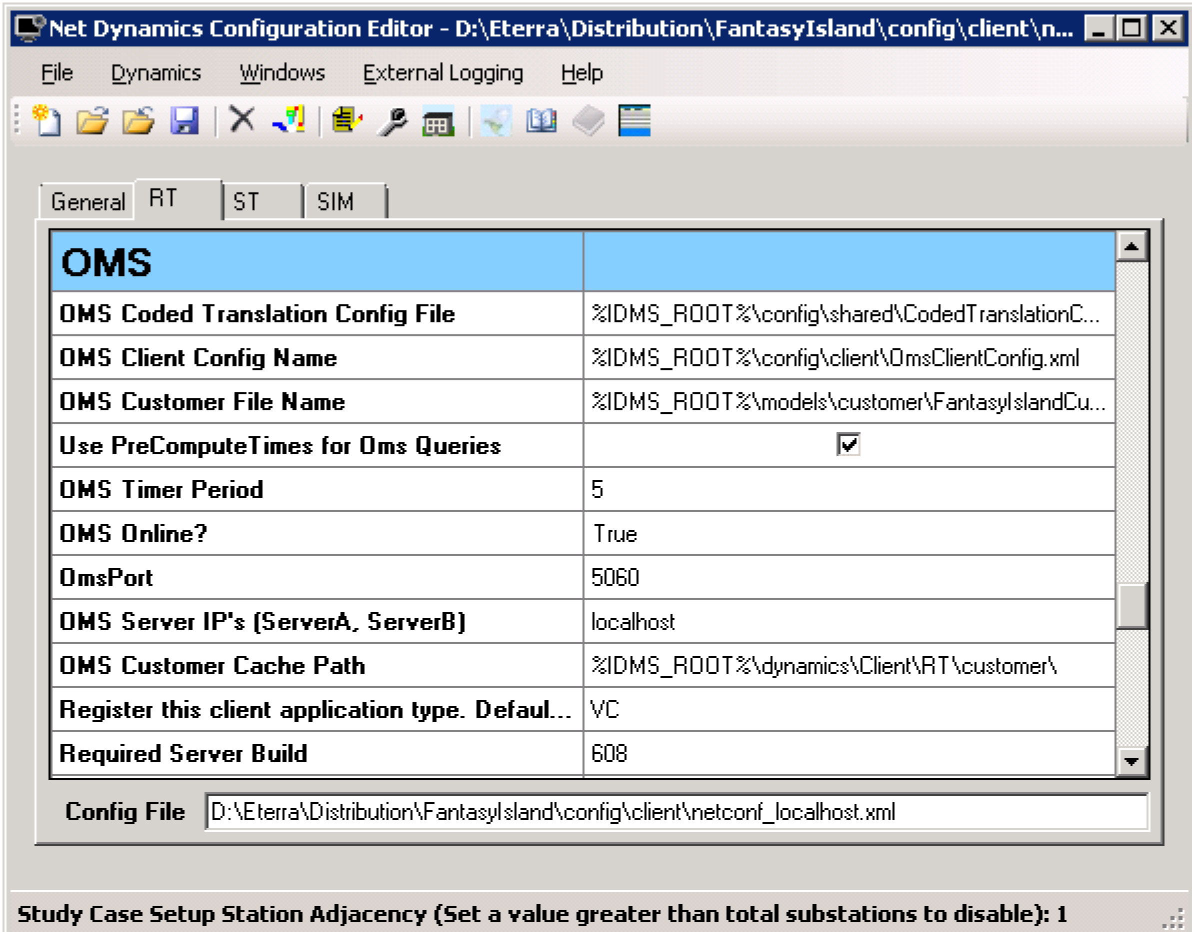
To configure the OMS parameters for RT, Study, or SIM dynamics:

1. Click Start> Programs> Eterra> Distribution> Tools> Configuration Tools> Net Dynamics Configuration Editor.

   The network configuration from the netconf_localhost.xml file, where <*localhost*> is a server name, is uploaded. Three new tabs named "RT"(real-time), "ST"(study), and "SIM"(training simulator) appear, as shown below:

2. Select the desired Dynamics tab and navigate to the OMS section.



The dynamics parameters associated with a selected configuration are shown.

Now you can modify the dynamics parameters, as described in the next step.

> **Note**
>
> All the path fields can be defined using absolute or relative paths. The product sample files use relative paths, so that the same configuration files can be used even if the installer chooses a non-default location.

3. Update the settings on the selected dynamics view. For more details about the settings, refer to the *Net Dynamics Configuration Editor User Guide*.

26. System Configuration

System configuration includes two important aspects: configuring the Outage Management System server parameters in the Net Server configuration file, and creating a separate instance of the dynamics using the Net Dynamics Configuration Editor (as described in Configuring the Dynamics Instance).

You can modify the Outage Management System server parameter settings using the `NetServerMainConfig_localhost.txt` configuration file, where *localhost* is a server name.

The default path to the configuration file is %IDMS_ROOT%\config\server\.

For the description of all OMS-related parameters, refer to the *ADMS Software Installation and Maintenance Guide*.

26.1. Modifying the Momentary Interruption Setting

1. Locate the OMS_MOMENTARY_OUTAGE_DURATION parameter in the NetServerMainConfig_localhost.txt configuration file.
2. Edit the time period for Momentary Interruption (the default setting is 60minutes).
3. Save the changes and close the file.

   The new parameter is applied after restarting the server.

26.2. Providing Crew Locations

The WFM_AVL_CAPABLE parameter in the NetServer configuration file (for example, `NetServerOmsConfig_localhost.txt`) specifies whether WFM is capable of providing crew locations.

WFM_AVL_CAPABLE = 0 (default value) means:

- When a crew is assigned, the location is instantly set to an incident's location.
- When an incident is moved, assigned crews of that incident are moved to the location near the incident.
- NetServer does not send VehicleQuery messages to CrewServer.

WFM_AVL_CAPABLE = 1 means:

- Crew location is provided by WFM.
- When an incident is moved, only the "assigned point" field is updated on the crew's side (i.e., crew is not moved).
- NetServer communicates with CrewServer normally.

26.3. OMS Polygon Groups

OMS polygon groups are defined in the oparea_polygongroup.xml file, located at %IDMS_ROOT%\models\RT. This XML file contains an area name, a service center name, and *X,Y* coordinates for each polygon group.

When the OMS system starts, it reads the file and draws the polygons on the geographic overview display of the OMS display plug-in. Also, when an unassociated incident occurs, the system determines the Area and the Service Center name for the incident based on the data in the `oparea_polygongroup.xml` file.

The path to the `oparea_polygongroup.xml` file is specified in the OMS Data Files section of the NetServer configuration file (`NetServerOmsConfig_localhost.txt`).

The OMS Polygon Group data can be modified directly in the XML file. If the shape file is available, the OMS Polygon Group data can be edited in the Customer List Compiler tool.

26.4. Provisioning STORM ETR for Planned Outage

STORM ETR behavior for planned outages is configured using the **DISABLE_STORM_AREANOTE_ETR_FOR_PLANNED_OUTAGES** flag in the NetServer configuration file

Configuration file:

```
C:\Eterra\Distribution\CurrentDev\FantasyIsland\config\server\NetServerOmsConfig_localhost.txt
```

Flag values

The flag supports the following values:

- 0 (Disabled – default)

  STORM ETR applies to planned outages.

  All incident types, including planned incidents, follow the STORM ETR.

- 1 (Enabled)

  STORM ETR does not apply to planned outages.

  Planned outages follow their prescribed ETR without deviation.

  All other incident types, except planned incidents, follow the STORM ETR.

Logging and auditing

All scenarios are logged for audit purposes.

Log location:

```
C:\Eterra\Distribution\CurrentDev\FantasyIsland\log\NetOMSServer
```

Each log entry captures the following details:

- Incident ID
- Timestamp
- Indicator showing whether the incident ETR was overridden

27. Adding a Custom Incident Working Status

Utilities may need to define a custom incident working status (IWS) to represent an incident status that is not defined in OMS by default or to add more granularity to an existing status.

> **Note**
>
> Additional incident working statuses are intended to provide additional information to operators and crews. However, the system does not support status transitions or other related logic for custom incident working statuses.

27.1. Defining a Custom Incident Working Status

The list of incident working statuses is defined using the IncidentWorkingStatus coded translation item in `CodedTranslationConfig.xml`. You can review the list using the Coded Translation Editor tool.

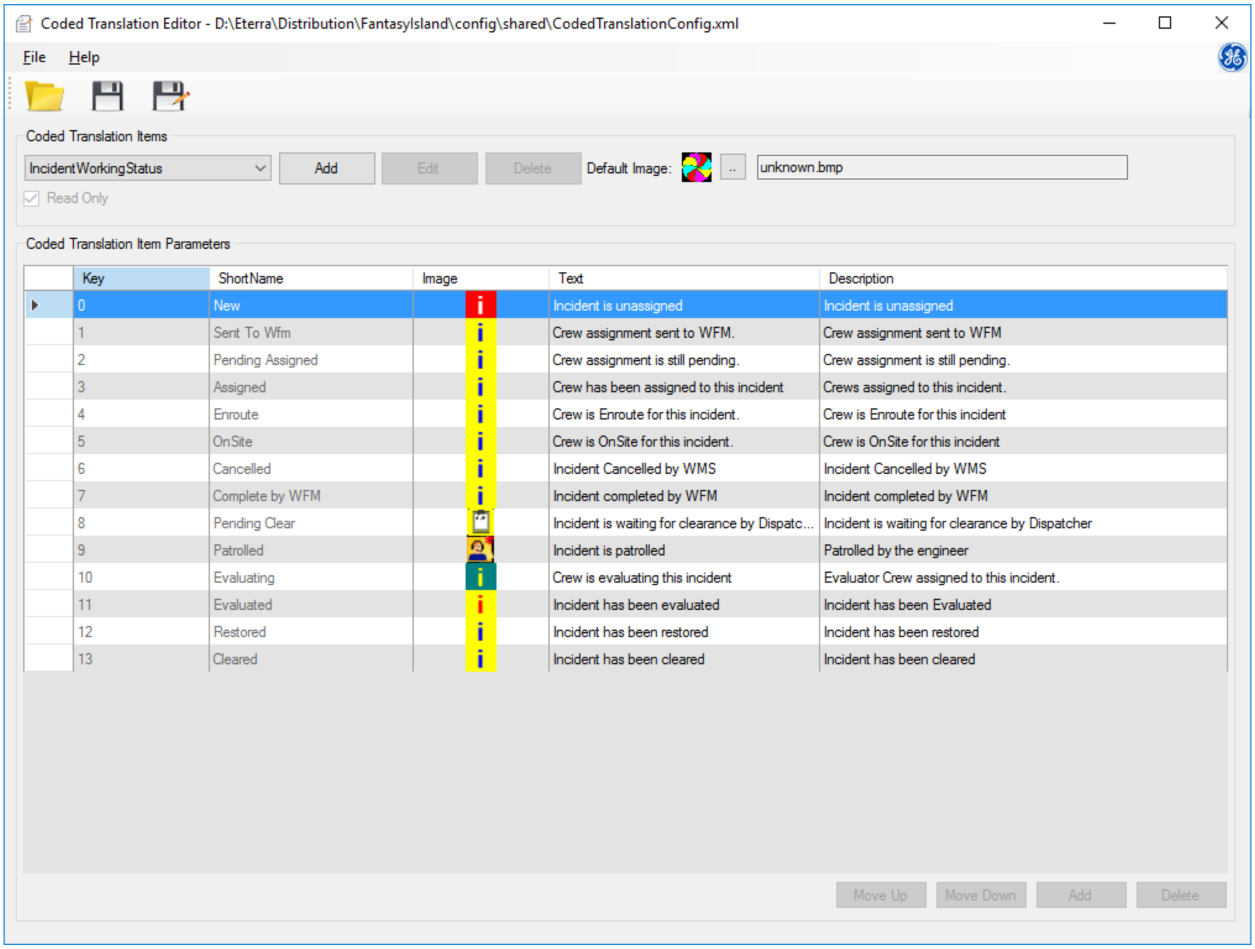


Figure 39. IncidentWorkingStatus Coded Translation Item

To add a custom incident working status:

1. Navigate to CodedTranslationConfig.xml in %IDMS_ROOT%\config\shared.
2. Open the file using a text editor. (The IncidentWorkingStatus coded translation item is read-only and cannot be edited using the Coded Translation Editor).
3. Add a new coded translation entry under the IncidentWorkingStatus coded translation item.

   The <Key> value assigned to the incident working status is important. Key values "0" to "14" and "100" to "110" are reserved for OMS required statuses. Custom incident working status must use key values between "14" and "100". We recommend that you start your custom incident working statuses with the key value "50", which provides many unused status key values for OMS future use.

   For example, to add a NonEvaluatorCrewPatrolled status with a key value of 50:

```
CodedTranslation>
  Name>IncidentWorkingStatus</Name>
  DefaultImageName>unknown.bmp</DefaultImageName>
  ReadOnly>true</ReadOnly>
  Entries>
...
    CodedTranslationEntry>
      Key>50</Key>
      ImageName>NonEvaluatorCrewPatrolled.bmp</ImageName>
      ShortName>Patrolled by non-evaluator crew</ShortName>
      Text>Incident has been patrolled by non-evaluator crew</Text>
      Text1>Incident has been patrolled by non-evaluator crew</Text1>
    /CodedTranslationEntry>
  /Entries>
/CodedTranslation>
```

Key	ShortName	Image	Text	Description
0	New		Incident is unassigned	Incident is unassigned
1	Sent To Wfm		Crew assignment sent to WFM.	Crew assignment sent to WFM
2	Pending Assigned		Crew assignment is still pending.	Crew assignment is still pending.
3	Assigned		Crew has been assigned to this incident	Crews assigned to this incident.
4	Enroute		Crew is Enroute for this incident.	Crew is Enroute for this incident
5	OnSite		Crew is OnSite for this incident.	Crew is OnSite for this incident
6	Cancelled		Incident Cancelled by WMS	Incident Cancelled by WMS
7	Complete by WFM		Incident completed by WFM	Incident completed by WFM
8	Pending Clear		Incident is waiting for clearance by Dispatc...	Incident is waiting for clearance by Dispatcher
9	Patrolled		Incident is patrolled	Patrolled by the engineer
10	Evaluating		Crew is evaluating this incident	Evaluator Crew assigned to this incident.
11	Evaluated		Incident has been evaluated	Incident has been Evaluated
12	Restored		Incident has been restored	Incident has been restored
13	Cleared		Incident has been cleared	Incident has been cleared
50	Patrolled by non-evaluator crew		Incident has been patrolled by non-evaluat...	Incident has been patrolled by non-evaluator crew

27.2. Using Custom Incident Working Statuses

A custom incident working status can be set for an incident using the Working Status dropdown list on the Update Incident display.

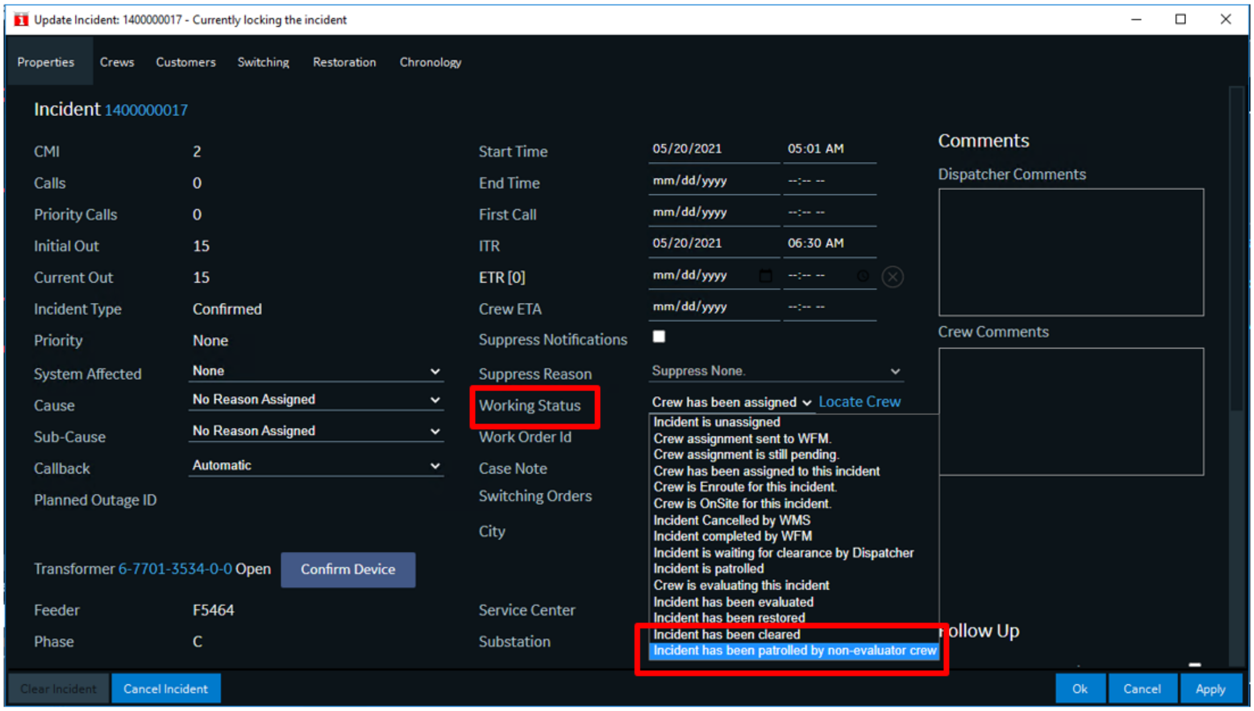


Figure 40. Setting a Custom Incident Working Status from the Update Incident Display

27.3. Integration with the Work Force Management System

The crews managed by the Workforce Management System can set a custom incident working status using the IncidentStatus message type.

To set the custom incident working status, the StatusCode value in the IncidentStatus payload must be set to the custom incident working status key defined in the `CodedTranslationConfig.xml`.

```json
{
    "StatusCode": 50,
    "IncidentId": "1390000009",
    "StatusName": "WmqPatrolled",
    "CrewId": "",
    "ErrorNumber": "",
    "ErrorCode": "",
    "Comment": "",
    "ActualRestorationTime": "0001-01-01T00:00:00",
    "CauseGroup": 0,
    "Cause": 0,
    "OutageType": 0,
    "OutageTypeName": "NotSelected",
    "PhasesOut": 0,
    "PhasesOutName": "ABC",
    "FailureType": 0,
    "FailureMode": 0,
    "FailedEquipment": 0,
    "PowerQualityType": 0,
```


Figure 41. Setting a Custom Incident Working Status from the WFM